

Orthotropic Plasticity Model for Mineralised Tissues: Complete Implementation Guide for MOOSE

Material Repository Documentation

October 7, 2026

Abstract

This document describes a finite element framework for mineralised tissues, implemented as a custom MOOSE application. It covers the constitutive equations, their numerical implementation, the input parameters, installation and build, and the verification tests that accompany the code.

Bulk constitutive model. The bulk material is elastic-viscoplastic with an orthotropic quadric yield surface [1], an associated flow rule, six post-yield laws and a continuous Perzyna viscosity. The response is selected by a material-model flag. Six bone models reproduce the material sets of the reference Abaqus user material [5], including density- and fabric-based elasticity and strength [2]; a coral model describes orthotropic aragonite single crystals with explicit constants; a legacy mode leaves every constant to the user. All models share one yield function, one two-stage return mapping (condensed Newton iteration with a primal closest-point projection as fallback) and one consistent tangent. Per-element Euler angles, read for example from Exodus files, orient the material frame in polycrystalline volume elements.

Cohesive zone model. Grain-boundary failure is described by `HomogenizedExponentialCZM`, an exponential traction-separation law with a separate damage variable, parameterised from molecular dynamics simulations of protein-aragonite interfaces [3]. Contact heterogeneity within an interface point is represented by a quality factor integrated with five-point Gauss-Hermite quadrature, which is deterministic and independent of the element area. A deterministic spatial quality field desynchronises failure between interface points.

Verification. Three sets of automated tests are documented. `run_tests.sh` checks the analytical derivatives against finite differences, compares the consistent tangent with an elastic-tangent reference solution, forces the fallback stage of the return mapping, and measures the convergence of the global Newton iteration up to a bone specimen with 2.4 million degrees of freedom, on which quadratic convergence is obtained. `run_examples.sh` runs 57 small inputs covering every material model and three interface loading modes against predictions computed from the parameters. `verify_presets.py` checks all model presets against the formulas of the reference implementation without running MOOSE. A gold-file suite (`run_validation.sh`, 25 tests) guards against unintended changes. Comparison with experiments is not part of this document.

Implementation. C++ material classes `OrthotropicPlasticityStressUpdate`, `ComputeElasticityTensorCoupled`, `ComputeFabricElasticityTensor` and `HomogenizedExponentialCZM` for the MOOSE solid mechanics module.

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Chapter 1

Introduction

1.1 Overview

Coral aragonite exhibits extreme mechanical anisotropy at the single-crystal level, with:

- Young's moduli varying by factor of 2 ($E_1 = 140$ GPa vs $E_3 = 63$ GPa)
- Yield stresses varying by 30% across loading directions
- Direction-dependent post-yield softening behaviour
- Complex grain boundary interactions in polycrystals

This implementation provides a continuum-scale finite element model that captures these features, bridging molecular dynamics simulations to engineering-scale analysis.

1.2 Model Capabilities

Elastic behaviour:

- Orthotropic elasticity with nine independent constants, given explicitly (`ComputeElasticityTensorCoupled`) or derived from bone volume fraction and fabric (`ComputeFabricElasticityTensor`, Zysset and Curnier [2])
- Isotropic, transversely isotropic and fabric-based presets for trabecular and compact bone (Section 1.4)
- Optional density scaling of explicit constants by a power law
- Per-element orientation of the material frame by Euler angles

Yield behaviour:

- Quadric yield surface [1] with three tensile, three compressive and three shear strengths and three interaction parameters
- Tension-compression asymmetry through the linear term of the quadric
- Strengths given explicitly or derived from density and fabric, with the cortical correction for dense bone
- Convexity of the surface checked at start-up
- Associated flow rule

Post-yield behaviour:

- Perfect plasticity
- Linear and exponential hardening
- Three softening laws: the peak-type law of the reference bone implementation, exponential softening and piecewise linear softening with a configurable onset
- Initial yield ratio, so that the input strengths can be ultimate strengths

Rate dependence:

- Continuous Perzyna viscosity with five viscosity functions, or rate-independent

Numerical treatment:

- Two-stage return mapping: condensed Newton iteration, then primal closest-point projection with Armijo line search
- Consistent (algorithmic) tangent; quadratic convergence of the global iteration verified (Chapter 7)
- A failed local solve cuts the time step instead of stopping the run
- Diagnostic output: return-mapping stage, iteration counts, fallback counters

Polycrystalline microstructures:

- Per-element Euler angles through AuxVariables, read from Exodus files
- Cohesive interfaces between grains (Chapter 3)
- The angle convention differs from the Bunge convention of EBSD software; see Section 2.11 before importing measured orientations

Implemented but not verified:

- Isotropic damage coupled to the plastic strain (Section 2.10)

1.3 Material Classes Overview

Class	Purpose	Key Features
<code>OrthotropicPlasticityStressUpdate</code>	Bulk plasticity	Quadric yield, fabric/density scaling, return mapping
<code>ComputeElasticityTensorCoupled</code>	Explicit elasticity	Per-element Euler angles, density scaling
<code>ComputeFabricElasticityTensor</code>	Fabric elasticity	Zysset-Curnier model [2], trabecular bone
<code>HomogenizedExponentialCZM*</code>	Interface failure	MD-calibrated traction-separation

*Chapter 3

1.4 Material Models

The bulk response is selected by a single name, the *material model*, which fixes the elastic formulation, the yield surface and the post-yield law together and supplies default constants. Eight models are available:

- **Six bone models**, identical to the material sets of the reference Abaqus user material [5]: trabecular bone as isotropic, transversely isotropic, fabric-based transversely isotropic and fabric-based orthotropic (`trabecular_iso`, `trabecular_ti`, `trabecular_fabric_ti`, `trabecular_fabric_ortho`), and compact bone as isotropic and transversely isotropic (`compact_iso`, `compact_ti`). They take the bone volume fraction and, for the fabric models, the fabric eigenvalues as input.
- **coral**: full orthotropy for aragonite single crystals, with nine elastic constants, nine strengths and three interaction parameters given explicitly. It is normally combined with cohesive interfaces between grains.
- **legacy** (the default): the behaviour before the models were introduced, in which every constant is set by hand. Existing input files run unchanged.

All models share one yield function, one return-mapping algorithm and one tangent; they differ only in how the elastic constants and the twelve numbers that define the yield surface are obtained. The formulations are given in Chapter 2 (Sections 2.4 and 2.5.7), the names, constants and input rules in Chapter 6.

1.5 Application Domains

This framework supports distinct application domains with appropriate material models:

1.5.1 Dense Coral Aragonite

Material characteristics:

- Single-crystal orthorhombic structure
- Near-full density ($\rho \approx 1.0$)
- Strong elastic anisotropy ($E_1/E_2 \approx 2$)
- Moderate yield strength variation (30% across directions)

Recommended models:

- `material_model = coral` (Table 6.1); the three interaction parameters must be given
- Elasticity: `ComputeFabricElasticityTensor` with the `model` flag, or `ComputeElasticityTensorCoupled` with explicit constants
- Plasticity: `OrthotropicPlasticityStressUpdate`
- Interfaces: `HomogenizedExponentialCZM` for organic interface behaviour
- Microstructure: Per-element Euler angles for polycrystalline RVEs

1.5.2 Porous Coral Skeletons

Material characteristics:

- Hierarchical porosity (sclerodermites + inter-grain voids)
- Variable density ($\rho = 0.6\text{--}0.9$)
- Anisotropic pore structure following crystal orientation

Recommended models:

- `legacy` mode with explicit constants and density scaling; no preset exists for porous coral
- Elasticity: `ComputeElasticityTensorCoupled` with `density_rho` and `elasticity_density_exponent` ($k \approx 2$)
- Plasticity: `OrthotropicPlasticityStressUpdate` with `yield_density_exponent` ($p \approx 1.5\text{--}2$)
- Scaling: power law ($E = E_0\rho^k$, $\sigma = \sigma_0\rho^p$)

1.5.3 Trabecular Bone (Extended Application)

Material characteristics:

- Porous foam structure
- Low density ($\rho = 0.1\text{--}0.4$ typical)
- Fabric tensor from μ CT imaging
- Mild to strong anisotropy ($m_1/m_3 = 1.2\text{--}3$)

Recommended models:

- `material_model = trabecular_fabric_ortho` (or one of the other five bone models) in `GlobalParams`, so that elasticity and plasticity use the same preset
- Elasticity: `ComputeFabricElasticityTensor`
- Plasticity: `OrthotropicPlasticityStressUpdate`
- Input: bone volume fraction and fabric eigenvalues (m_1, m_2, m_3); the tissue constants, exponents and interaction parameters come from the preset and can be overridden individually
- The older switch `use_fabric_scaling` belongs to the `legacy` mode and is not needed with a bone model

Fabric tensor acquisition: Fabric eigenvalues m_i obtained from:

1. μ CT scan of trabecular bone
2. Mean intercept length (MIL) or volume orientation (VO) analysis
3. Principal directions and eigenvalues satisfy $m_1 + m_2 + m_3 = 3$
4. Eigenvalues range: $m_i \in [0.1, 2.9]$ (extreme anisotropy) to $m_i = 1.0$ (isotropy)

1.6 Scope and Structure

This document describes the equations, the code and its verification. It does not report a comparison with experiments.

- **Chapter 2:** Bulk material: kinematics, elasticity, yield surface, post-yield laws, viscosity, return mapping, consistent tangent, orientation
- **Chapter 3:** Cohesive zone model
- **Chapter 4:** MOOSE installation
- **Chapter 5:** Building the application and running the gold-file tests
- **Chapter 6:** Material models, preset constants, parameter reference, input templates and solver settings
- **Chapter 7:** Verification: automated tests, earlier studies, example inputs
- **Chapter 8:** Troubleshooting

A reader who wants to run a simulation can start with Chapters 4 and 5, take a template from Section 6.7 and the solver settings from Section 6.8. A reader who changes the code should start with Chapter 2 and Section 7.5.

Chapter 2

Mathematical Formulation

This chapter states the bulk constitutive model as it is implemented in three classes:

- `OrthotropicPlasticityStressUpdate`: yield surface, post-yield evolution, return mapping and tangent;
- `ComputeFabricElasticityTensor`: elasticity from material-model presets, density and fabric;
- `ComputeElasticityTensorCoupled`: elasticity from explicit constants.

Where the implementation departs from the published model, the departure is stated. The cohesive zone model is treated separately in Chapter 3.

2.1 Sources, Scope and Notation

2.1.1 Sources

The bulk model is a port of an existing constitutive model, not a new one. Its parts come from the following sources:

- **Yield criterion:** the generalized anisotropic quadric of Schwiedrzik et al. [1], including the fabric- and density-based scaling of its coefficients.
- **Constitutive framework:** the elastic-viscoplastic damage model for bone of Schwiedrzik and Zysset [4], as documented for the reference implementation by Zysset and Schwiedrzik [5]. That document defines the internal variables, the continuous Perzyna formulation, the post-yield functions and the six bone material sets used here.
- **Application to coral:** Wolfram et al. [6] (supplementary Section S7) apply the same framework to cold-water coral skeleton, without damage and under a stated small-deformation assumption.
- **Fabric elasticity:** Zysset and Curnier [2].
- **Reference code:** the Abaqus user material `UMAT_QUADRIC_PRIMAL_Major.f`. The return mapping, the line search and the six bone presets are transcribed from it; `MaterialModelPresets.h` records the UMAT line each constant was taken from.
- **Framework:** the MOOSE solid mechanics module [7], which supplies the kinematics, the global Newton solve and the stress-update interface.

Two parts of the UMAT are deliberately not carried over: its densification law and its damage law. Orientation is extended: besides the UMAT's main-axis flag, the material frame can be rotated per element by Euler angles (Section 2.11).

2.1.2 Material Models Covered

The chapter describes one constitutive model in eight parameterisations, the material models of Table 6.1. They fall into four families:

Table 2.1: Families of material models and where each part is defined.

Family	Models	Input	Symmetry
Density-based	<code>trabecular_iso</code> , <code>compact_iso</code> , <code>trabecular_ti</code> , <code>compact_ti</code>	volume fraction ρ ; main axis for TI	isotropic or trans- versely isotropic
Fabric-based	<code>trabecular_fabric_ti</code> , <code>trabecular_fabric_ortho</code>	ρ and fabric eigenval- ues m_1, m_2, m_3	transversely isotropic or or- thotropic
Explicit	<code>coral</code>	nine stiffnesses, nine strengths, three ζ_{ij}	orthotropic
Legacy	<code>legacy</code>	all constants by hand; optional fabric scal- ing	as specified

What differs between the models is confined to two places: how the elastic stiffness is built (Section 2.4) and how the strengths and interaction parameters of the yield surface are obtained (Section 2.5.7). Everything else in this chapter, that is the kinematics, the yield function, the post-yield laws, the viscoplastic regularization, the return mapping and the tangent, is common to all of them.

2.1.3 Tensor Notation

Second-order tensors are written in bold ($\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$, $\boldsymbol{N}$), fourth-order tensors in blackboard bold ($\mathbb{C}$, $\mathbb{F}$, $\mathbb{A}$). The double contraction is $\boldsymbol{A} : \boldsymbol{B} = A_{ij}B_{ij}$ and $(\mathbb{C} : \boldsymbol{A})_{ij} = C_{ijkl}A_{kl}$, the dyadic product is $(\boldsymbol{A} \otimes \boldsymbol{B})_{ijkl} = A_{ij}B_{kl}$, and $\|\boldsymbol{A}\| = \sqrt{\boldsymbol{A} : \boldsymbol{A}}$ is the Frobenius norm. All tensors that enter the constitutive equations are symmetric, and all fourth-order tensors have both minor symmetries. Major symmetry is *not* assumed: the algorithmic operators of Sections 2.8 and 2.9 lack it.

Unless stated otherwise, components refer to the *material frame*, whose axes 1, 2, 3 are the orthotropy axes (crystal a , b , c for aragonite; fabric principal axes for bone). Stresses and strengths are in MPa.

2.1.4 Six-Component Representations

Tensor notation is used throughout this chapter. The code, however, stores symmetric tensors as six-component arrays in two different conventions, and confusing them resulted in an error in earlier versions (see the box at the end of this section). Both conventions use the index order

$$p = 0, \dots, 5 \leftrightarrow (ij) = 11, 22, 33, 23, 13, 12. \quad (2.1)$$

Converting a derivative taken with respect to the plain components back to a tensor requires the weights w : since $dY = \sum_p (\partial Y / \partial s_p) ds_p = \boldsymbol{G} : d\boldsymbol{\sigma}$ and each shear component appears twice

Table 2.2: Six-component conventions in the implementation. $w = (1, 1, 1, 2, 2, 2)$ and $c = (1, 1, 1, \sqrt{2}, \sqrt{2}, \sqrt{2})$.

Convention	Definition	Where it is used
Plain component	$s_p = \sigma_{ij}$, no factors. Quadric stored so that $\boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma} = \sum_{pq} s_p \mathbf{F}_{pq} s_q$, hence $\mathbf{F}_{55} = 1/\tau_{12}^2$.	<code>F.matrix</code> , <code>f.lin.vector</code> ; evaluation of the yield function, its gradient and its Hessian.
Mandel	$\hat{a}_p = c_p a_{ij}$ and $\hat{A}_{pq} = c_p c_q A_{ijkl}$. Matrix product, inverse, dot product and Euclidean norm then coincide with the tensor operations.	<code>MandelHelpers.h</code> ; the 7×7 system of the second return-mapping stage and the consistent tangent.
Tensor	<code>RankTwoTensor</code> , <code>RankFourTensor</code>	Interface to MOOSE; first return-mapping stage; rotations.

in the double contraction,

$$G_{ij} = \frac{1}{w_p} \frac{\partial Y}{\partial s_p}, \quad H_{ijkl} = \frac{1}{w_p w_q} \frac{\partial^2 Y}{\partial s_p \partial s_q}. \quad (2.2)$$

The reference UMAT works in Mandel notation internally, with the shear slots in the order 12, 13, 23, that is, slots 4 and 6 interchanged relative to Eq. (2.1). Its quadric entry `FFFF(4, 4) = 0.5/\tau^2` is the Mandel image of the same tensor whose plain-component entry is $1/\tau^2$, so shear strengths transfer between the two codes without a conversion factor.

The elastic input `C.ijkl` with `fill_method = symmetric9` takes true tensor components in the order C_{1111} , C_{1122} , C_{1133} , C_{2222} , C_{2233} , C_{3333} , C_{2323} , C_{1313} , C_{1212} , so the last three entries are the shear moduli G_{23} , G_{13} , G_{12} .

Corrections relative to earlier versions of this document and code

- The former `VoigtHelpers.h` copied tensor components into 6×6 arrays without any factors. Every shear compliance entering the tangent was wrong by a factor of 4. It is replaced by `MandelHelpers.h`.
- The yield gradient and Hessian were returned without the weights of Eq. (2.2), i.e. twice the true value in every shear slot (four times in the shear-shear block of the Hessian). The yield *surface* was unaffected, but the plastic flow direction was wrong in every state that combines normal and shear stress in the material frame. Uniaxial and pure-shear single-element tests cannot detect this.
- The convexity check, the viscosity derivatives and the tangent itself were corrected (Sections 2.5.3, 2.7.2 and 2.9).

Results for rotated crystals, polycrystals and bone specimens produced before these corrections (September 2026) were computed with the earlier flow rule and are superseded.

2.2 Kinematics and Stress Update

2.2.1 The Model as Published

Zysset and Schwiedrzik [5] formulate the model with the Green-Lagrange strain $\mathbf{E}$ and its work-

conjugate second Piola-Kirchhoff stress $\mathbf{S}$, using an additive (Green-Naghdi) split of the total strain,

$$\mathbf{E} = \mathbf{E}^e + \mathbf{E}^p, \quad \mathbf{S} = (1 - D(\kappa)) \mathbb{C} : (\mathbf{E} - \mathbf{E}^p), \quad (2.3)$$

with the plastic strain $\mathbf{E}^p$ and the accumulated plastic strain κ as internal variables and a scalar damage $D(\kappa)$. Wolfram et al. [6] use the same split for coral, without damage, and state the assumption of small deformations under which it is used.

2.2.2 What the Implementation Does

The implementation does not evaluate Eq. (2.3) in total form. It uses the incremental, corotational update of the MOOSE solid mechanics module, which is also how the reference UMAT is driven by Abaqus when geometric nonlinearity is active. With `strain = FINITE`:

1. `ComputeFiniteStrain` forms the incremental deformation gradient of the step and decomposes it into a strain increment $\Delta\boldsymbol{\varepsilon}$ and a rotation increment $\Delta\mathbf{R}$, by default with the approximation of Rashid [8].
2. `ComputeMultipleInelasticStress` forms the trial stress in the configuration at the start of the step. For an anisotropic elasticity tensor this is

$$\boldsymbol{\sigma}^{\text{tr}} = \boldsymbol{\sigma}_n + \mathbb{C} : \Delta\boldsymbol{\varepsilon}. \quad (2.4)$$

3. `OrthotropicPlasticityStressUpdate::updateState` returns the admissible stress, the inelastic strain increment and the tangent (Sections 2.8 and 2.9).
4. The stress, the elastic strain, the inelastic strain and the tangent are rotated forward with the rotation increment, $\boldsymbol{\sigma}_{n+1} \leftarrow \Delta\mathbf{R} \boldsymbol{\sigma}_{n+1} \Delta\mathbf{R}^T$.

The stress carried by the implementation is therefore the Cauchy stress $\boldsymbol{\sigma}$, updated by a hypoelastic, incrementally objective rule, and $\boldsymbol{\varepsilon}$ denotes the accumulated strain increments. For small strains and rotations this coincides with Eq. (2.3), with $\mathbf{E} \rightarrow \boldsymbol{\varepsilon}$ and $\mathbf{S} \rightarrow \boldsymbol{\sigma}$, and the remainder of this chapter is written in these variables. With `strain = SMALL` steps 1 and 4 reduce to the linearised strain and no rotation.

Two consequences should be kept in mind.

- The orthotropy axes do not follow the material rotation. MOOSE rotates the stress and strain tensors with $\Delta\mathbf{R}$ but not the elasticity tensor, and the Euler angles that define the material frame (Section 2.11) are constant in time. The model is appropriate while the accumulated rotation of a material point remains small. There is no lattice reorientation.
- The property `plastic_strain` declared by `OrthotropicPlasticityStressUpdate` holds the plastic strain in the material frame and is not rotated with $\Delta\mathbf{R}$. MOOSE's own `inelastic_strain` is the rotated, global-frame quantity.

The global Jacobian needs one setting in addition to the material tangent, see Section 2.9.3.

2.2.3 Material Frame

The return mapping is carried out in the material frame. With the orientation matrix $\mathbf{g}$ of Section 2.11,

$$\boldsymbol{\sigma}^{\text{mat}} = \mathbf{g}^T \boldsymbol{\sigma} \mathbf{g}, \quad \Delta\boldsymbol{\varepsilon}^{\text{mat}} = \mathbf{g}^T \Delta\boldsymbol{\varepsilon} \mathbf{g}, \quad C_{ijkl}^{\text{mat}} = g_{mi} g_{nj} g_{ok} g_{pl} C_{mnop}, \quad (2.5)$$

and the converged stress, plastic strain increment and tangent are transformed back with the inverse relations. The elasticity tensor arrives from the elasticity material already rotated to the global frame; it is rotated back here so that the two materials use one and the same orientation.

2.2.4 Accumulated Plastic Strain

Following Zysset and Schwiedrzik [5] and Wolfram et al. [6] (supplementary Eq. S16), the accumulated plastic strain is the time integral of the norm of the plastic strain rate,

$$\kappa = \int_0^t \|\dot{\epsilon}^p\| d\tau, \quad \Delta\kappa = \|\Delta\epsilon^p\|. \quad (2.6)$$

It is stored as the material property `effective_plastic_strain`.

Definition of κ

κ is the plain Frobenius norm of the plastic strain, not the von Mises equivalent plastic strain, which carries a factor $\sqrt{2/3}$ and presumes isochoric flow. Plastic flow on the quadric surface is in general not isochoric (Section 2.7), so there is no fixed conversion between the two. Do not compare κ with an equivalent plastic strain from a J_2 model, and note that all post-yield laws of Section 2.6 are functions of κ as defined here.

2.3 Orthotropic Elasticity

2.3.1 Constitutive Relation

In the material frame the elastic law is $\sigma = \mathbb{C} : \epsilon^e$ with an orthotropic stiffness tensor of nine independent constants. Written for the six components in the order of Eq. (2.1), with tensor shear strains,

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix} = \begin{bmatrix} C_{1111} & C_{1122} & C_{1133} & 0 & 0 & 0 \\ C_{1122} & C_{2222} & C_{2233} & 0 & 0 & 0 \\ C_{1133} & C_{2233} & C_{3333} & 0 & 0 & 0 \\ 0 & 0 & 0 & 2C_{2323} & 0 & 0 \\ 0 & 0 & 0 & 0 & 2C_{1313} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2C_{1212} \end{bmatrix} \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \epsilon_{23} \\ \epsilon_{13} \\ \epsilon_{12} \end{bmatrix}. \quad (2.7)$$

The factor 2 in the shear block is the one that a plain copy of tensor components omits. The shear moduli are $G_{23} = C_{2323}$, $G_{13} = C_{1313}$, $G_{12} = C_{1212}$, so that $\sigma_{12} = 2G_{12} \epsilon_{12} = G_{12} \gamma_{12}$ with the engineering shear strain $\gamma_{12} = 2\epsilon_{12}$.

2.3.2 Aragonite Constants

The default constants of the `coral` material model are, in GPa,

$$\begin{aligned} C_{1111} &= 171.8, & C_{2222} &= 106.7, & C_{3333} &= 84.2, \\ C_{1122} &= 57.5, & C_{1133} &= 30.2, & C_{2233} &= 46.9, \\ C_{2323} &= 42.1, & C_{1313} &= 31.1, & C_{1212} &= 46.6. \end{aligned} \quad (2.8)$$

They originate from the molecular dynamics calibration and are marked provisional in the presets header. The engineering constants that follow by inverting the stiffness are

$$\begin{aligned} E_1 &= 140.4 \text{ GPa}, & E_2 &= 70.3 \text{ GPa}, & E_3 &= 63.4 \text{ GPa}, \\ G_{12} &= 46.6 \text{ GPa}, & G_{13} &= 31.1 \text{ GPa}, & G_{23} &= 42.1 \text{ GPa}, \\ \nu_{12} &= 0.505, & \nu_{13} &= 0.077, & \nu_{23} &= 0.466. \end{aligned} \quad (2.9)$$

Note that $E_i = 1/(\mathbb{C}^{-1})_{iiii}$ is the uniaxial-stress modulus and is smaller than the uniaxial-strain stiffness C_{iiii} that is given as input; a single-element tension test with free lateral faces measures E_i .

2.4 Density- and Fabric-Based Elasticity

`ComputeFabricElasticityTensor` builds the orthotropic stiffness from a small set of constants, the bone volume fraction ρ (BV/TV) and, for the fabric models, the eigenvalues m_1, m_2, m_3 of the normalised fabric tensor ($m_1 + m_2 + m_3 = 3$). Which constants are used is selected by the material-model flag (Chapter 6). The stiffness is always assembled from the orthotropic compliance

$$S_{iiii} = \frac{1}{E_i}, \quad S_{iijj} = -\frac{\nu_{ij}}{E_i} \quad (i \neq j), \quad S_{ijij} = \frac{1}{4G_{ij}} \quad (i \neq j), \quad (2.10)$$

which is inverted in closed form. For the models defined by engineering constants, positive definiteness is checked and the run stops if it fails.

2.4.1 Density Function

All bone models scale with the piecewise power function of Zysset and Schwiedrzik [5],

$$f(\rho; k, \delta) = \begin{cases} \rho^k & \text{if } \rho \leq \frac{1}{2}, \\ \rho^k + (\delta - 1)(2\rho - 1)^k & \text{if } \rho > \frac{1}{2}, \end{cases} \quad (2.11)$$

called TSFU in the UMAT. The second branch is a correction towards compact bone; $\delta = 1$ (`delta_cortical`, the default and the value of every preset) disables it. The same function with the exponent p in place of k scales the strengths (Section 2.5.7).

2.4.2 Elastic Models

With $f = f(\rho; k, \delta)$ and $G_{\text{iso}} = E_0/[2(1 + \nu_0)]$:

Isotropic (`trabecular_iso`, `compact_iso`). Two constants E_0, ν_0 :

$$E_i = E_0 f, \quad G_{ij} = G_{\text{iso}} f, \quad S_{iijj} = -\frac{\nu_0}{E_0 f}. \quad (2.12)$$

Transversely isotropic (`trabecular_ti`, `compact_ti`). Five constants: E_0, ν_0 in the transverse plane, and E_a, ν_a, G_a for the axis a selected by `main_direction`:

$$E_a^{\text{eff}} = E_a f, \quad E_t = E_0 f, \quad G_{at} = G_a f, \quad G_{tt'} = G_{\text{iso}} f, \quad S_{aatt} = -\frac{\nu_a}{E_a f}, \quad S_{ttt't'} = -\frac{\nu_0}{E_0 f}. \quad (2.13)$$

Fabric-based orthotropic (`trabecular_fabric_ortho`, and `legacy`). The model of Zysset and Curnier [2] with three constants E_0, ν_0, G_0 and the fabric exponent l :

$$E_i = E_0 f m_i^{2l}, \quad G_{ij} = G_0 f (m_i m_j)^l, \quad S_{iijj} = -\frac{\nu_0}{E_0 f (m_i m_j)^l}, \quad (2.14)$$

equivalently $\nu_{ij} = \nu_0(m_i/m_j)^l$. For $m_1 = m_2 = m_3 = 1$ and $G_0 = G_{\text{iso}}$ the tensor is isotropic. In `legacy` mode G_0 defaults to G_{iso} and the exponents to $k = 2, l = 1$; the bone presets carry their own G_0, k and l .

Fabric-based transversely isotropic (`trabecular_fabric_ti`). As Eq. (2.14), after the two eigenvalues transverse to `main_direction` are replaced by their mean, and with G_{iso} instead of G_0 for the transverse plane, which is what keeps that plane isotropic.

Coral (coral). No density, no fabric: the nine constants C_{ijkl} are given explicitly (Eq. (2.7)).

A corrected UMAT entry. For the fabric-based transversely isotropic set with the axis in direction 3, the reference UMAT assigns the 23 shear stiffness without the density factor f , while the corresponding compliance entry carries it. The UMAT stress therefore uses a shear stiffness too high by $1/f$, a factor of 21.6 at $\rho = 0.15$. The implementation here uses the consistent value. Anyone comparing against the original UMAT for that combination will see a difference in that shear component.

2.4.3 Density Scaling of Explicit Constants

The class `ComputeElasticityTensorCoupled` accepts explicit constants, in the same way as MOOSE's `ComputeElasticityTensor`, and offers an optional plain power-law scaling for porous variants,

$$\mathbb{C}(\rho) = \rho^k \mathbb{C}_0, \quad (2.15)$$

with $k = \text{elasticity_density_exponent}$. The scaling is applied only if $k > 0$ and $\text{density_rho} < 1$, and it does not use the two-branch function of Eq. (2.11).

2.5 Quadric Yield Surface

2.5.1 Yield Function

The yield function of Schwiedrzik et al. [1] is

$$Y(\boldsymbol{\sigma}, \kappa) = \sqrt{\boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma} + \mathbf{F} : \boldsymbol{\sigma} - r(\kappa)}, \quad (2.16)$$

with a fourth-order tensor $\mathbb{F}$ (units MPa^{-2}), a second-order tensor $\mathbf{F}$ (units MPa^{-1}) and the dimensionless post-yield function $r(\kappa)$ of Section 2.6. Y is dimensionless. The elastic domain is $Y < 0$. The quadratic term gives an ellipsoidal surface in stress space; the linear term shifts it and produces the tension-compression asymmetry.

2.5.2 Orthotropic Coefficients

In the material frame the orthotropic surface is defined by twelve numbers: three tensile strengths σ_i^+ , three compressive strengths σ_i^- (entered as positive values), three shear strengths τ_{ij} and three interaction parameters ζ_{ij} . Written out,

$$\begin{aligned} \boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma} = & F_{11}\sigma_{11}^2 + F_{22}\sigma_{22}^2 + F_{33}\sigma_{33}^2 \\ & + 2F_{12}\sigma_{11}\sigma_{22} + 2F_{13}\sigma_{11}\sigma_{33} + 2F_{23}\sigma_{22}\sigma_{33} \\ & + \frac{\sigma_{23}^2}{\tau_{23}^2} + \frac{\sigma_{13}^2}{\tau_{13}^2} + \frac{\sigma_{12}^2}{\tau_{12}^2}, \end{aligned} \quad (2.17)$$

$$\mathbf{F} : \boldsymbol{\sigma} = F_1\sigma_{11} + F_2\sigma_{22} + F_3\sigma_{33}, \quad (2.18)$$

with

$$F_{ii} = \left[\frac{\sigma_i^+ + \sigma_i^-}{2\sigma_i^+\sigma_i^-} \right]^2, \quad (2.19)$$

$$F_i = \frac{1}{2} \left(\frac{1}{\sigma_i^+} - \frac{1}{\sigma_i^-} \right), \quad (2.20)$$

$$F_{ij} = -\zeta_{ij} F_{ii}, \quad i < j. \quad (2.21)$$

Each shear stress appears once in Eq. (2.17). In terms of tensor components this means $\mathbb{F}_{iijj} = F_{ij}$ and $\mathbb{F}_{ijji} = \mathbb{F}_{ijjj} = \mathbb{F}_{jiii} = \mathbb{F}_{jiji} = 1/(4\tau_{ij}^2)$ for $i \neq j$.

Inserting a uniaxial stress $\sigma_{ii} = s$ into Eq. (2.16) gives $Y = s/\sigma_i^+ - r$ for $s > 0$ and $Y = |s|/\sigma_i^- - r$ for $s < 0$, and a single shear stress gives $Y = |\sigma_{ij}|/\tau_{ij} - r$. The surface at $r = 1$ therefore passes through the nine strengths. If $\sigma_i^- = \sigma_i^+$ in all three directions then $\mathbf{F} = \mathbf{0}$ and the surface is an ellipsoid centred at the origin.

The interaction term is referenced to the lower index. Equation (2.21) is not symmetric in i and j : F_{12} and F_{13} are referenced to F_{11} , F_{23} to F_{22} . The matrix of coefficients is symmetric ($F_{ji} = F_{ij}$), but exchanging two axes changes the value of ζ that describes the same surface. This matters for the transversely isotropic models below.

Meaning of the sign. For $\zeta_{ij} > 0$ the coefficient F_{ij} is negative. Two normal stresses of the same sign then reduce the quadratic term, so yield occurs at *higher* stresses than each would reach alone, while stresses of opposite sign yield earlier. $\zeta_{ij} = 0$ decouples the two directions.

2.5.3 Convexity

$\sqrt{\boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma}}$ is convex if and only if $\mathbb{F}$ is positive semi-definite. The shear part is diagonal and positive, so the condition reduces to the 3×3 matrix of normal coefficients,

$$\begin{bmatrix} F_{11} & -\zeta_{12}F_{11} & -\zeta_{13}F_{11} \\ -\zeta_{12}F_{11} & F_{22} & -\zeta_{23}F_{22} \\ -\zeta_{13}F_{11} & -\zeta_{23}F_{22} & F_{33} \end{bmatrix} \succeq 0. \quad (2.22)$$

The constructor tests all principal minors: the three 2×2 minors, which give

$$|\zeta_{ij}| \leq \sqrt{\frac{F_{jj}}{F_{ii}}}, \quad i < j, \quad (2.23)$$

and the determinant, which couples the three parameters. A violation stops the run with an error naming the offending parameter. Note the square root in Eq. (2.23): for strengths that differ between directions the admissible range is not $|\zeta| \leq 1$.

2.5.4 Isotropy Constraint

If all normal strengths are equal ($F_{ii} = S_0^2$, with $S_0 = (\sigma_0^+ + \sigma_0^-)/(2\sigma_0^+\sigma_0^-)$) and $\zeta_{ij} = \zeta_0$, the quadratic form is isotropic only if the shear strength satisfies

$$\tau_{\text{iso}} = \frac{1}{S_0 \sqrt{2(1 + \zeta_0)}}. \quad (2.24)$$

The isotropic and transversely isotropic bone models do not take the corresponding shear strength as input but derive it from Eq. (2.24), as the UMAT does.

2.5.5 Determining ζ from a Biaxial Test

For an equibiaxial test $\sigma_{11} = \sigma_{22} = s_b$ at initial yield ($r = 1$), Eq. (2.16) with Eq. (2.21) gives

$$\zeta_{12} = \frac{1}{2F_{11}} \left[F_{11} + F_{22} - \left(\frac{1 - (F_1 + F_2)s_b}{s_b} \right)^2 \right], \quad (2.25)$$

with F_{11} , F_{22} from Eq. (2.19) and F_1 , F_2 from Eq. (2.20). The other two parameters follow by permuting indices, keeping the lower index as the reference in the prefactor.

2.5.6 Aragonite Strengths

The default strengths of the `coral` model, from molecular dynamics and marked provisional in `MaterialModelPresets.h`, are

$$\begin{aligned} \sigma_1^+ &= 4980 \text{ MPa}, & \sigma_2^+ &= 4100 \text{ MPa}, & \sigma_3^+ &= 5340 \text{ MPa}, \\ \tau_{12} &= 4510 \text{ MPa}, & \tau_{13} &= 5080 \text{ MPa}, & \tau_{23} &= 5060 \text{ MPa}, \end{aligned} \quad (2.26)$$

with compressive strengths equal to the tensile ones unless given. There is no calibrated value of ζ_{ij} for aragonite. The `coral` model has no default for `zeta12`, `zeta13` and `zeta23` and stops if they are not set; the value 0.27 used in the examples is carried over from bone and is a placeholder.

2.5.7 How Each Material Model Defines the Surface

Every material model only supplies the twelve numbers; the coefficients are then assembled by one and the same code from Eqs. (2.19)–(2.21). With $f = f(\rho; p, \delta)$ from Eq. (2.11) and the axis $a = \text{main_direction}$:

Table 2.3: Strengths and interaction parameters per material model. t, t' denote the two directions transverse to a .

Model	$\sigma_i^\pm$	τ_{ij}	$\zeta_{ij} \ (i < j)$
<code>*_iso</code>	$\sigma_0^\pm f$	$\tau_{\text{iso}} f$	ζ_0
<code>*_ti</code>	$\sigma_a^\pm f$ on a , $\sigma_0^\pm f$ otherwise	$\tau_a f$ for planes containing a , $\tau_{\text{iso}} f$ for tt'	$\zeta_a F_{aa}/F_{ii}$ for pairs containing a , ζ_0 for tt'
<code>trabecular_fabric_ortho;</code> <code>legacy with use_fabric_scaling</code>	$\sigma_0^\pm f m_i^{2q}$	$\tau_0 f (m_i m_j)^q$	$\zeta_0 (m_i/m_j)^{2q}$
<code>trabecular_fabric_ti</code>	as above, with the transverse eigenvalues averaged	as above, $\tau_{\text{iso}} f (m_t m_{t'})^q$ for tt'	but as above for
<code>coral, legacy explicit</code>	input	input	input

Fabric scaling. The fabric rows are the scaling of Schwiedrzik et al. [1]. With it, $F_{ij} = -\zeta_0 S_0^2 / (f^2 m_i^{2q} m_j^{2q})$, which is symmetric in i and j , so for these models the choice of reference index in Eq. (2.21) is immaterial.

Conversion of the axial interaction parameter. The UMAT writes the interaction between the axis and a transverse direction as $F_{aj} = -\zeta_a F_{aa}$, referenced to the *axial* coefficient. In the convention of Eq. (2.21) the same surface requires

$$\zeta_{ij} = \zeta_a \frac{F_{aa}}{F_{ii}} \quad \text{for } i < j \text{ with } a \in \{i, j\}, \quad (2.27)$$

which is the identity only if a is the lower index, that is, for `main_direction` = 1. Omitting this conversion gives a different, and possibly non-convex, surface for `main_direction` 2 and 3.

Explicit strengths with density. In legacy explicit mode the strengths can be scaled by a plain power law, $\sigma \rightarrow \rho^p \sigma$ with $p = \text{yield_density_exponent}$, applied if $p > 0$ and $\text{density_rho} < 1$.

Consistency between the two materials. Density, fabric eigenvalues and main direction are parameters of both the elasticity and the plasticity object. Setting `material_model` and these values once in `[GlobalParams]` keeps the two from drifting apart.

2.5.8 Gradient and Hessian

With $Q = \boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma}$, the derivatives of Eq. (2.16) with respect to stress are

$$\mathbf{G} \equiv \frac{\partial Y}{\partial \boldsymbol{\sigma}} = \frac{\mathbb{F} : \boldsymbol{\sigma}}{\sqrt{Q}} + \mathbf{F}, \quad (2.28)$$

$$\frac{\partial^2 Y}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}} = \frac{\mathbb{F}}{\sqrt{Q}} - \frac{(\mathbb{F} : \boldsymbol{\sigma}) \otimes (\mathbb{F} : \boldsymbol{\sigma})}{Q^{3/2}}. \quad (2.29)$$

The code evaluates $\mathbf{F}$ s and $Q = \mathbf{s} \cdot \mathbf{F}$ s with the plain-component arrays of Table 2.2 and converts to tensors with Eq. (2.2). For example $(\mathbb{F} : \boldsymbol{\sigma})_{12} = \sigma_{12}/(2\tau_{12}^2)$, half of the corresponding entry of $\mathbf{F}$ s. Q is bounded below by 10^{-16} to keep the derivatives finite at the origin.

Two scalar quantities derived from the gradient are used throughout:

$$H = \|\mathbf{G}\|, \quad \mathbf{N} = \frac{\mathbf{G}}{H}, \quad \frac{\partial H}{\partial \boldsymbol{\sigma}} = \frac{1}{H} \frac{\partial^2 Y}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}} : \mathbf{G}. \quad (2.30)$$

H has units MPa^{-1} . For a uniaxial stress on a symmetric surface without interaction ($\mathbf{F} = \mathbf{0}$, $\zeta_{ij} = 0$) it is $H = 1/\sigma_i^+$.

Self-check. With `debug_checks = true` the constructor compares Eqs. (2.28) and (2.29) with central finite differences of the yield function at three stress states that all contain shear, scaled to the surface of the material at hand, and stops if the relative error exceeds 10^{-5} . States with shear are essential: a missing weight in Eq. (2.2) is invisible in a purely normal state.

2.5.9 Input Parameters of the Surface

Table 2.4: Strength parameters, the stress component they refer to, and their position in the six-component arrays (Eq. (2.1)).

Symbol	Input parameter	Component	Slot p
σ_1^+, σ_1^-	<code>sigma_xx_tension</code> , <code>sigma_xx_compression</code>	σ_{11}	0
σ_2^+, σ_2^-	<code>sigma_yy_tension</code> , <code>sigma_yy_compression</code>	σ_{22}	1
σ_3^+, σ_3^-	<code>sigma_zz_tension</code> , <code>sigma_zz_compression</code>	σ_{33}	2
τ_{23}	<code>tau_yz_max</code>	σ_{23}	3
τ_{13}	<code>tau_xz_max</code>	σ_{13}	4
τ_{12}	<code>tau_xy_max</code>	σ_{12}	5
$\zeta_{12}, \zeta_{13}, \zeta_{23}$	<code>zeta12</code> , <code>zeta13</code> , <code>zeta23</code>		
$\sigma_0^\pm, \tau_0, \zeta_0$	<code>sigma_0_tension</code> , <code>sigma_0_compression</code> , <code>tau_0</code> , <code>zeta_0</code>	base values	
$\sigma_a^\pm, \tau_a, \zeta_a$	<code>sigma_a_tension</code> , <code>sigma_a_compression</code> , <code>tau_a</code> , <code>zeta_a</code>	axial values (TI)	
p, q, δ	<code>exponent_p</code> , <code>exponent_q</code> , <code>delta_cortical</code>	scaling	

Here x, y, z in the parameter names are the material axes 1, 2, 3 (crystal a, b, c), not the global axes. Compressive strengths are entered as positive numbers and default to the tensile ones. Parameters that the selected material model does not use are reported with a warning. The complete parameter reference is in Chapter 6.

2.6 Post-Yield Evolution

The function $r(\kappa)$ scales the size of the yield surface. Six laws are available through `postyieldmode`. In the following, r_0 is `initial_yield_ratio`; h and r_{res} both stand for `residual_strength`, which acts as a hardening amount in one law and as a residual level in two others; k_s is `kslope`, κ_{max} is `kmax`, κ_{min} is `kmin` and k_w is `kwidth`. Despite the names, $\kappa_{\text{max}} < \kappa_{\text{min}}$: the names refer to the maximum and minimum of r , as in the UMAT.

`perfect`.

$$r(\kappa) = 1. \quad (2.31)$$

`linear_hardening`.

$$r(\kappa) = 1 + k_s \kappa. \quad (2.32)$$

k_s is dimensionless (per unit plastic strain); the hardening modulus in direction i is $k_s \sigma_i^+$.

`exp_hardening`.

$$r(\kappa) = r_0 + (1 - r_0 + h) \left[1 - e^{-k_s \kappa} \right]. \quad (2.33)$$

The surface starts at r_0 and saturates at $1 + h$. With $h = 0$ this is the UMAT law for compact bone (`PYFL` = 1), which saturates at the input strengths. With $r_0 = 1$ it is the earlier form $1 + h[1 - e^{-k_s \kappa}]$. Note that the parameter default `residual_strength` = 0.7 then means saturation at 1.7 times the input strengths; set it explicitly when using this mode outside a preset.

`simple_softening`.

$$r(\kappa) = r_0 + (1 - r_0) \left[\exp\left(-\frac{(\kappa - \kappa_{\text{max}})^2}{k_w \kappa_{\text{max}}^2}\right) - \exp\left(-\frac{1}{k_w} - k_s \kappa\right) \right]. \quad (2.34)$$

This is the UMAT law for trabecular bone (`PYFL` = 2). It starts exactly at $r(0) = r_0$, rises to a maximum near κ_{max} and decays back to r_0 . It is smooth everywhere. It requires $\kappa_{\text{max}} > 0$ and $k_w > 0$. With the bone preset values ($r_0 = 0.7$, $k_s = 100$, $\kappa_{\text{max}} = 0.015$, $k_w = 8$) the maximum is $r = 0.963$ at $\kappa = 0.023$, and $r(\kappa_{\text{max}}) = 0.941$; the surface never reaches $r = 1$.

`exp_softening` (the parameter default).

$$r(\kappa) = \begin{cases} 1 & \text{if } \kappa < \kappa_{\text{max}}, \\ r_{\text{res}} + (1 - r_{\text{res}}) \exp[-k_s(\kappa - \kappa_{\text{max}})] & \text{if } \kappa \geq \kappa_{\text{max}}. \end{cases} \quad (2.35)$$

`piecewise_softening`.

$$r(\kappa) = \begin{cases} 1 & \text{if } \kappa < \kappa_{\text{max}}, \\ 1 - (1 - r_{\text{res}}) \frac{\kappa - \kappa_{\text{max}}}{\kappa_{\text{min}} - \kappa_{\text{max}}} & \text{if } \kappa_{\text{max}} \leq \kappa < \kappa_{\text{min}}, \\ r_{\text{res}} & \text{if } \kappa \geq \kappa_{\text{min}}. \end{cases} \quad (2.36)$$

Table 2.5: Parameters read by each post-yield law. All have defaults, so none is formally required; parameters not listed for a law are ignored by it.

postyield_mode	Reads	$r(0)$ and limit
perfect	none	1, constant
linear_hardening	kslope	1, unbounded
exp_hardening	initial_yield_ratio, residual_strength, kslope	$r_0 \rightarrow 1 + h$
simple_softening	initial_yield_ratio, kslope, kmax, kwidth	$r_0 \rightarrow \text{peak} \rightarrow r_0$
exp_softening	residual_strength, kslope, kmax	$1 \rightarrow r_{\text{res}}$
piecewise_softening	residual_strength, kmax, kmin	$1 \rightarrow r_{\text{res}}$

2.6.1 Which Parameters Each Law Reads

An `initial_yield_ratio` different from 1 is ignored, with a warning, by the four laws that start at $r(0) = 1$. It must lie in $(0, 1]$.

2.6.2 Yield Strength and Ultimate Strength

With $r_0 < 1$ the strengths of Section 2.5 are *ultimate* strengths: first yield occurs on the surface $r = r_0$, at r_0 times the strength. All six bone presets use $r_0 = 0.7$, the UMAT value, so a bone preset yields at 70% of its tabulated strengths. This is the most common source of surprise when comparing a computed curve with a strength table. For the trabecular presets the peak of a computed curve is at most 96% of the tabulated strength, see Eq. (2.34).

2.6.3 Preset Values

A material-model preset sets the post-yield law and its scalars together: the trabecular bone models use `simple_softening`, the compact bone models `exp_hardening` with $h = 0$, both with $r_0 = 0.7$, $k_s = 100$, $\kappa_{\text{max}} = 0.015$, $k_w = 8$; the coral model uses `exp_softening` with $r_{\text{res}} = 0.7$, $k_s = 30$, $\kappa_{\text{max}} = 0.001$. The preset scalars belong to the preset law. If `postyield_mode` is set explicitly for a preset model, the scalars fall back to the plain parameter defaults, because for example `kmax` is a peak position in one law and a softening onset in another.

2.6.4 Smoothness

`exp_softening` has a discontinuous derivative at κ_{max} , and `piecewise_softening` at κ_{max} and κ_{min} . For the coral values r' jumps from 0 to $-(1 - r_{\text{res}})k_s = -9$. r itself is continuous, and the return mapping converges across the kink because it is controlled by residuals. The global Jacobian, however, changes discontinuously whenever a quadrature point crosses the threshold, which can cost Newton iterations when many points cross in the same step. The two laws used by the bone presets are smooth. With `debug_checks = true` the derivative r' is compared with finite differences away from the kinks, and the continuity of r at κ_{max} and κ_{min} is tested.

2.7 Flow Rule and Viscoplastic Regularization

2.7.1 Associated Flow

Plastic flow is associated. In rate form $\dot{\boldsymbol{\varepsilon}}^p = \dot{\lambda} \mathbf{G}$ with the plastic multiplier $\dot{\lambda} \geq 0$. Because κ is the norm of the plastic strain (Eq. (2.6)), $\dot{\kappa} = \dot{\lambda} H$, and the flow rule is used in the form

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\kappa} \mathbf{N}, \quad \mathbf{N} = \frac{\mathbf{G}}{\|\mathbf{G}\|}, \quad (2.37)$$

with κ rather than λ as the unknown. $\mathbf{G}$ is the unnormalised gradient of Eq. (2.28); $\mathbf{N}$ is the unit flow direction.

When the linear term $\mathbf{F}$ is non-zero the flow direction has a volumetric part even for a pure shear stress. A test that prescribes all displacement components, and thereby forbids that dilatation, develops normal reaction stresses. This is a property of the model, not an error, and it is the reason why the shear single-element examples of Chapter 7 are not states of pure shear stress.

2.7.2 Continuous Perzyna Formulation

Rate dependence follows the continuous Perzyna formulation of Zysset and Schwiedrzik [5]. With an invertible overstress function ψ and a viscosity η , $\dot{\lambda} = \psi(Y)/\eta$, which is inverted to a generalised yield condition that holds during plastic flow,

$$\bar{Y} = Y(\boldsymbol{\sigma}, \kappa) + V = 0, \quad V = -\psi^{-1}(\eta \dot{\lambda}), \quad (2.38)$$

together with $\bar{Y} \leq 0$, $\dot{\kappa} \geq 0$, $\dot{\kappa} \bar{Y} = 0$. For $\eta \rightarrow 0$ the rate-independent model $Y = 0$ is recovered. In the time-discrete setting $\dot{\lambda} = \Delta\kappa/(H \Delta t)$, and the term V is a function of the dimensionless quantity

$$x = \frac{\eta}{\Delta t} \frac{\Delta\kappa}{H}. \quad (2.39)$$

Since $\Delta\kappa$ is dimensionless and H has units MPa^{-1} , η has units of sMPa^{-1} if the simulation time is in seconds. In a quasi-static analysis the time is a load parameter, and η is then a regularization parameter relative to that pseudo-time, not a material viscosity.

Table 2.6: Viscosity modes (`viscosity_mode`). `VISCFL` is the corresponding flag of the reference UMAT.

Mode	$V(x)$	Role of m	VISCFL
<code>rate_independent</code>	0	unused	0
<code>linear</code> (default)	$-x$	unused	1
<code>logarithmic</code>	$-\frac{\ln(1+x)}{m}$	divisor; $V \approx -x/m$ for small x	2
<code>exponential</code>	$-\frac{e^x - 1}{m}$	divisor; $V \approx -x/m$ for small x	3
<code>polynomial</code>	$\frac{m}{2} - \sqrt{\frac{m^2}{4} + x}$	$V \approx -x/m$ for $x \ll m^2/4$	4
<code>powerlaw</code>	$-\text{sign}(\Delta\kappa) x ^{1/m}$	reciprocal exponent	5

The parameter `m` means something different in each mode. In the logarithmic, exponential and polynomial modes the small- x limit is $V \approx -x/m$, so these modes behave like the linear one with an effective viscosity η/m . With the parameter default $m = 0.001$ that is a thousand times the nominal η . In the power-law mode m is a reciprocal exponent: x is typically of order 10^{-3} , so $m = 0.001$ gives x^{1000} , which underflows to zero and silently yields rate-independent behaviour. The power-law mode needs m of order one and reproduces the linear mode for $m = 1$; a warning is issued for $m < 0.5$. The bone presets use the linear mode, which does not read `m`.

Naming. The mode names describe the form of V , i.e. of ψ^{-1} . The UMAT documentation names the overstress function ψ instead, so that its “exponential” law (`VISCFL = 2`) is the **logarithmic** mode here and its “logarithmic” law (`VISCFL = 3`) is the **exponential** mode. Transfer a UMAT parameter set by flag number.

Magnitude. For the linear mode and a uniaxial stress on a symmetric surface without interaction, $H = 1/\sigma_i^+$ and Eq. (2.38) gives an overstress on the yield condition of

$$\Delta\sigma \approx \eta \dot{\kappa} (\sigma_i^+)^2. \quad (2.40)$$

With the coral values $\eta = 0.002 \text{ s MPa}^{-1}$ and $\sigma_1^+ = 4980 \text{ MPa}$, a plastic strain rate of $5 \times 10^{-4} \text{ s}^{-1}$ gives about 25 MPa, or 0.5% of the strength. The dependence on the square of the strength means that a value of η tuned for aragonite is not transferable to bone, and vice versa; the bone presets use $\eta = 10^{-4}$.

Derivatives. The return mapping and the tangent need

$$\frac{\partial V}{\partial \Delta\kappa} = V'(x) \frac{x}{\Delta\kappa}, \quad \frac{\partial V}{\partial \boldsymbol{\sigma}} = -V'(x) \frac{x}{H} \frac{\partial H}{\partial \boldsymbol{\sigma}}, \quad (2.41)$$

with $\partial H/\partial \boldsymbol{\sigma}$ from Eq. (2.30). For the linear mode $\partial V/\partial \Delta\kappa = -\eta/(\Delta t H)$. On a plateau of $r(\kappa)$ this term is the only resistance to plastic flow in the consistency condition; without it the tangent of Section 2.9 would be singular there.

2.8 Return Mapping Algorithm

All quantities in this section are in the material frame.

2.8.1 Elastic Predictor

The trial stress is Eq. (2.4). The step is elastic if

$$Y(\boldsymbol{\sigma}^{\text{tr}}, \kappa_n) \leq \text{tol}, \quad (2.42)$$

with $\text{tol} = \text{absolute_tolerance}$. The viscous term vanishes in this check because $\Delta\kappa = 0$. In an elastic step the trial stress is accepted, the internal variables are unchanged and the tangent is $\mathbb{C}$.

2.8.2 Local Equations

Otherwise the stress $\boldsymbol{\sigma}$ and the increment $\Delta\kappa$ at the end of the step are the solution of seven nonlinear equations (implicit Euler, closest point projection), here without damage:

$$\mathcal{R}(\boldsymbol{\sigma}, \Delta\kappa) = \mathbb{C}^{-1} : (\boldsymbol{\sigma}^{\text{tr}} - \boldsymbol{\sigma}) - \Delta\kappa \mathbf{N}(\boldsymbol{\sigma}) = \mathbf{0}, \quad (2.43)$$

$$\bar{Y}(\boldsymbol{\sigma}, \Delta\kappa) = Y(\boldsymbol{\sigma}, \kappa_n + \Delta\kappa) + V(\boldsymbol{\sigma}, \Delta\kappa) = 0. \quad (2.44)$$

The first states that the elastic strain released by the stress correction equals the plastic strain increment $\Delta\boldsymbol{\varepsilon}^p = \Delta\kappa \mathbf{N}$; the second is the generalised yield condition. The derivatives are

$$\frac{\partial \mathcal{R}}{\partial \boldsymbol{\sigma}} = -\mathbb{C}^{-1} - \Delta\kappa \frac{\partial \mathbf{N}}{\partial \boldsymbol{\sigma}}, \quad \frac{\partial \mathbf{N}}{\partial \boldsymbol{\sigma}} = \frac{1}{H^2} \left[H \frac{\partial^2 Y}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}} - \mathbf{G} \otimes \frac{\partial H}{\partial \boldsymbol{\sigma}} \right], \quad (2.45)$$

$$\frac{\partial \mathcal{R}}{\partial \Delta\kappa} = -N, \quad \frac{\partial \bar{Y}}{\partial \boldsymbol{\sigma}} = \mathbf{G} + \frac{\partial V}{\partial \boldsymbol{\sigma}}, \quad \frac{\partial \bar{Y}}{\partial \Delta\kappa} = -r'(\kappa) + \frac{\partial V}{\partial \Delta\kappa}. \quad (2.46)$$

$\partial \mathbf{N} / \partial \boldsymbol{\sigma}$ has the minor symmetries but not the major one, because $\mathbf{N}$ is a normalised gradient.

Convergence requires both

$$\|\mathcal{R}\| \leq \text{tol} \quad \text{and} \quad |\bar{Y}| \leq \text{tol}. \quad (2.47)$$

Both residuals are dimensionless ($\mathcal{R}$ is a strain), and the test is absolute; there is no relative criterion. The tolerance should be chosen well below the strains of interest: a loose tolerance leaves noise in $\boldsymbol{\sigma}(\boldsymbol{\varepsilon})$ that the global Newton iteration cannot converge through. The production value is 10^{-8} (Chapter 6).

2.8.3 Stage 1: Newton Iteration

Starting from $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\text{tr}}$, $\Delta\kappa = 0$, the linearised system is solved by condensation. With

$$\mathbb{A} = \left[\mathbb{C}^{-1} + \Delta\kappa \frac{\partial \mathbf{N}}{\partial \boldsymbol{\sigma}} \right]^{-1} = \left[-\frac{\partial \mathcal{R}}{\partial \boldsymbol{\sigma}} \right]^{-1}, \quad (2.48)$$

the corrections are

$$\delta\kappa = -\frac{\bar{Y} + \frac{\partial \bar{Y}}{\partial \boldsymbol{\sigma}} : \mathbb{A} : \mathcal{R}}{\frac{\partial \bar{Y}}{\partial \boldsymbol{\sigma}} : \mathbb{A} : \frac{\partial \mathcal{R}}{\partial \Delta\kappa} + \frac{\partial \bar{Y}}{\partial \Delta\kappa}}, \quad (2.49)$$

$$\delta\boldsymbol{\sigma} = \mathbb{A} : \left[\mathcal{R} + \frac{\partial \mathcal{R}}{\partial \Delta\kappa} \delta\kappa \right]. \quad (2.50)$$

The inverse in Eq. (2.48) is a general 6×6 inversion that assumes minor symmetry only. If an update produces $\Delta\kappa < 0$ the iteration restarts from the trial state. The stage fails if the denominator of Eq. (2.49) vanishes, if H vanishes, or after `max_iterations_newton` iterations.

Equation (2.49) uses the full derivative $\partial \bar{Y} / \partial \boldsymbol{\sigma}$. The UMAT contracts with $\mathbf{N}$ and divides by the norm instead, which is the same thing only when $\partial \bar{Y} / \partial \boldsymbol{\sigma}$ is parallel to $\mathbf{N}$, that is, without viscosity.

2.8.4 Stage 2: Primal Closest Point Projection

If stage 1 fails and `use_primal_cpp = true`, the same seven equations are solved by the primal closest point projection algorithm with line search of Armero and Pérez-Foguet [9] and Pérez-Foguet and Armero [10], transcribed from the UMAT. The unknowns are collected as $\mathbf{x} = (\hat{\boldsymbol{\sigma}}, \Delta\kappa)$ and the residuals as $\boldsymbol{\rho} = (\mathcal{R}, \bar{Y})$ in Mandel form, with the 7×7 Jacobian $\mathbf{J} = \partial \boldsymbol{\rho} / \partial \mathbf{x}$ assembled from Eqs. (2.45) and (2.46).

1. **Search direction.** Normally the Newton direction $\mathbf{d} = -\mathbf{J}^{-1} \boldsymbol{\rho}$. If the iterate sits on the constraint $\Delta\kappa = 0$ with $\mathbf{N} : (\boldsymbol{\sigma} - \boldsymbol{\sigma}^{\text{tr}}) > 0$, the projected direction $\mathbf{d} = -\tilde{\mathbf{D}} \mathbf{J}^T \boldsymbol{\rho}$ is used, where $\tilde{\mathbf{D}}$ is $\mathbf{J}^{-1} \mathbf{J}^{-T}$ with the entries that couple stress and $\Delta\kappa$ set to zero.
2. **Merit function.** $M = \frac{1}{2} \|\mathcal{R}\|^2 + \frac{1}{2} \bar{Y}^2$, evaluated before and after the step.

3. **Line search.** The full step $\alpha = 1$ is accepted if $M(\alpha) \leq (1 - 2\beta\alpha) M_0$ for a Newton direction, or $M(\alpha) \leq M_0 + \beta\alpha M'_0$ for a projected one, with $M'_0 = \boldsymbol{\rho} \cdot \mathbf{J}\mathbf{d}$. Otherwise the step is reduced to

$$\alpha \leftarrow \max \left\{ \eta_{\text{ls}} \alpha, \frac{-\alpha^2 M'_0}{2 [M(\alpha) - M_0 - \alpha M'_0]} \right\}, \quad (2.51)$$

the larger of a fixed contraction and the minimiser of the quadratic interpolant, for at most 20 reductions and while $\alpha \geq \beta$. Here $\beta = \text{line_search_beta}$ and $\eta_{\text{ls}} = \text{line_search_eta}$.

4. **Constraint.** $\Delta\kappa$ is projected onto $\Delta\kappa \geq 0$ after every update.

The stage ends when Eq. (2.47) holds or after `max\_iterations\_primal` iterations.

2.8.5 Update, Diagnostics and Failure

After convergence

$$\Delta\boldsymbol{\varepsilon}^p = \Delta\kappa \mathbf{N}(\boldsymbol{\sigma}_{n+1}), \quad \kappa_{n+1} = \kappa_n + \Delta\kappa, \quad (2.52)$$

and the stress and the plastic strain increment are rotated to the global frame.

Table 2.7: Material properties declared by `OrthotropicPlasticityStressUpdate`.

Property	Content
<code>effective_plastic_strain</code>	κ
<code>plastic_strain</code>	$\boldsymbol{\varepsilon}^p$ in the material frame
<code>damage_variable</code>	D (zero unless <code>use_damage = true</code>)
<code>yield_function</code>	value of the yield function at the last evaluation of the point
<code>return_mapping_stage</code>	−1 elastic; 2 converged in stage 1; 4 converged in stage 2. The values 1 and 3 mark a stage that was entered and did not converge.
<code>return_mapping_iterations</code>	local iterations of the stage that converged

If both stages fail, a `MooseException` is thrown. MOOSE then rejects the global step and cuts the time step, which is the counterpart of the UMAT requesting a smaller increment from Abaqus; the run is not aborted. The switch from stage 1 to stage 2 and every fallback of the tangent (Section 2.9) are counted. The individual warnings are limited to the first five occurrences, and cumulative totals are printed at the start of each time step when they have changed, so the totals, not the number of warning lines, are the record. A healthy run reports no switches and no fallbacks.

2.9 Consistent Tangent

The global Newton iteration converges quadratically only if the material returns the derivative of the stress *as computed by the algorithm* with respect to the strain increment. The plasticity class reports the tangent method `FULL` and returns this operator.

2.9.1 Derivation

At a converged plastic state Eqs. (2.43) and (2.44) hold for any strain increment, so their total differentials vanish. With $\mathbb{C}^{-1} : d\boldsymbol{\sigma}^{\text{tr}} = d\boldsymbol{\varepsilon}$ and $\mathbb{A}$ from Eq. (2.48),

$$d\boldsymbol{\sigma} = \mathbb{A} : \left[d\boldsymbol{\varepsilon} + \frac{\partial \mathcal{R}}{\partial \Delta\kappa} d\Delta\kappa \right], \quad (2.53)$$

$$0 = \frac{\partial \bar{Y}}{\partial \boldsymbol{\sigma}} : d\boldsymbol{\sigma} + \frac{\partial \bar{Y}}{\partial \Delta\kappa} d\Delta\kappa. \quad (2.54)$$

Eliminating $d\Delta\kappa$ gives

$$\mathbb{C}^{\text{ep}} = \mathbb{A} - \frac{\left(\mathbb{A} : \frac{\partial \mathcal{R}}{\partial \Delta\kappa}\right) \otimes \left(\mathbb{A}^T : \frac{\partial \bar{Y}}{\partial \boldsymbol{\sigma}}\right)}{\frac{\partial \bar{Y}}{\partial \boldsymbol{\sigma}} : \mathbb{A} : \frac{\partial \mathcal{R}}{\partial \Delta\kappa} + \frac{\partial \bar{Y}}{\partial \Delta\kappa}}, \quad (2.55)$$

all quantities evaluated at the converged state. Three points distinguish it from the textbook expression.

- **$\mathbb{A}$, not $\mathbb{C}$.** The operator $\mathbb{A}$ contains the change of the flow direction over the finite step. For $\Delta\kappa \rightarrow 0$ it tends to $\mathbb{C}$ and Eq. (2.55) to the continuum tangent.
- **The transpose.** $\mathbb{A}$ lacks major symmetry, so $\mathbb{A}^T : \partial \bar{Y} / \partial \boldsymbol{\sigma}$ is not $\mathbb{A} : \partial \bar{Y} / \partial \boldsymbol{\sigma}$.
- **The viscous terms.** $\partial \bar{Y} / \partial \boldsymbol{\sigma}$ and $\partial \bar{Y} / \partial \Delta\kappa$ include the derivatives of V (Eq. (2.41)), exactly as in the return mapping. The tangent is then consistent with the equations that were actually solved.

Without viscosity, $\partial \mathcal{R} / \partial \Delta\kappa = -\mathbf{N}$ and $\partial \bar{Y} / \partial \boldsymbol{\sigma} = H\mathbf{N}$, and Eq. (2.55) reduces to

$$\mathbb{C}^{\text{ep}} = \mathbb{A} - \frac{(\mathbb{A} : \mathbf{N}) \otimes (\mathbb{A}^T : \mathbf{N})}{\mathbf{N} : \mathbb{A} : \mathbf{N} + r'(\kappa)/H}. \quad (2.56)$$

$\mathbb{C}^{\text{ep}}$ is in general not symmetric and is not symmetrised.

2.9.2 Evaluation

The tangent is computed in the same call that returns the stress, from the quantities of the converged state, and only when MOOSE requests it. The algebra is carried out on 6×6 Mandel matrices (Table 2.2), with a general inversion for $\mathbb{A}$. The result is rotated to the global frame, $C_{ijkl}^{\text{ep}} \leftarrow g_{im}g_{jn}g_{ko}g_{lp}C_{mnop}^{\text{ep}}$, and MOOSE applies the rotation increment of the step. In an elastic step the tangent is $\mathbb{C}$.

The elastic tensor is substituted, and the event counted, if $\mathbb{A}$ cannot be formed, if the denominator of Eq. (2.55) vanishes or is not finite, or if any entry of the result is not finite or exceeds ten times the largest elastic entry. These are safeguards; in the verification runs of Chapter 7 none of them was triggered.

2.9.3 Requirements in the Input File

- `use_finite_deform_jacobian = true` in the `QuasiStatic` physics block is required with `strain = FINITE` and `solve_type = NEWTON`. Without it MOOSE's Jacobian omits the terms from the rotation increment, and global convergence remains linear however exact the material tangent is.
- **A line search** (`line_search = bt`) is required for multi-element problems with spreading plasticity. An undamped Newton step across the elastic-plastic transition can invert elements.
- For comparison, `tangent_operator = elastic` on `ComputeMultipleInelasticStress` makes MOOSE ignore the returned tangent. The converged solution must not depend on this setting; only the number of iterations does. This is how the tangent is verified (Chapter 7).

2.10 Optional Damage Coupling

`use_damage = true` activates a scalar damage $D(\kappa)$ that grows linearly beyond a threshold and saturates,

$$D(\kappa) = \min\{d \langle \kappa - \kappa_{\text{crit}} \rangle, 0.99\}, \quad (2.57)$$

with $d = \text{damage_rate}$ and $\kappa_{\text{crit}} = \text{damage_critical}$. It enters the strain residual in the form taken from the UMAT,

$$\mathcal{R} = -\frac{\mathbb{C}^{-1} : (\boldsymbol{\sigma} - \boldsymbol{\sigma}^{\text{tr}})}{1 - D} - \frac{D - D_n}{1 - D} \mathbb{C}^{-1} : \boldsymbol{\sigma}^{\text{tr}} - \Delta\kappa \mathbf{N}, \quad (2.58)$$

which reduces to Eq. (2.43) for $D = D_n = 0$, with the corresponding additional terms in $\partial\mathcal{R}/\partial\Delta\kappa$ and a factor $(1 - D + D_n)/(1 - D)$ on the tangent. The yield function is evaluated with the stress $\boldsymbol{\sigma}$ itself, not an effective stress.

This option is not verified. No test in the verification or example sets runs with `use_damage = true`, and the damage terms of the tangent have not been checked against finite differences. The law of Eq. (2.57) is also not the damage law of the reference UMAT, which is not ported. Softening of the bulk should be modelled through $r(\kappa)$, which is verified; treat `use_damage` as experimental.

2.11 Orientation of the Material Frame

2.11.1 Orientation Matrix

The orientation of the material frame is given by three angles (ϕ_1, Φ, ϕ_2) in degrees, supplied per element through coupled variables: `euler_angle_1/2/3` for the plasticity and `coupled_euler_angle_1/2/3` for the elasticity. From them MOOSE's `RotationTensor` builds the matrix

$$\mathbf{g} = \begin{bmatrix} c_1 c_2 - s_1 s_2 c & s_1 c_2 + c_1 s_2 c & s_2 s \\ -c_1 s_2 - s_1 c_2 c & -s_1 s_2 + c_1 c_2 c & c_2 s \\ s_1 s & -c_1 s & c \end{bmatrix}, \quad \begin{aligned} c_1 &= \cos \phi_1, & s_1 &= \sin \phi_1, \\ c_2 &= \cos \phi_2, & s_2 &= \sin \phi_2, \\ c &= \cos \Phi, & s &= \sin \Phi, \end{aligned} \quad (2.59)$$

the Bunge (ZXZ) matrix $\mathbf{g} = \mathbf{R}_Z(\phi_2) \mathbf{R}_X(\Phi) \mathbf{R}_Z(\phi_1)$.

2.11.2 How the Matrix Is Applied

Both elasticity materials rotate the material-frame stiffness to the global frame with

$$C_{ijkl} = g_{im} g_{jn} g_{ko} g_{lp} C_{mnop}^{\text{mat}}, \quad (2.60)$$

which is what `RankFourTensor::rotate` does in MOOSE's `ComputeElasticityTensor`. The plasticity class uses the matching relations of Eq. (2.5), $\boldsymbol{\sigma}^{\text{mat}} = \mathbf{g}^T \boldsymbol{\sigma} \mathbf{g}$ and $\boldsymbol{\sigma} = \mathbf{g} \boldsymbol{\sigma}^{\text{mat}} \mathbf{g}^T$. The elastic and the plastic response therefore share one orientation, provided both objects are coupled to the same three variables. In this usage the material axis i , expressed in global coordinates, is the i -th *column* of $\mathbf{g}$.

Relation to Bunge angles from texture software

In texture analysis $\mathbf{g}$ transforms sample components into crystal components, so the crystal axes are the *rows* of $\mathbf{g}$. Equation (2.60) applies the same matrix in the opposite sense. A triple (ϕ_1, Φ, ϕ_2) entered here therefore describes the inverse of the orientation that the same triple denotes in the Bunge convention. To reproduce a Bunge orientation (ϕ_1, Φ, ϕ_2) , for example from EBSD, enter

$$(180^\circ - \phi_2, \Phi, 180^\circ - \phi_1),$$

which gives the transposed matrix. For rotations about a single axis the difference is only the sense of rotation, which is why tests with 0° and 90° do not reveal it.

2.11.3 Checks

- $(0^\circ, 0^\circ, 0^\circ)$ gives $\mathbf{g} = \mathbf{I}$: material axes equal global axes, crystal c along global z .
- $(90^\circ, 0^\circ, 0^\circ)$ rotates the frame about z : a global x load acts along material axis 2, so an aragonite crystal yields at $\sigma_2^+ = 4100$ MPa instead of $\sigma_1^+ = 4980$ MPa.
- With `debug_checks = true` the Mandel conversion of the rotated elasticity tensor is compared with MOOSE's own inverse on the first stress update.

`ComputeElasticityTensorCoupled` inherits the scalar parameters `euler_angle_1/2/3` of MOOSE's `ComputeElasticityTensor`. If they are non-zero, that rotation is applied first and the coupled one on top of it; leave them unset when coupled angles are used. Section 5.4 describes the use for polycrystals.

Chapter 3

Cohesive Zone Model for Protein-Aragonite Interfaces

The cohesive zone model (CZM) governs the mechanical response of protein-water interfaces and grain boundary debonding in coral aragonite structures. This Chapter provides the mathematical formulation; implementation details and parameter reference are given in Chapter 6.

3.1 Relationship to Bulk Plasticity Model

3.1.1 Overview

The cohesive zone model (CZM) operates **independently** from the bulk plasticity model described in Chapter 2; the two are coupled only through the displacement field. This section clarifies the interaction between bulk and interface behaviour.

3.1.2 Mechanical Coupling

Sequential Solution: Within each Newton iteration:

1. **Bulk stress update:**

- Bulk elements (grains) compute stress using `OrthotropicPlasticityStressUpdate`
- Plastic strain accumulates: $\kappa^{n+1} = \kappa^n + \Delta\lambda$
- Internal variables updated in material frame

2. **Interface traction update:**

- CZM elements compute traction using `HomogenizedExponentialCZM`
- Displacement jump extracted: $\llbracket \mathbf{u} \rrbracket = \mathbf{u}^+ - \mathbf{u}^-$
- Damage is updated from the largest opening reached so far (Section 3.4)

3. **Residual assembly:**

- Bulk residual: $\mathbf{R}_{\text{bulk}} = \int_{\Omega} \nabla \mathbf{w} : \boldsymbol{\sigma} dV - \mathbf{F}_{\text{ext}}$
- Interface residual: $\mathbf{R}_{\text{int}} = \int_{\Gamma} \mathbf{w} \cdot \mathbf{T} dS$
- Total: $\mathbf{R} = \mathbf{R}_{\text{bulk}} + \mathbf{R}_{\text{int}}$

4. **Convergence check:**

- If $\|\mathbf{R}\| < \text{tol}$: Accept step
- Else: Update displacement, repeat

Frame Handling:

- **Bulk plasticity:** Computed in **material (crystal) frame**
- **CZM tractions:** Computed in **local interface frame**
- **Coupling:** Both transformed to **global frame** for residual assembly

Interface local frame definition:

- $\mathbf{n}$: Normal to interface (outward from grain 1)
- $\mathbf{s}$: First tangent (in-plane)
- $\mathbf{t}$: Second tangent (out-of-plane)
- Orthonormal: $\mathbf{n} \cdot \mathbf{s} = 0$, $\mathbf{n} \cdot \mathbf{t} = 0$, $\mathbf{s} \cdot \mathbf{t} = 0$

3.1.3 No Direct Coupling of Internal Variables

Important: Bulk plastic strain κ and interface damage D are **independent**:

- Plastic strain does NOT directly affect interface damage
- Interface damage does NOT directly affect yield strength
- Coupling is purely kinematic (through displacement field)

Physical justification:

- Bulk plasticity: Dislocation motion within grains
- Interface damage: Bond rupture at grain boundaries
- Different physical mechanisms, different length scales

3.1.4 Indirect Coupling Through Stress State

While internal variables are independent, stress state couples behaviours:

1. Plasticity increases interface loading:

- Bulk yielding $\rightarrow$ stress redistribution
- Stresses concentrate at grain boundaries
- Increased interface displacement jumps
- Earlier interface damage initiation

2. Interface damage increases bulk stress:

- Interface softening $\rightarrow$ load transfer to bulk
- Remaining grains carry higher load
- May trigger additional yielding

Competition: Intergranular vs Transgranular Failure The final failure mode depends on relative strengths:

- **Strong interfaces, weak bulk:** Transgranular fracture
 - Interface: $\sigma_{\text{int}} = 700$ MPa, $D < 0.1$
 - Bulk: $\sigma_y = 400$ MPa, $\kappa > 0.05$ (extensive plasticity)
 - Failure: Through grains
- **Weak interfaces, strong bulk:** Intergranular fracture
 - Interface: $\sigma_{\text{int}} = 400$ MPa, $D > 0.8$
 - Bulk: $\sigma_y = 700$ MPa, $\kappa < 0.01$ (minimal plasticity)
 - Failure: Along grain boundaries
- **Matched strengths:** Mixed mode
 - Competing mechanisms
 - Failure path determined by microstructure and loading

3.1.5 Numerical Considerations

Convergence:

- CZM softening can cause non-convergence if too abrupt
- A small softening exponent `eta` and, if needed, viscous regularization (`damage_viscosity`) help
- Bulk viscoplasticity (`eta`, `m`) also improves convergence
- Timestep adaptation essential during damage onset

Mesh Requirements:

- **Bulk mesh:** Sufficient refinement to capture plastic zones
- **Interface mesh:** One-to-one correspondence with bulk boundaries
- **Compatibility:** Nodes are duplicated along grain boundaries (`BreakMeshByBlockGenerator`); the mesh generator chain must run in the input file. Hexahedral and tetrahedral meshes both work.

3.1.6 Input File Structure

Bulk and interface materials coexist in the same input file:

```
[Materials]
# ===== BULK PLASTICITY =====
[plasticity]
  type = OrthotropicPlasticityStressUpdate
  # ... bulk parameters ...
  block = 'grain_1 grain_2 grain_3' # Apply to grain blocks
[]

# ===== INTERFACE CZM =====
[czm_interface]
```

```

type = HomogenizedExponentialCZM
# ... interface parameters ...
boundary = 'grain_1_grain_2 grain_2_grain_3' # Apply to interfaces
[]
[]

```

Key difference:

- Plasticity uses `block` = (3D elements)
- CZM uses `boundary` = (2D interface elements)

3.1.7 Summary

Table 3.1: Bulk plasticity vs CZM comparison

Aspect	Bulk Plasticity	CZM
Location	Within grains (3D)	At interfaces (2D)
Frame	Material (crystal)	Local interface
Internal variable	Plastic strain κ	Damage D
Failure mechanism	Dislocation motion	Bond rupture
Direct coupling	None	None
Indirect coupling	Via stress field	Via stress field
Element type	Bulk (hex, tet)	Surface (quad, tri)

The following sections describe the CZM formulation in detail.

3.2 Probabilistic Homogenization

3.2.1 Quality Factor

An interface quadrature point represents a very large number of molecular contacts of unequal quality. Instead of sampling a discrete contact population, which would make the response depend on the interface area and on a random seed, the heterogeneity is described by a dimensionless **quality factor**

$$f \sim \mathcal{N}(\bar{f}, \sigma_q^2), \quad (3.1)$$

with mean $\bar{f}$ (Section 3.5; $\bar{f} = 1$ without spatial variation) and standard deviation $\sigma_q = \text{quality_std_dev}$ (default 0.10). The quality factor scales the peak strengths and the characteristic openings together,

$$\sigma_n^u(f) = f \sigma_n^u, \quad \tau_s^u(f) = f \tau_s^u, \quad \tau_t^u(f) = f \tau_t^u, \quad (3.2)$$

$$\delta_0(f) = f \delta_0, \quad \delta_c(f) = f \delta_c, \quad (3.3)$$

so that a weaker contact also fails at a smaller opening.

3.2.2 Gauss-Hermite Quadrature

The traction at the quadrature point is the expectation of the damaged single-contact traction over the quality distribution, evaluated with a 5-point Gauss-Hermite rule:

$$T = \sum_{k=1}^5 w_k (1 - D_k) T^{\text{und}}(\delta_{\text{eff}}; f_k), \quad f_k = \max\{\bar{f} + \sigma_q z_k, 0.1\}. \quad (3.4)$$

Nodes z_k and weights w_k are listed in Table 3.2. Each class k carries its own damage D_k and its own history variable, both stored per quadrature point. The factor $(1 - D_k)$ is inside the sum, so the result is the mean of the damaged tractions and not the product of a mean traction and a mean damage. The reported material property **damage** is $\sum_k w_k D_k$.

Table 3.2: 5-point Gauss-Hermite rule for a standard normal variable (weights sum to 1). f_k is shown for $\bar{f} = 1$, $\sigma_q = 0.10$.

k	z_k	f_k	w_k
1	-2.857	0.714	0.01126
2	-1.356	0.864	0.22208
3	0	1.000	0.53333
4	+1.356	1.136	0.22208
5	+2.857	1.286	0.01126

Because the weights sum to one, the traction depends neither on the interface area nor on a contact count, and there is no parameter for either. With $\sigma_q = 0$ all five classes coincide and the law is deterministic.

3.3 Exponential Traction-Separation Law

3.3.1 Kinematics

The displacement jump in the local interface frame has a normal component δ_n and two tangential components δ_s, δ_t . Only opening contributes to the cohesive law. The Macaulay bracket is replaced by a smooth approximation,

$$\langle \delta_n \rangle = \frac{1}{2} \left(\delta_n + \sqrt{\delta_n^2 + \epsilon^2} \right), \quad \epsilon = 10^{-6} \text{ length units}, \quad (3.5)$$

so that traction and Jacobian derive from the same differentiable function. The effective opening and the mode ratio are

$$\delta_{\text{eff}} = \sqrt{\langle \delta_n \rangle^2 + \delta_s^2 + \delta_t^2}, \quad \varphi = \frac{\sqrt{\delta_s^2 + \delta_t^2}}{\max\{\langle \delta_n \rangle, 10^{-6} \delta_{0,n}\}}. \quad (3.6)$$

$\varphi = 0$ is pure opening, $\varphi \rightarrow \infty$ pure sliding. The smoothing width ϵ is a fixed number in the length unit of the mesh, not a parameter; see the remark on units in Section 3.7.

3.3.2 Single-Contact Law

For a contact of quality f , with $r = \delta_{\text{eff}}/\delta_0(f)$, the undamaged traction magnitude is

$$T^{\text{und}} = T_{\text{peak}} e r^\nu \exp(-r^\nu), \quad \nu = \begin{cases} \mu & r < 1 \quad (\text{loading}), \\ \eta & r \geq 1 \quad (\text{softening}), \end{cases} \quad (3.7)$$

after Xu and Needleman [11] and Wang et al. [12]. The factor e places the maximum T_{peak} at $r = 1$, where both branches have zero slope, so the law is continuously differentiable. The exponents are **mu** and **eta** (both default 1).

The traction vector is collinear with the jump,

$$\mathbf{T} = \frac{T}{\delta_{\text{eff}}} (\langle \delta_n \rangle, \delta_s, \delta_t), \quad (3.8)$$

with T from Eq. (3.4).

3.3.3 Mixed-Mode Peak Strength

The peak strength for a given jump direction is an elliptic interpolation between the pure-mode strengths on the direction cosines of the jump:

$$T_{\text{peak}} = \sqrt{\left(\sigma_n^u \frac{\langle \delta_n \rangle}{\delta_{\text{eff}}}\right)^2 + \left(\tau_s^u \frac{\delta_s}{\delta_{\text{eff}}}\right)^2 + \left(\tau_t^u \frac{\delta_t}{\delta_{\text{eff}}}\right)^2}. \quad (3.9)$$

For a single tangential direction and in terms of the mode ratio this reads

$$T_{\text{peak}}(\varphi) = \sqrt{\frac{(\sigma_n^u)^2 + \varphi^2 (\tau_s^u)^2}{1 + \varphi^2}}. \quad (3.10)$$

This is not the stress-space ellipse

Equation (3.9) is sometimes confused with the quadratic stress criterion $(\sigma_n/\sigma_n^u)^2 + (\tau_s/\tau_s^u)^2 + (\tau_t/\tau_t^u)^2 = 1$ [13, 14]. The two agree in the pure modes and differ in between (Table 3.3). Because the traction is collinear with the jump, the stress-space ellipse would give $T_{\text{peak}} = [\sum_i (r_i/\sigma_i^u)^2]^{-1/2}$ with the direction cosines r_i , which is lower. Predictions for mixed-mode tests must be made with Eq. (3.9), the form that is implemented. Earlier versions of this document quoted stress-space values.

Table 3.3: Peak traction magnitude for the protein interface ($\sigma_n^u = 626$ MPa, $\tau_s^u = \tau_t^u = 374$ MPa) under three rules. Only the last column is implemented.

Loading	$\langle \delta_n \rangle : \delta_s : \delta_t$	Linear, stress	Ellipse, stress	Eq. (3.9)
Pure opening	1:0:0	626	626	626 MPa
Pure sliding	0:1:0	374	374	374 MPa
Mixed, $\varphi = 1$	1:1:0	331	454	516 MPa

The linear criterion in the first column is the Mohr-Coulomb-type form used to characterise single interfaces in the molecular dynamics study [3]. The continuum law does not reproduce it; it only shares the pure-mode strengths. Whether the elliptic interpolation is the appropriate homogenised form has not been tested against mixed-mode molecular dynamics data.

3.3.4 Mixed-Mode Characteristic Opening

The characteristic opening interpolates between the normal value $\delta_{0,n} = \text{delta_0_normal}$ and the tangential value $\delta_{0,t} = \text{delta_0_tangent}$ following Wang et al. [12]:

$$\delta_0(\varphi) = \frac{\delta_{0,n} \delta_{0,t} \sqrt{1 + \varphi^2}}{\sqrt{\delta_{0,t}^2 + \varphi^2 \delta_{0,n}^2}}. \quad (3.11)$$

Its limits and special cases are

$$\delta_0(0) = \delta_{0,n}, \quad (3.12)$$

$$\lim_{\varphi \rightarrow \infty} \delta_0(\varphi) = \delta_{0,t}, \quad (3.13)$$

$$\delta_0(\varphi) = \delta_0 \quad \text{for all } \varphi \text{ if } \delta_{0,n} = \delta_{0,t} = \delta_0. \quad (3.14)$$

For $\varphi = 1$ and the protein values (191 pm, 217 pm) the result is 203 pm. For equal values the threshold does not depend on the mode: damage then starts when δ_{eff} reaches δ_0 , whatever the direction of the jump. The function returns $\delta_{0,n}$ for $\varphi < 10^{-12}$ and $\delta_{0,t}$ for $\varphi > 10^6$.

Both parameters are required; there is no single `delta_0` parameter.

3.4 Damage Evolution

3.4.1 History Variable

Each class keeps the largest effective opening it has experienced,

$$\kappa_k = \max_{t' \leq t} \delta_{\text{eff}}(t'), \quad r_k = \frac{\kappa_k}{\delta_0(f_k)}, \quad (3.15)$$

stored as `qp_kappa`. Damage is a function of κ_k , not of the current opening, so it cannot heal on unloading. This κ is an opening and is unrelated to the accumulated plastic strain of Chapter 2.

3.4.2 Damage Law

Following Wang et al. [12],

$$D^{\text{law}}(r) = 1 - r^{\eta-1} \exp(1 - r^{\eta}), \quad (3.16)$$

which is zero at $r = 1$ and tends to one for $r \rightarrow \infty$. The hard switch at $r = 1$ is replaced by a smooth activation,

$$D_k^{\text{target}} = D^{\text{law}}(r_k) \frac{1}{2} \left[1 + \tanh \frac{r_k - 1}{0.02} \right], \quad (3.17)$$

which removes a kink in the Jacobian at damage initiation. The width 0.02 is fixed in the code.

In the softening branch the traction of a class is therefore the product of the two expressions,

$$(1 - D^{\text{law}}) T^{\text{und}} = T_{\text{peak}} e^2 r^{2\eta-1} \exp(-2r^{\eta}), \quad r > 1, \quad (3.18)$$

when the opening increases monotonically and without viscous delay. This is the law as implemented and it is intended: damage is kept as a separate variable, and its effect on the softening branch is that of the traction law with the exponent doubled. When choosing η , keep in mind that the post-peak decay is governed by 2η .

3.4.3 Viscous Regularization

With $\tau_d = \text{damage_viscosity} > 0$ the damage approaches its target at a finite rate,

$$D_k^{n+1} = D_k^n + \frac{\Delta t}{\tau_d} \left(\max\{D_k^{\text{target}}, D_k^n\} - D_k^n \right), \quad (3.19)$$

bounded to $[0, 1]$. The default is $\tau_d = 0$, for which D_k follows the target directly. The update is an explicit relaxation: it is meaningful for $\Delta t \leq \tau_d$, and a time step larger than τ_d overshoots the target. Choose τ_d with the time step in mind. When the target does not exceed the old value the derivative of the update is set to zero, so the Jacobian matches the response.

3.4.4 Complete Failure

A class is set to $D_k = 1$, bypassing the viscous delay, once $\kappa_k \geq \delta_c(f_k)$. If `delta_c` is not given it is computed as the opening at which the undamaged law has decayed to `failure_traction_ratio` (default 0.01) of the peak, i.e. the root $r_c > 1$ of $e r^{\eta} \exp(-r^{\eta}) = 0.01$, times the larger of the two characteristic openings.

Limitation. The root search is capped at $r_c = 5000$. For small η the law decays so slowly that the cap is reached (for η below about 0.24), δ_c becomes $5000 \delta_0$, and `failure_traction_ratio` has no effect. This happens silently. In that regime complete failure is governed by Eq. (3.18) alone, or by an explicit `delta_c`.

3.4.5 Role of the Softening Exponent

μ shapes the loading branch and is fitted to the molecular dynamics traction curves. η controls how fast the traction decays after the peak. It is used here mainly as a *regularization* parameter and not as a fitted material constant: a small η gives a slow decay, so an interface that has passed its peak keeps carrying load while neighbouring interfaces take over. This allows the load to redistribute through the interface network and avoids the simultaneous, catastrophic loss of many interfaces on which the global solve diverges. The value fitted to the first debonding event in molecular dynamics ($\eta \approx 1.07$) describes a single abrupt event and is not suitable for this purpose. Values between 0.17 and 0.45 have been used in production which allows to include multiple peaks from MD curves representing gradual bond rupture events. The relation is always $\eta < \mu$; the precise value has little influence on the peak, which is set by the strengths.

The consequence is stated plainly: the peak traction, which controls failure initiation, is physical, while the post-peak branch and hence the dissipated energy are numerical choices. The code does not enforce any relation between η and μ .

3.5 Spatial Variation of Interface Quality

If every quadrature point had the same mean quality, all points of a uniformly loaded interface would reach their peak in the same increment. To avoid this, the mean $\bar{f}$ of Eq. (3.1) is drawn once per quadrature point,

$$\bar{f} = \max\{f_{\text{sp}}, 0.1\}, \quad f_{\text{sp}} \sim \mathcal{N}(1, \sigma_{\text{sp}}^2), \quad (3.20)$$

with $\sigma_{\text{sp}} = \text{spatial_quality_std_dev}$ (default 0.15). The draw is deterministic: the generator is seeded with `spatial\_random\_seed + 100 × (element ID) + (quadrature point index)`, so the field is the same in every run and no random state is stored. The within-point spread is added to this mean, $f_k = \bar{f} + \sigma_q z_k$; the two are not multiplied.

```
[czm]
  type = HomogenizedExponentialCZM
  quality_std_dev      = 0.10    # within a quadrature point
  spatial_quality_std_dev = 0.15 # between quadrature points
  spatial_random_seed  = 1234
[]
```

Set `spatial\_quality\_std\_dev = 0` for single-interface verification tests, where a uniform interface is wanted. For polycrystals the default 0.15 is the value used in production. The two parameters act differently: σ_q rounds the peak of the local law with almost no change in peak traction, whereas σ_{sp} lowers and delays the peak of the *interface-averaged* response, because weak points soften while strong points are still loading (Chapter 7).

3.6 Compression, Initial Stiffness and Jacobian

Contact in compression. The cohesive traction vanishes smoothly for $\delta_n < 0$ because of Eq. (3.5). Interpenetration is resisted by a penalty on the negative part of the normal jump,

$$T_n \leftarrow T_n - K \langle -\delta_n \rangle, \quad K = \frac{\sigma_n^u}{\delta_{0,n}}, \quad (3.21)$$

with the same smooth bracket. Sliding under compression is not penalised and carries no friction.

Zero opening. For $\delta_{\text{eff}} < 10^{-12}$ the Jacobian is set to the diagonal stiffness $\sigma_n^u/\delta_{0,n}$, $\tau_s^u/\delta_{0,t}$, $\tau_t^u/\delta_{0,t}$, the normal entry multiplied by the derivative of the smooth bracket.

Jacobian. The derivative of the traction with respect to the jump is analytic for the dependence through δ_{eff} , including damage growth and the smooth bracket. The dependence of δ_0 on the mode ratio is added as a rank-one term whose derivative $d\delta_0/d\varphi$ is taken by finite differences. The dependence of T_{peak} on the jump direction, Eq. (3.9), is not included. The Jacobian is therefore exact when the jump keeps its direction, that is, when $\delta_n : \delta_s : \delta_t$ stays constant while the magnitude grows (pure opening, pure sliding, or a fixed mix), and approximate when the direction of the jump changes during a step, as it does at most interfaces of a polycrystal. The traction itself is always evaluated exactly; an approximate Jacobian costs Newton iterations, not accuracy.

Outputs. The material declares `damage` (weighted mean over the classes), `delta_eff`, and the per-class vectors `qp_damage` and `qp_kappa`.

3.7 MD-Derived Interface Parameters

Strengths and characteristic openings are taken directly from the molecular dynamics simulations of Kvashin et al. [3]. The openings are separations measured from the equilibrium configuration of the interface: the organic layer itself is about 1 nm thick, and $\delta_0 \approx 0.2$ nm is the additional separation at peak traction. The cohesive element has zero thickness and represents only this additional separation. The values are provisional and serve as the reference parameter set of this document.

Table 3.4: MD-derived CZM parameters [3].

	Protein	Water	Unit
σ_n^u	626 ± 53	742 ± 29	MPa
τ_s^u	374 ± 39	354 ± 61	MPa
$\delta_{0,n}$	191	224	pm
$\delta_{0,t}$	217	182	pm
μ	0.916	1.224	—

Use micrometres as the length unit with this class

The class has no unit system of its own: openings are in the length unit of the mesh, and tractions in the stress unit of the bulk model (MPa). The coral models are meshed in micrometres, and the protein values are entered as `delta_0_normal` = 1.91e-4 and `delta_0_tangent` = 2.17e-4.

Do not enter the MD openings in millimetres. The smoothing width $\epsilon = 10^{-6}$ of Eq. (3.5) is a fixed number in mesh units and must be small compared with δ_0 . In micrometres $\epsilon/\delta_0 \approx 0.005$. In millimetres $\delta_0 = 1.9 \times 10^{-7}$ is five times *smaller* than ϵ : the bracket $\langle \delta_n \rangle$ then returns about $\epsilon/2$ at zero jump, more than twice δ_0 , so every interface would be past its peak and damaged before any load is applied. The code does not check this. As a rule, keep $\delta_0 \geq 100 \epsilon = 10^{-4}$ in mesh units.

The start-up banner of the class prints the openings in nanometres on the assumption that the mesh is in micrometres.

Resolution. What matters numerically is the ratio of δ_0 to the displacement increment at the interface. With δ_0 three orders of magnitude below the size of an element that resolves a grain, the load step must be chosen so that the interface is not carried past its peak in a single increment; the smooth activation of Eq. (3.17), a small η and, if needed, $\tau_d > 0$ are the means

that make this tractable. Where a model instead uses a characteristic opening enlarged towards the element size, the peak traction remains physical but the opening and the dissipated energy do not, and this must be stated with the results.

Chapter 4

MOOSE Installation

4.1 Why Conda MOOSE?

Traditional manual build: 2–4 hours

- Build PETSc from source: 1–2 hours
- Build libMesh: 30–60 minutes
- Build MOOSE test suite: 30–60 minutes
- Requires compiler toolchain, MPI, dependencies

Conda installation: 5–10 minutes

- Pre-compiled binaries for all dependencies
- Consistent environment across machines
- Easy updates and version management
- No complex dependency resolution

4.2 Installation Steps

4.2.1 Step 1: Install Miniconda

Linux / Windows WSL:

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.
sh
bash Miniconda3-latest-Linux-x86_64.sh
# Follow prompts, answer 'yes' to initialization
source ~/.bashrc
```

macOS:

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64
.sh
bash Miniconda3-latest-MacOSX-x86_64.sh
# Follow prompts, answer 'yes' to initialization
source ~/.bash_profile
```

Verify:

```
conda --version
# Should show: conda 24.x.x or later
```

4.2.2 Step 2: Create MOOSE Environment

```
# Add INL's public channel
conda config --add channels https://conda.software.inl.gov/public
# Create conda environment and install MOOSE dependencies
conda create -n moose moose-dev -c conda-forge
```

This installs:

- MOOSE framework (pre-compiled)
- PETSc (pre-configured)
- libMesh (pre-configured)
- MPI compilers (mpicxx, mpicc, mpifort)
- All required dependencies

Time: 5–10 minutes depending on network.

4.2.3 Step 3: Activate Environment

```
conda activate moose
```

CRITICAL: Always activate this environment before any MOOSE work!

4.2.4 Step 4: Verify Installation

```
# Check tools are available
which mpicxx
which mpirun

# Check versions
mpicxx --version
python --version

# Test MPI
mpirun -n 2 echo "MPI works!"
# Should print "MPI works!" twice
```

4.2.5 Step 5: (Important) Clone MOOSE for building references

```
cd ~/projects
git clone https://github.com/idaholab/moose.git
cd moose
git checkout master # or specific release tag

# Set MOOSE_DIR (optional, for reference only)
echo "export MOOSE_DIR=~/projects/moose" >> ~/.bashrc
source ~/.bashrc
```

Note: This is optional for general users. Conda package includes everything needed to build apps.

4.3 Environment Management

4.3.1 Daily Commands

```
# Activate MOOSE environment (ALWAYS FIRST!)
conda activate moose

# Check you're in the right environment
echo $CONDA_DEFAULT_ENV
# Should output: moose

# Deactivate when done
conda deactivate
```

4.3.2 Installing Additional Packages

```
# Install IN the moose environment, not base!
conda activate moose
conda install matplotlib numpy pandas scipy

# Or using pip if needed
pip install some-package
```

4.3.3 Updating MOOSE

```
conda activate moose
conda update moose-dev -c conda-forge
```

4.3.4 Managing Multiple Versions

```
# Stable version for production
conda create -n moose-stable moose-dev=<version> -c conda-forge

# Latest version for development
conda create -n moose-dev moose-dev -c conda-forge

# Switch between them
conda activate moose-stable
conda activate moose-dev
```

4.4 Shell Configuration

Add to ~/.bashrc (Linux/WSL) or ~/.bash_profile (macOS):

```
# MOOSE conda environment
alias moose-on='conda activate moose'
alias moose-off='conda deactivate'

# Optional: auto-activate in project directories
```

Reload:

```
source ~/.bashrc # or ~/.bash_profile on macOS
```

Chapter 5

Building Your Application

5.1 Overview

This chapter covers two scenarios for building and using the aragonite application:

1. **Using the Git repository** (recommended) – Clone, build, test
2. **Building from source files** (for development) – Use MOOSE’s `stork.sh` script to create application skeleton

Note: The Git repository approach is recommended once the code is published. During development, use Scenario 2.

5.2 Scenario 1: Using the Git Repository

Recommended for: Users who want to use the published materials

Prerequisites:

- Conda MOOSE environment installed and activated (Chapter 4)
- Git installed (usually included with conda)

5.2.1 Clone Repository

```
# Activate conda environment
conda activate moose

# Clone repository
cd ~/projects
git clone https://gitlab.tu-clausthal.de/iww/tuc-fe-moose
cd aragonite

# Repository contents:
#   - Makefile (pre-configured)
#   - Source files (src/materials/*.C)
#   - Header files (include/materials/*.h)
#   - Test suite (test/validation/*)
#   - Example inputs (examples/*)
#   - Documentation (doc/*)
```

5.2.2 Build

```
# Build application (one command!)
make -j8

# Expected output:
#   ...
#   Linking Library...
#   Linking Executable .../projects/aragonite/aragonite-opt

# Verify build
ls -lh aragonite-opt
./aragonite-opt --version
```

Build time:

- First build: 2–5 minutes
- Incremental builds: 10–30 seconds

5.2.3 Run Validation Tests

After building, verify the installation by running the validation test suite:

```
# Navigate to scripts directory
cd scripts
./run_validation.sh

./run_validation.sh -c ortho
./run_validation.sh -c czm

# Run all tests
./run_validation.sh           % all categories

# Or run specific category
./run_validation.sh -c czm     % CZM only
./run_validation.sh -c fabric  % fabric orthotropy tests

# With parallel execution
./run_validation.sh -p 8       % 8 MPI ranks
```

Expected summary:

Category: rve

Description: Representative volume element tests

Testing: dry ... [PASS] Matches gold file

Testing: grain_interfaces ... [PASS] Matches gold file

Category: postyield

Description: Post-yield behavior modes

Testing: exp_hardening ... [PASS] Matches gold file

Testing: exp_softening ... [PASS] Matches gold file

Testing: linear_hardening ... [PASS] Matches gold file

Testing: perfect_plasticity ... [PASS] Matches gold file

Testing: piecewise_softening ... [PASS] Matches gold file

Category: czm

Description: Cohesive zone model tests

Testing: complete_separation ... [PASS] Matches gold file

Testing: mixed_mode ... [PASS] Matches gold file

Testing: mode_i ... [PASS] Matches gold file

Testing: mode_ii ... [PASS] Matches gold file

Category: ortho

Description: Orthotropic plasticity single-element tests

Testing: x_tension ... [PASS] Matches gold file

Testing: xy_shear ... [PASS] Matches gold file

Testing: xz_shear ... [PASS] Matches gold file

Testing: y_tension ... [PASS] Matches gold file

Testing: yz_shear ... [PASS] Matches gold file

Testing: z_tension ... [PASS] Matches gold file

Category: asymmetry

Description: Tension-compression asymmetry tests

Testing: x_compression ... [PASS] Matches gold file

Testing: yield_surface_shift ... [PASS] Matches gold file

Category: fabric

Description: Fabric-scaling elasticity + plasticity tests

Testing: explicit_plasticity ... [PASS] Matches gold file

Testing: fabric_plasticity ... [PASS] Matches gold file

Category: grains

Description: Multi-grain bicrystal tests

Testing: bicrystal_0deg ... [PASS] Matches gold file

Testing: bicrystal_45deg ... [PASS] Matches gold file

Testing: bicrystal_90deg ... [PASS] Matches gold file

Testing: twin ... [PASS] Matches gold file

=====
Test Summary

=====

Total: 25

Passed: 25

Failed: 0

Skipped: 0

Note: Segmentation faults during cleanup (after CSV generation) are harmless and do not indicate test failure. If all tests show [PASS], the validation succeeded.

Test Categories

The validation suite includes multiple test categories:

Table 5.1: Validation test categories

Category	Tests	Purpose
ortho	6	Orthotropic elasticity & yield (6 directions)
asymmetry	2	Tension-compression asymmetry
postyield	5	Hardening/softening behaviour
grains	4	Multi-grain load redistribution
czm	4	GH cohesive zone traction-separation
fabric	2	Two simulations on identical mesh, BCs, and plasticity: <code>fabric_plasticity</code> via <code>ComputeFabricElasticityTensor</code> (Zysset-Curnier) and <code>explicit_plasticity</code> via <code>ComputeElasticityTensorCoupled</code> with the equivalent <code>symmetric9</code> tensor. Verifies that both routes give identical elastic slope and yield stress.
rve	2	(1) grain plasticity only; (2) grain plasticity + GH interfaces

RVE pair: Both runs use the same mesh and bulk properties. Run 1 disables CZM to isolate grain plasticity; run 2 activates GH cohesive interfaces. The difference quantifies the interface contribution. Not included in the plot set.

Fabric tests: Two MOOSE simulations (`fabric_plasticity`, `explicit_plasticity`) on an identical single-element mesh with identical boundary conditions and plasticity parameters. The plot overlays both stress-strain curves to verify they coincide through elastic slope and yield point, confirming that `ComputeFabricElasticityTensor` and the equivalent `ComputeElasticityTensorCoupled` give the same result. Gold files are generated by `compute_fabric_referen`

Command Options

```
# Show help
./run_validation.sh --help

# Run specific category
./run_validation.sh -c ortho
./run_validation.sh -c czm

# Verbose output (show simulation details)
./run_validation.sh -v -c czm

# Generate new gold files (after code changes)
./run_validation.sh --generate-gold -c fabric

# List all available tests
./run_validation.sh --list
```

5.2.4 Visualizing Validation Results

After running validation tests, generate publication-quality figures using the plotting script:

```
# Navigate to scripts directory
cd scripts

# Generate all validation plots
./plot_validation.sh

# Or plot specific category
./plot_validation.sh -c ortho
./plot_validation.sh -c grains
./plot_validation.sh -c czm
```

Output files: Figures are saved to `test/validation/figures/`

Table 5.2: Generated validation figures

File	Content
ortho.png	All 6 orthotropic directions (X/Y/Z tension, XY/XZ/YZ shear) on single panel
asymmetry.png	Tension-compression asymmetry + biaxial yield surface shift
postyield.png	All 5 post-yield evolution modes (perfect/hardening/softening) on single panel
grains.png	Multi-grain orientation effects with load redistribution annotation
czm.png	All 4 CZM modes (Mode I/II/mixed/complete separation) on single panel

Plot Interpretation

Orthotropic validation (ortho.png) Six stress-strain curves demonstrating:

- Different elastic moduli: $E_1 > E_3 > E_2$ (slopes vary)
- Anisotropic yield strengths: Different plateaus for each direction

Post-yield validation (postyield.png) Five evolution modes showing:

- **Perfect plasticity:** Flat plateau after yield
- **Exponential hardening:** Asymptotic approach to $r_\infty \sigma_y$
- **Linear hardening:** Linear stress increase with plastic strain
- **Exponential softening:** Post-peak stress decrease to $r_{\text{res}} \sigma_y$
- **Piecewise softening:** Bilinear stress reduction

Multi-grain validation (grains.png) Demonstrates grain-level heterogeneity:

- **Color grouping:** One colour per misorientation angle (0° , 45° , 90° , 63.8° Twin)
- **Line style:** Solid = Grain 1 (fixed), Dashed = Grain 2 (pulled)
- **Load redistribution:** Apparent strain decrease in fixed grain indicates elastic unloading as pulled grain yields and softens

- **Orientation effects:** Larger misorientations produce greater stress heterogeneity

Physical interpretation: When Grain 2 (pulled) yields first due to unfavorable orientation, it accumulates plastic strain and softens. As its load-bearing capacity decreases, Grain 1 (fixed) elastically unloads, releasing stored elastic energy. This manifests as decreasing strain in the non-yielding grain.

CZM validation (czm.png) Four traction-separation curves:

- **Mode I (Opening):** normal traction vs. normal separation
- **Mode II (Shear):** tangential traction vs. tangential separation
- **Mixed mode:** Combined normal and shear loading
- **Complete separation:** Load-unload-reload cycle demonstrating damage irreversibility

5.2.5 Update from Repository

```
# Pull latest changes
git pull

# Rebuild
make -j8

# Run tests to verify
cd scripts && ./run_validation.sh
```

That's it. The repository includes everything pre-configured.

5.3 Scenario 2: Building from Source Files

Recommended for: Developers/interested in creating the application from source files

Prerequisites:

- Conda MOOSE environment installed and activated (Chapter 4)
- MOOSE repository cloned (for `stork.sh` script)
- Material source files: `*.C` and `*.h`

5.3.1 Step 1: Clone MOOSE (for `stork.sh` script)

Important: Even with conda MOOSE, you need the MOOSE repository for the `stork.sh` application generator script.

```
# Activate environment
conda activate moose

# Clone MOOSE repository (just for scripts)
cd ~/projects
git clone https://github.com/idaholab/moose.git
cd moose
git checkout master

# You only need this for the stork.sh script
# The actual MOOSE framework comes from conda
```

5.3.2 Step 2: Create Application Skeleton with stork.sh

The `stork.sh` script generates a complete MOOSE application structure with proper Makefiles and registration boilerplate.

```
# Navigate to projects directory
cd ~/projects

# Run stork.sh to create application
./moose/scripts/stork.sh aragonite
```

```
# Script creates:
aragonite/
├── Makefile          (auto-generated!)
├── src/
├── include/
├── test/
├── doc/
└── ... (standard MOOSE app structure)
```

Important: The application name (`aragonite`) will be used in material registration. If you use a different name, you may need to update material registration in source files (see Section 5.3.5).

5.3.3 Step 3: Enable Solid Mechanics Module

Edit the auto-generated `Makefile` to enable the solid mechanics module:

```
cd aragonite
vim Makefile # or your preferred editor
```

Find the block after:

```
ALL_MODULES := no
```

Change to:

```
ALL_MODULES := no
...
SOLID_MECHANICS := yes
```

Enabling all modules increases build time. We only need solid mechanics for this application. Save and close the file.

5.3.4 Step 4: Copy Material Source Files

Copy your material implementation files to the generated directories:

```
# Copy header files
cp /path/to/OrthotropicPlasticityStressUpdate.h \
include/materials/
cp /path/to/ComputeElasticityTensorCoupled.h \
include/materials/
cp /path/to/ComputeFabricElasticityTensor.h \
include/materials/
cp /path/to/HomogenizedExponentialCZM.h \
include/materials/
cp /path/to/MaterialModelPresets.h \
```

```

include/materials/
cp /path/to/MandelHelpers.h \
include/materials/

# Copy source files
cp /path/to/OrthotropicPlasticityStressUpdate.C \
src/materials/
cp /path/to/ComputeElasticityTensorCoupled.C \
src/materials/
cp /path/to/ComputeFabricElasticityTensor.C \
src/materials/
cp /path/to/HomogenizedExponentialCZM.C \
src/materials/

```

Directory structure after copying:

```

aragonite/
├── Makefile
├── include/
│   └── materials/
│       ├── OrthotropicPlasticityStressUpdate.h
│       ├── ComputeElasticityTensorCoupled.h
│       ├── ComputeFabricElasticityTensor.h
│       ├── HomogenizedExponentialCZM.h
│       ├── MaterialModelPresets.h
│       └── MandelHelpers.h
├── src/
│   └── materials/
│       ├── OrthotropicPlasticityStressUpdate.C
│       ├── ComputeElasticityTensorCoupled.C
│       ├── ComputeFabricElasticityTensor.C
│       └── HomogenizedExponentialCZM.C
├── test/
└── ... (add tests later)

```

Header-only Files

Two of the files have no corresponding .C file; all their functions are `inline`.

MandelHelpers.h. Conversion between tensors and their six-component Mandel form: a symmetric rank-2 tensor becomes a 6-vector and a rank-4 tensor with minor symmetries a 6×6 matrix, with the factors $\sqrt{2}$ and 2 on the shear entries (Section 2.1.2). With these factors the matrix product, the inverse and the scalar product of the six-component objects coincide with the tensor operations, which is what the 7×7 system of the second return-mapping stage relies on (Section 2.8). It is required by `OrthotropicPlasticityStressUpdate` and replaces the former `VoigtHelpers.h`, which must not be used any more: it copied components without the shear factors.

MaterialModelPresets.h. The constants and derivations behind the `material_model` flags (Section 6.2). It is included by both `ComputeFabricElasticityTensor` and `OrthotropicPlasticityStressUpdate`, so that the elastic and the plastic side of a preset come from one place.

5.3.5 Step 5: Verify Material Registration

Check that material classes are registered with the correct application name:

```
grep -n "registerMooseObject" src/materials/*.C
```

Each .C file should have at the top (after includes):

```
// Option 1: Register with aragonite app
registerMooseObject("aragoniteApp", OrthotropicPlasticityStressUpdate);

// Option 2: Register with SolidMechanics (also works)
registerMooseObject("SolidMechanicsApp",
    OrthotropicPlasticityStressUpdate);
```

If you used a different app name in `stork.sh` (e.g., `myapp`), replace `aragoniteApp` with `myappApp`.

Note: Registering with `SolidMechanicsApp` should work regardless of application name.

5.3.6 Step 6: Build Application

```
# Ensure environment is activated
conda activate moose

# Build (use number of CPU cores available)
make -j8

# Expected output:
#   Building CXX object ...
#   ...
#   Linking aragonite-opt...
```

Build time:

- First build: 2–5 minutes
- Incremental builds (after editing): 10–30 seconds

5.3.7 Step 7: Test Build

Verify the executable works:

```
# Check executable exists
ls -lh aragonite-opt

# Run version check
./aragonite-opt --version

# Should display MOOSE version and build info
```

Create a minimal test input:

```
mkdir -p test/basic
cat > test/basic/build_test.i << 'EOF'
[Mesh]
type = GeneratedMesh
dim = 3
nx = 1
ny = 1
nz = 1
```

```

[]

[Problem]
  solve = false # Just test that app runs
[]

[Executioner]
  type = Steady
[]
EOF

# Run test
./aragonte-opt -i test/basic/build_test.i

# Should complete without errors

```

5.3.8 Step 8: Run Validation Tests

After successful build, run the validation test suite to verify implementation:

```

# Copy test suite scripts if not already present
mkdir -p scripts
cp /path/to/run_validation.sh scripts/
cp /path/to/compare_with_gold.py scripts/
chmod +x scripts/run_validation.sh

# Copy test inputs and gold files
cp -r /path/to/test/validation test/

# Run validation
cd scripts
./run_validation.sh -c ortho

```

If tests pass, the implementation is verified.

Minimum test suite for basic validation:

Create at least these 3 tests before running validation:

- ortho/x_tension - Primary loading direction
- ortho/y_tension - Anisotropy verification
- ortho/xy_shear - Shear behaviour

See Section 5.2.3 for detailed minimal validation procedures and Chapter 7.

5.4 Additional Material for Polycrystals

For polycrystalline simulations with per-element Euler angles, you need an additional material: `ComputeElasticityTensorCoupled`.

5.4.1 Why This is Needed

Standard MOOSE's `ComputeElasticityTensor` only accepts scalar parameters for Euler angles:

```

# Standard version - same angle for ALL elements
[Materials]
  [elasticity]

```

```

    type = ComputeElasticityTensor
    euler_angle_1 = 45.0 # Scalar - applies to all elements
[]
[]

```

For polycrystals, each element needs different angles. This requires coupling to AuxVariables:

```

# Coupled version - different angle PER element
[Materials]
[elasticity]
    type = ComputeElasticityTensorCoupled
    coupled_euler_angle_1 = euler_phi1 # AuxVariables, degrees
    coupled_euler_angle_2 = euler_Phi
    coupled_euler_angle_3 = euler_phi2
[]
[]

```

5.4.2 Implementation

Files: `ComputeElasticityTensorCoupled.C` and `ComputeElasticityTensorCoupled.h`

The class is derived from the logic of MOOSE's `ComputeElasticityTensor` and adds three optional coupled variables, `coupled_euler_angle_1`, `coupled_euler_angle_2` and `coupled_euler_angle_3` (degrees; give all three or none, Table in Section 6.3). When they are given, the rotation is built at every quadrature point from the local values of the three variables and applied to the material-frame elasticity tensor with the same convention as the standard class (Section 2.11). The scalar parameters `euler_angle_1/2/3` of the standard class remain available for a uniform orientation. `ComputeFabricElasticityTensor` accepts the same three coupled variables.

The plasticity class reads its orientation from its own coupled variables `euler_angle_1/2/3`. Both materials must be given the same three variables; nothing in the code enforces this.

5.4.3 When to Use Each Version

Use Case	Material	Euler Angles
Single crystal	<code>ComputeElasticityTensor</code>	Scalar parameters
Multiple crystals, same orientation	<code>ComputeElasticityTensor</code>	Scalar parameters
Twin grains (2 blocks)	<code>ComputeElasticityTensor</code>	Different per block
Polycrystal (many orientations)	<code>ComputeElasticityTensorCoupled</code>	AuxVariables
EBSD-based RVE	<code>ComputeElasticityTensorCoupled</code>	From exodus file

5.5 Summary

5.5.1 Scenario Comparison

Table 5.3 compares both installation scenarios.

5.5.2 Key Files

The application requires these material files:

Table 5.3: Build scenario comparison

Aspect	From Source	Git Clone
Setup time	15–30 minutes	5 minutes
MOOSE clone	Required (for <code>stork.sh</code>)	Not required
<code>stork.sh</code>	Manual run	Not needed
Makefile edit	Manual (enable module)	Pre-configured
File copying	Manual	Included
Build command	<code>make -j8</code>	<code>make -j8</code>
Tests	Manual setup	Included
Updates	Manual	<code>git pull</code>
Recommended for	Developers	Users

Table 5.4: Required material files

File	Purpose
<code>OrthotropicPlasticityStressUpdate.C/.h</code>	Orthotropic plasticity model
<code>ComputeElasticityTensorCoupled.C/.h</code>	Coupled elasticity (per-element Euler angles)
<code>ComputeFabricElasticityTensor.C/.h</code>	Density- and fabric-based elasticity, model flags
<code>HomogenizedExponentialCZM.C/.h</code>	Cohesive zone model with homogenization
<code>MaterialModelPresets.h</code>	Constants of the material-model presets (header-only)
<code>MandelHelpers.h</code>	Tensor–Mandel conversion utilities (header-only)

5.5.3 Key Points

- `stork.sh` creates proper application structure automatically
- Enable `SOLID_MECHANICS := yes` in Makefile
- `MandelHelpers.h` and `MaterialModelPresets.h` are header-only and required by the plasticity class
- Material registration must match app name (or use `SolidMechanicsApp`)
- Incremental builds are fast (10–30 seconds)
- Multiple build types can coexist (opt, dbg, oprof)

5.5.4 Troubleshooting

For build errors, compilation issues, and common problems, see Section 8.1 (Troubleshooting). Common issues include:

- `stork.sh`: `command not found` – MOOSE repository not cloned
- `ModuleNotFoundError` – Conda environment not activated
- Undefined reference errors – Material registration issues
- Compilation errors – Missing header files or version mismatches

5.5.5 Next Steps

After successful build:

1. **Run validation tests** (Chapter 7): Verify implementation
2. **Try examples** (Section 7.25): Check usage patterns
3. **Create simulations**: Apply to your material system

Chapter 6

Complete Parameter Reference

This chapter lists every input parameter of the four material classes with its default, the constants behind each material-model preset, three input templates and the solver settings. The meaning of the parameters is explained in Chapters 2 and 3. Defaults are those of the source code; where a default is a poor choice this is said.

6.1 Material-Model Flags

A material model is selected by name. It fixes which elastic and plastic formulation is used and supplies default values for the constants.

Table 6.1: Material models. IDs 0 to 5 are the flags of the reference UMAT.

ID	Name	Elasticity	Yield surface	Post-yield
0	trabecular_iso	isotropic	isotropic	simple_softening
1	trabecular_ti	transversely isotropic	transversely isotropic	simple_softening
2	trabecular_fabric_ti	fabric, TI	fabric, TI	simple_softening
3	trabecular_fabric_ortho	fabric, orthotropic	fabric, orthotropic	simple_softening
4	compact_iso	isotropic	isotropic	exp_hardening
5	compact_ti	transversely isotropic	transversely isotropic	exp_hardening
6	coral	9 constants C_{ijkl}	9 strengths, 3 ζ_{ij}	exp_softening
7	legacy (default)	behaviour before the flags existed; nothing is preset		

Three parameters carry the flag:

- `material_model` is accepted by both `ComputeFabricElasticityTensor` and `OrthotropicPlasticityStressUpdate`. Set it once in `[GlobalParams]`; both objects follow it.
- `elastic_model` (elasticity object) and `plastic_model` (plasticity object) override it for that one block. Mixing two different models is allowed and produces a warning.

Rules that apply to every model other than `legacy`:

- **Every preset value is a default.** Setting the parameter explicitly overrides it. The resolved values are printed at start-up.
- **Unused parameters are reported.** A parameter that the selected model does not read produces a warning, not an error.

- **Bone models require `density_rho`.** The fabric models also require `fabric_m1`, `fabric_m2`, `fabric_m3`. The TI and fabric-TI models read `main_direction` (default 1).
- **coral requires `zeta12`, `zeta13`, `zeta23`.** They have no default.
- **`use_fabric_scaling` is rejected.** It belongs to `legacy` only.

`ComputeElasticityTensorCoupled` does not take the flag. It is the elasticity object for legacy inputs with explicit constants.

6.2 Preset Constants

All values are defined in `MaterialModelPresets.h`, which also records the UMAT line each one was taken from. The coral values come from the aragonite molecular dynamics runs and are marked provisional there.

Table 6.2: Elastic presets. Moduli in MPa. k is the density exponent, l the fabric exponent; $\delta = 1$ throughout.

Model	E_0	ν_0	G_0	E_a	ν_a	G_a	k	l
<code>trabecular_iso</code>	8534.64	0.246					1.63	
<code>trabecular_ti</code>	6561.94	0.323		18661.3	0.32	3739.02	1.63	
<code>trabecular_fabric_ti</code>	9994.65	0.2278	3361.14				1.62	1.1
<code>trabecular_fabric_ortho</code>	9994.65	0.2278	3361.14				1.62	1.1
<code>compact_iso</code>	19327.0	0.3434					1.63	
<code>compact_ti</code>	15079.6	0.4620		24578.5	0.354	6578.01	1.63	

The coral elastic default is `C_ijkl = '171800 57500 30200 106700 46900 84200 42100 31100 46600'` (MPa, `symmetric9` order).

Table 6.3: Plastic presets. Strengths in MPa. Empty shear entries are derived from the isotropy constraint, Eq. (2.24). $\delta = 1$ throughout.

Model	σ_0^+	σ_0^-	σ_a^+	σ_a^-	τ_0	τ_a	ζ_0	ζ_a	p	q
<code>trabecular_iso</code>	61.43	89.24					0.1876		1.686	
<code>trabecular_ti</code>	50.63	68.07	98.68	165.80		51.62	0.4707	0.1809	1.69	
<code>trab._fabric_ti</code>	66.012	98.876			41.889		0.2182		1.686	1.05
<code>trab._fabric_ortho</code>	66.012	98.876			41.889		0.218		1.69	1.05
<code>compact_iso</code>	144.7	234.2					0.49		1.69	
<code>compact_ti</code>	56.54	201.28	176.40	268.00		82.55	0.0074	1.4045	1.686	

The small differences between the two fabric rows (ζ_0 , p) are in the UMAT and are reproduced as they are. The coral strength defaults are $\sigma^+ = 4980$, 4100, 5340 MPa (xx, yy, zz), compression equal to tension, and $\tau = 4510$, 5080, 5060 MPa (xy, xz, yz).

The post-yield scalars of a preset apply only while its `postyield_mode` is in use, and likewise `eta` and `m` with its `viscosity_mode` (Section 2.6).

Table 6.4: Post-yield and viscosity presets.

	Trabecular (0–3)	Compact (4, 5)	Coral (6)
postyield_mode	simple_softening	exp_hardening	exp_softening
initial_yield_ratio	0.7	0.7	1.0
residual_strength	(not read)	0	0.7
kslope	100	100	30
kmax	0.015	(not read)	0.001
kmin	(not read)	(not read)	(not read)
kwidth	8	(not read)	(not read)
viscosity_mode	linear	linear	linear
eta	10^{-4}	10^{-4}	0.002

6.3 ComputeFabricElasticityTensor

Parameter	Default	Description
material_model	legacy	Shared model flag (Section 6.1).
elastic_model	unset	Model for this block only; overrides <code>material_model</code> .
E_0, nu_0	preset	Base (isotropic or transverse) Young’s modulus [MPa] and Poisson’s ratio. Required for <code>legacy</code> .
G_0	preset	Base shear modulus [MPa], fabric models. In <code>legacy</code> : $E_0/[2(1 + \nu_0)]$.
E_a, nu_a, G_a	preset	Axial modulus, axial Poisson’s ratio ($S_{at} = -\nu_a/E_a$) and shear modulus of the planes containing the axis. TI models.
C_ijkl	aragonite	Nine constants in <code>symmetric9</code> order [MPa]. <code>coral</code> only.
density_rho	1.0	Bone volume fraction ρ in $(0, 1]$. Must be set for bone models.
fabric_m1/m2/m3	1.0	Fabric eigenvalues, normalised to a sum of 3. A warning is issued if the sum differs from 3 by more than 0.01.
exponent_k	2.0 or preset	Density exponent k .
exponent_l	1.0 or preset	Fabric exponent l .
delta_cortical	1.0	δ of Eq. (2.11); 1 disables the correction for $\rho > 0.5$.
main_direction	1	Axis of transverse isotropy (1, 2 or 3).
coupled_euler_angle_1/2/3	none	Orientation variables in degrees (Section 2.11). Give all three or none.

6.4 ComputeElasticityTensorCoupled

This class adds the following to the parameters of MOOSE’s `ComputeElasticityTensor` (`C_ijkl`, `fill_method`, and the scalar `euler_angle_1/2/3`, which should be left unset).

Parameter	Default	Description
coupled_euler_angle_1/2/3	none	Orientation variables in degrees.
density_rho	1.0	Relative density in $(0, 1]$.
elasticity_density_exponent	0.0	k in $\mathbb{C} = \rho^k \mathbb{C}_0$; 0 disables the scaling.

6.5 OrthotropicPlasticityStressUpdate

The orientation variables `euler_angle_1`, `euler_angle_2`, `euler_angle_3` are **required** for every model, including the bone models. For a material whose axes coincide with the global axes, couple them to variables that are zero or write the constant 0.

6.5.1 Model and Yield Surface

Parameter	Default	Description
material_model	legacy	Shared model flag.
plastic_model	unset	Model for this block only.
main_direction	1	Axis of transverse isotropy.
sigma_xx/yy/zz_tension	none, or coral	Tensile strengths along material axes 1, 2, 3 [MPa]. Required in legacy explicit mode.
sigma_xx/yy/zz_compr.	= tension	Compressive strengths, positive numbers [MPa].
tau_xy/xz/yz_max	none, or coral	Shear strengths [MPa]. Required in legacy explicit mode.
zeta12, zeta13, zeta23	0.0	Interaction parameters, Eq. (2.21). Required for coral . In legacy the zero is an initialisation value, not a recommendation: set them explicitly. Bounds: Eq. (2.23).
yield_density_exponent	0.0	legacy explicit mode: strengths $\times \rho^p$; 0 disables.
use_fabric_scaling	false	legacy only: derive strengths from base values, fabric and density.
sigma_0_tension	preset	Base tensile strength σ_0^+ [MPa].
sigma_0_compression	preset	Base compressive strength σ_0^- ; in legacy defaults to σ_0^+ .
tau_0	preset	Base shear strength (fabric models).
zeta_0	0.22 or preset	Base interaction parameter.
sigma_a_tension, _compression	preset	Axial strengths (TI models).
tau_a	preset	Shear strength of planes containing the axis (TI).
zeta_a	preset	Axial interaction parameter in the UMAT convention; converted with Eq. (2.27).
density_rho	1.0	Bone volume fraction. Must be set for bone models.
fabric_m1/m2/m3	1.0	Fabric eigenvalues.
exponent_p	1.6 or preset	Density exponent of the strengths.
exponent_q	1.0 or preset	Fabric exponent of the strengths.
delta_cortical	1.0	δ of Eq. (2.11).

6.5.2 Post-Yield Law

Parameter	Default	Description
postyieldmode	exp_softening	perfect, exp_hardening, linear_hardening, simple_softening, exp_softening, piecewise_softening.
initial_yield_ratio	1.0	r_0 , in $(0, 1]$. Read by exp_hardening and simple_softening . With $r_0 < 1$ the strengths are ultimate strengths.
residual_strength	0.7	Residual level r_{res} in the softening laws; hardening amount h in exp_hardening (saturation at $1 + h$).
kslope	10.0	Rate k_s , per unit plastic strain.
kmax	0.001	Softening onset (exp_softening , piecewise_softening); peak position (simple_softening).
kmin	0.015	End of the linear drop (piecewise_softening). Must exceed kmax .
kwidth	8.0	Peak width (simple_softening).

Table 2.5 lists which law reads which parameter.

6.5.3 Viscoplastic Regularization

Parameter	Default	Description
viscosity_mode	linear	rate_independent, linear, exponential, logarithmic, polynomial, powerlaw (Table 2.6).
eta	10^{-3}	Viscosity η in sMPa^{-1} (per unit of simulation time). The overstress scales with the square of the strength, Eq. (2.40), so a value is specific to a material.
m	0.001	Shape parameter with a mode-dependent meaning. Not read by linear . In the exponential, logarithmic and polynomial modes the effective viscosity is η/m . In powerlaw it is a reciprocal exponent and must be of order one.

Choosing η . Start with **linear**. Estimate the overstress from Eq. (2.40) and keep it well below one percent of the strength unless rate dependence is intended. Check that halving the time step at a fixed loading rate does not change the result, and that the result approaches the **rate_independent** one as η is reduced.

6.5.4 Local Solver

Parameter	Default	Description
absolute_tolerance	10^{-6}	Tolerance on both local residuals, Eq. (2.47). Use 10^{-8} : with 10^{-6} the returned stress carries a relative error of order 10^{-4} in rotated states, which the global Newton iteration cannot converge through.
max_iterations_newton	10	Iterations of stage 1 before stage 2 is tried. Use 50.
use_primal_cpp	true	Enable stage 2.
max_iterations_primal	1000	Iterations of stage 2.
line_search_beta	10^{-4}	β of the stage-2 line search.
line_search_eta	0.25	Contraction factor η_{ls} , Eq. (2.51).
debug_checks	false	Run the finite-difference and Mandel self-checks at start-up (Sections 2.5.8, 2.6). Cheap; switch off in production because it prints.

Relaxing **absolute_tolerance** is not a remedy for convergence problems. If the local solve fails, MOOSE cuts the time step; frequent failures point to the load step, not to the tolerance.

6.5.5 Damage (Experimental)

Parameter	Default	Description
use_damage	false	Activates Eq. (2.57). Not verified, see Section 2.10.
damage_critical	0.05	κ_{crit} .
damage_rate	0.1	d .

6.6 HomogenizedExponentialCZM

For water interfaces (001 and 100 faces): $\sigma_n^u = 742 \text{ MPa}$, $\tau_s^u = 354 \text{ MPa}$, $\delta_{0,n} = 224 \text{ pm}$, $\delta_{0,t} = 182 \text{ pm}$, $\mu = 1.224$. The water-010 interface is excluded from the averages [3].

Only the strengths, the openings and μ are derived from molecular dynamics. The two quality parameters and τ_d are modelling and regularization choices and should be reported as such. Although η is treated as a regularization parameter as well, using value ranging from 0.15 to 0.45 allows to include up to 5 sequential bond rupture peaks characteristic of protein-mediated softening.

Table 6.5: CZM parameters. Lengths are in the unit of the mesh (Section 3.7).

Parameter	Default	Description
normal_strength	required	Peak normal traction σ_n^u [MPa].
shear_strength_s, _t	required	Peak shear tractions τ_s^u, τ_t^u [MPa].
delta_0_normal	required	Characteristic opening $\delta_{0,n}$.
delta_0_tangent	required	Characteristic sliding $\delta_{0,t}$.
mu	1.0	Loading exponent μ .
eta	1.0	Softening exponent η . The default gives an abrupt post-peak drop; for polycrystals use a small value (Section 3.4).
quality_std_dev	0.10	σ_q , within a quadrature point; in $[0, 1]$.
spatial_quality_std_dev	0.15	σ_{sp} , between quadrature points; 0 for a uniform interface.
spatial_random_seed	1234	Seed of the spatial field.
damage_viscosity	0.0	τ_d in simulation time units; 0 is rate independent. Keep $\Delta t \leq \tau_d$.
delta_c	computed	Opening of complete failure.
failure_traction_ratio	0.01	Defines the computed δ_c ; without effect for η below about 0.24.

Table 6.6: Parameter set for the protein interface in a mesh in micrometres.

Parameter	Value	Origin
normal_strength	626	MD [3]
shear_strength_s, _t	374	MD
delta_0_normal	1.91e-4 (191 pm)	MD
delta_0_tangent	2.17e-4 (217 pm)	MD
mu	0.92	MD fit of the loading branch (0.916)
eta	0.27	regularization, not fitted
quality_std_dev	0.10	assumed
spatial_quality_std_dev	0.15	assumed; chosen to desynchronise interface failure
damage_viscosity	problem dependent	regularization

6.7 Input Templates

6.7.1 Bone Model from a Preset

```
[GlobalParams]
displacements = 'disp_x disp_y disp_z'
material_model = trabecular_fabric_ortho
density_rho = 0.2 # BV/TV
fabric_m1 = 0.85 # sum of the three = 3
fabric_m2 = 0.95
fabric_m3 = 1.20
[]

[Physics/SolidMechanics/QuasiStatic]
[all]
strain = FINITE
incremental = true
add_variables = true
use_finite_deform_jacobian = true # required, Section 2.9.3
[]
[]

[Materials]
```

```

[elasticity]
  type = ComputeFabricElasticityTensor
[]
[stress]
  type = ComputeMultipleInelasticStress
  inelastic_models = 'plasticity'
[]
[plasticity]
  type = OrthotropicPlasticityStressUpdate
  euler_angle_1 = 0
  euler_angle_2 = 0
  euler_angle_3 = 0
  absolute_tolerance = 1e-8
  max_iterations_newton = 50
[]
[]

```

Nothing else is needed: constants, post-yield law and viscosity come from the preset. To change one constant, add it to the block concerned, for example `sigma_0_tension = 70` in `[plasticity]`. For the TI and fabric-TI models add `main.direction` to `[GlobalParams]`.

6.7.2 Coral Grains with Cohesive Interfaces

```

[GlobalParams]
  displacements = 'disp_x disp_y disp_z'
  material_model = coral
[]

[Physics/SolidMechanics]
  [QuasiStatic]
    [all]
      strain = FINITE
      incremental = true
      add_variables = true
      use_finite_deform_jacobian = true
    []
  []
  [CohesiveZone] # sibling of QuasiStatic, not nested in it
    [czm]
      boundary = 'grain_1_grain_2'
      strain = FINITE
    []
  []
[]

[Materials]
  [elasticity]
    type = ComputeFabricElasticityTensor # C_ijkl defaults to aragonite
    coupled_euler_angle_1 = euler_phi1
    coupled_euler_angle_2 = euler_Phi
    coupled_euler_angle_3 = euler_phi2
  []
  [stress]
    type = ComputeMultipleInelasticStress
    inelastic_models = 'plasticity'
  []

```

```

[plasticity]
  type = OrthotropicPlasticityStressUpdate # strengths default to
    aragonite
  zeta12 = 0.27      # no default; provisional value carried over from
    bone
  zeta13 = 0.27
  zeta23 = 0.27
  euler_angle_1 = euler_phi1
  euler_angle_2 = euler_Phi
  euler_angle_3 = euler_phi2
  absolute_tolerance = 1e-8
  max_iterations_newton = 50
[]
[czm_interface]
  type = HomogenizedExponentialCZM
  boundary = 'grain_1_grain_2'
  normal_strength = 626
  shear_strength_s = 374
  shear_strength_t = 374
  delta_0_normal = 1.91e-4      # mesh units
  delta_0_tangent = 2.17e-4
  mu = 0.92
  eta = 0.27
  quality_std_dev = 0.10
  spatial_quality_std_dev = 0.15
[]
[]

```

The Euler-angle variables are constant monomial auxiliary variables, set by initial conditions per block or read from the mesh file (Section 7.13.3). The interface must be created by **BreakMeshByBlockGenerator** inside the input file. The three coral examples in **scripts/examples/coral** are complete, runnable versions of this template.

6.7.3 Explicit Constants (Legacy)

```

[Materials]
[elasticity]
  type = ComputeElasticityTensorCoupled
  fill_method = symmetric9
  C_ijkl = '171800 57500 30200 106700 46900 84200 42100 31100 46600'
  coupled_euler_angle_1 = euler_phi1
  coupled_euler_angle_2 = euler_Phi
  coupled_euler_angle_3 = euler_phi2
[]
[stress]
  type = ComputeMultipleInelasticStress
  inelastic_models = 'plasticity'
[]
[plasticity]
  type = OrthotropicPlasticityStressUpdate
  sigma_xx_tension = 4980
  sigma_yy_tension = 4100
  sigma_zz_tension = 5340
  tau_xy_max = 4510
  tau_xz_max = 5080
  tau_yz_max = 5060
  # compression strengths default to tension

```

```

    zeta12 = <set>      # initialised to 0 if omitted; set all three
                        explicitly
    zeta13 = <set>
    zeta23 = <set>
    postyield_mode      = exp_softening
    residual_strength = 0.7
    kmax      = 0.001
    kslope    = 30
    viscosity_mode = linear
    eta       = 0.002
    euler_angle_1 = euler_phi1
    euler_angle_2 = euler_Phi
    euler_angle_3 = euler_phi2
    absolute_tolerance = 1e-8
    max_iterations_newton = 50
[]
[]

```

In legacy mode `zeta12`, `zeta13` and `zeta23` are initialised to zero, which decouples the normal stresses. Zero is only the initial value of the parameters. A production input should set all three explicitly, as the `coral` model enforces.

6.8 Solver Settings

The following settings were determined by measurement on a trabecular bone specimen (483 045 HEX8 elements, 2.4 million degrees of freedom, 32 processes) and are the recommended starting point for large problems.

```

[Executioner]
    type = Transient
    solve_type = NEWTON
    line_search = bt
    nl_rel_tol = 1e-8
    nl_abs_tol = 1e-6
    nl_max_its = 40
    petsc_options_iname = '-pc_type -pc_gamg_agg_nsmooths -ksp_type -
                          ksp_gmres_restart -ksp_max_it -ksp_rtol'
    petsc_options_value = 'gamg 2 gmres 300 200 1e-4'
    dt = 0.1
[]

```

Single-element and few-element problems can use `-pc_type lu` in place of the multigrid options. A healthy run shows no failed steps, no fallback of the tangent, no switch to the second return-mapping stage, and residuals that fall by more than an order of magnitude per iteration at the end of each step. In a step in which plasticity spreads, the first iterations converge slowly while the set of yielding points is still changing; this is expected and is not a defect of the tangent.

Table 6.7: Reasons for the solver settings.

Setting	Reason
<code>line_search = bt</code>	Required. Without a line search the Newton step across the elastic-plastic transition inverted elements and the run did not complete the first step, even after the time step had been cut 128 times. The 12 line search accepted steps that increased the residual.
<code>nl_rel_tol = 1e-8</code>	This is the criterion that ends the solve. With 10^{-5} the iteration stops one step before the quadratic phase and the tangent appears to converge only linearly.
<code>nl_abs_tol = 1e-6</code>	Backstop only.
<code>-ksp_rtol 1e-4</code>	Tolerances of 10^{-10} , 10^{-6} and 10^{-4} gave the same 23 Newton iterations and the same result in 4073, 1673 and 920 s. With 10^{-3} the final contraction degraded fivefold. About 92% of a step is the linear solve.
<code>dt = 0.1</code>	Quartering the step gave cleaner quadratic convergence per step but 2.4 times as many iterations in total for the same strain.
<code>use_finite_deform_jacobian</code>	Required with <code>strain = FINITE</code> (Section 2.9.3).
<code>absolute_tolerance = 1e-8</code>	Local tolerance of the plasticity block, see above.

Chapter 7

Verification Workflow and Examples

This chapter documents how the implementation is verified: the automated tests that accompany the code, the earlier single-element, multi-grain and cohesive-zone studies, and annotated example input files.

7.1 Scope and Status of This Chapter

Two words are kept apart in this chapter. *Verification* asks whether the code solves the equations of Chapters 2 and 3 correctly: derivatives against finite differences, single-element responses against closed-form predictions from the input parameters, the consistent tangent against a reference solution, and the convergence rate of the global Newton iteration. *Validation* asks whether the equations describe the material, which requires experimental data. This document contains verification only. No result in this chapter is compared with an experiment or with the output of the original UMAT binary; the bone presets are checked against the UMAT *formulas*, not against a UMAT run. Experimental validation is reported separately.

The chapter has three layers:

1. **Current automated verification** (Sections 7.2–7.4). These were run with the present code and are the tests to repeat after any change to the material classes.
2. **Earlier studies** (Section 7.7 onwards). Single-element, post-yield, sensitivity, multi-grain and cohesive-zone studies from the first version of this document. They are kept because the behaviour they illustrate is unchanged in kind. Their figures were produced before the flow-rule and tangent corrections and before the present cohesive parameter set, so the numbers in them are not reference values for the current code. Each affected section says so.
3. **Example input files** (Section 7.25).

The current tests live in two directories of the repository, with different purposes:

- **test/verification**: the driver `run_tests.sh` with its inputs and `analyse_run.py`. Developer tests of the tangent and the solver; the bone-cube targets need a parallel machine.
- **scripts/examples**: `run_examples.sh`, `analyse_examples.py`, the generated inputs, and the offline check `verify_presets.py`. User-facing: every model flag as a small runnable input with a predicted result, finished in under a minute on one core.

`run_validation.sh` (Section 5.2.3) is in `scripts`.

Table 7.1 lists what is covered by the current tests and what is not.

Table 7.1: Coverage of the current automated verification

Property	Tested by	Level
Yield gradient, Hessian, $dr/d\kappa$, Mandel mapping	<code>run_tests.sh fd, M</code>	in-code finite differences, four models
Preset constants and derived yield tensors, all model flags	<code>verify_presets.py</code>	offline, against UMAT formulas
Elastic modulus, yield onset, peak stress for every bone model	<code>run_examples.sh bone</code>	single element, 54 inputs
Consistent tangent does not change the solution	<code>run_tests.sh A B Bs Bf C</code>	single element and two grains, against elastic-tangent reference
Quadratic global convergence	<code>Bf, D1, D, Ddt</code>	single element and bone cube (2.4 million DOFs)
Viscosity: all modes, rate dependence, step-size independence	<code>V, Vr</code>	single element
Primal CPPA stage (7×7 system)	<code>P</code>	single element, forced, against the Newton stage
Legacy behaviour unchanged by the model flags	<code>A, M4</code>	against stored baseline
Cohesive peak traction, mode mixity, damage	<code>run_examples.sh coral</code>	two elements, one interface
<i>Not covered by a current test</i>		
Bone preset in a multi-element run		the bone cube uses explicit parameters
Isotropic damage coupling (Section 2.10)		not verified
Non-proportional interface loading (CZM Jacobian)		not verified
Polycrystal with interfaces at current parameters		earlier study only

7.2 Tangent and Solver Verification: `run_tests.sh`

The driver `test/verification/run_tests.sh` plays the same role for the constitutive update as `run_validation.sh` (Section 5.2.3) plays for the installation, with one difference in method. `run_validation.sh` compares each output with a stored gold file and so detects *change*. `run_tests.sh` compares each run with an independent reference computed by the same binary and so detects *error*.

The principle is that a tangent operator affects only the path of the global Newton iteration, never the converged solution. Every matrix test is therefore run twice:

- `<ID>_ref`: `tangent_operator = elastic` on `ComputeMultipleInelasticStress`. MOOSE ignores the tangent returned by the material. Convergence is slow but the answer does not depend on $\mathbb{C}^p$.
- `<ID>_full`: the code default, the consistent tangent of Eq. (2.55).

`analyse_run.py` then reports solver health for both logs (attempts, failed steps, Newton iterations per step, observed convergence order, counts of tangent fallbacks and of Newton-to-Primal switches) and compares the two CSV files column by column. A test passes when every column agrees within the relative tolerance `TOL` (default 10^{-5}). A wrong tangent shows up as more iterations; a wrong stress update shows up as a CSV difference.

7.2.1 Running the Tests

```

cd test/verification

# In-code derivative checks, one step
./run_tests.sh fd

# Single-element matrix: regression, rotated, two grains
./run_tests.sh A B C

# Variants of B: tight local tolerance, SMALL strain,
# FINITE strain with finite-deformation Jacobian
./run_tests.sh Bt Bs Bf

# Viscosity modes; rate dependence of one mode
./run_tests.sh V
MODE=exponential ./run_tests.sh Vr

# Primal CPPA stage, forced
./run_tests.sh P

# Four material-model presets
./run_tests.sh M

# Bone cube, production settings (32 MPI ranks by default)
NOREF=1 ./run_tests.sh D          # three steps, no elastic reference
./run_tests.sh D1                # one step
./run_tests.sh Ddt Dtol Dls      # step size, tolerance, no line search
KSPRTOL=1e-3 ./run_tests.sh Dksp

```

Several identifiers may be given in one call. The environment variables in Table 7.2 replace the defaults at the top of the script. Results are written to `results/<ID>_{ref,full}.{log,csv}` and `results/<ID>.summary`.

Table 7.2: Environment variables of `run_tests.sh`

Variable	Default	Meaning
APP	<code>~/projects/aragonite/aragonite-opt</code>	executable
MPI_SMALL	empty (serial)	launcher for single-element runs
MPI_BONE	<code>mpiexec -n 32</code>	launcher for the bone cube
TOL	<code>1e-5</code>	relative tolerance per CSV column
NOREF	<code>0</code>	<code>1</code> skips the elastic-tangent reference
FDJ	<code>0</code>	<code>1</code> adds <code>use_finite_deform_jacobian = true</code> to every run
MODE, M	linear, 0.001	viscosity mode and exponent for Vr
KSPRTOL	unset	linear tolerance for Dksp
RHO_M	<code>0.2</code>	density for the bone presets in M

The inputs are `linear_hardening.i` (one HEX8 element, legacy aragonite parameters, linear hardening with $k_s = 30$, linear viscosity with $\eta = 0.002$ s/MPa, uniaxial strain rate 10^{-3} s $^{-1}$ to a strain of 0.12 in 1200 steps), `twin.i` (two grains with different orientations, softening and viscosity), `material_flags_single_element.i` (one element with a `ComputeFabricElasticityTensor` block, so that the model flags can be switched from the command line) and `cube_tension.22.i` (bone cube, Section 7.2.3). A command-line override can set parameters of a block but cannot change its type, which is why the preset tests need their own input.

7.2.2 Expected Summary

A passing matrix test ends as follows (test A, shortened):

```
--- results/A_full.log
  attempts 1201   converged 1201   failed 0   time reached 120
  Newton its/step total 2401   histogram 1:1 2:1200
  C_ep fallbacks 0   Newton->Primal 0
--- results/A_ref.log [reference]
  Newton its/step total 5761   histogram 1:1 2:360 6:840
  convergence order p median 1.00 (linear)
--- Newton total: test 2401 vs reference 5761 (2.4x fewer)
--- CSV ref=results/A_ref.csv test=results/A_full.csv
ok  plastic_strain    max rel 1.26e-12 max abs 4.76e-14
ok  strain_xx         max rel 8.83e-14 max abs 1.00e-14
ok  stress_xx         max rel 1.64e-12 max abs 1.74e-08
ok  stress_yy         max rel 2.40e+00 max abs 4.11e-08 (abs floor)
VERDICT: PASS
```

The three lines to read are the verdict, the fallback counters (both must be zero in every test except P, which forces the switch) and the iteration total relative to the reference.

Table 7.3 collects the results obtained with the present code.

Table 7.3: Results of `run_tests.sh`. Iterations are global Newton iterations summed over the run; “ref” is the elastic-tangent reference.

ID	Configuration	full	ref	Outcome
A	axis-aligned, 1200 steps	2401	5761	pass; equals stored baseline
B	rotated (30°, 45°, 60°), rate-independent	6879	9655	pass; linear rate (no finite-deformation Jacobian)
Bs	B with <code>strain = SMALL</code>	1876	7932	pass
Bf	B with finite-deformation Jacobian	2403	8337	pass; 2 iterations per step
Bt	B with local tolerance 10^{-12}			differs from B by 3×10^{-4} : effect of the local tolerance
C	two grains, softening, viscosity	7085	11801	pass up to $t = 90$
V	five viscosity modes		2 per step	all modes converge
P	Primal stage forced, axis-aligned; “ref” is the Newton-stage run (rotated: 2402 each)	2401	2401	pass; agreement 10^{-13}
D1	bone cube, one step	23		order 2.29, no fallback
D	bone cube, three steps	70		no fallback
Dtol	D with <code>nl_abs_tol = 1e-8</code>	73		same solution
Ddt	bone cube, four steps of $\Delta t/4$	50		order 2.34

7.2.3 Individual Tests

fd: derivative checks

Purpose Confirm that the analytical derivatives used by the return mapping and the tangent are the derivatives of the implemented yield function, including the factors $1/w_p$ and $1/(w_p w_q)$ of the six-component storage (Section 2.5.8).

Run One step of `linear_hardening.i` with `debug_checks = true`. The constructor evaluates central finite differences at three stress states with non-zero shear; the first stress update checks the Mandel mapping.

Check Relative difference between analytical and finite-difference yield gradient, Hessian and $dr/d\kappa$; continuity of $r(\kappa)$ across the branch points of the post-yield law; tensor $\rightarrow$ Mandel $\rightarrow$ tensor round trip.

Result Table 7.4. The round trip is exact to 2.2×10^{-16} . The values are at the truncation level of the difference quotient.

Table 7.4: In-code finite-difference checks (relative differences), from tests M1–M3

Model	Gradient	Hessian	$dr/d\kappa$	r continuity
trabecular_fabric_ortho	1.2×10^{-10}	1.9×10^{-10}	4.7×10^{-10}	1.8×10^{-7}
compact_ti, main_direction = 3	2.2×10^{-10}	1.7×10^{-10}	1.0×10^{-9}	2.0×10^{-7}
coral	2.7×10^{-10}	1.7×10^{-10}	5.2×10^{-9}	2.0×10^{-7}

A: axis-aligned regression

Purpose Sentinel. With the material axes on the global axes and uniaxial loading there is no shear in the material frame, so the shear-factor corrections of Chapter 2 cannot act. The result must equal the result of the code before those corrections.

Run `linear_hardening.i` unchanged, full and reference.

Check Full against reference, and full against the stored `baseline_lh.csv`.

Result Pass. Full and reference agree to about 10^{-12} ; the baseline is reproduced. 2401 iterations (two per step) against 5761.

B, Bs, Bf, Bt: rotated crystal

Purpose The key tangent test. Euler angles ($30^\circ, 45^\circ, 60^\circ$) put all six stress components into the material frame, so every shear entry of the gradient, the Hessian and $\mathbb{C}^{\text{ep}}$ is exercised. Viscosity is switched off so that the rate-independent tangent is tested alone.

Run B: `strain = FINITE` as in the input. Bs: `strain = SMALL`. Bf: `FINITE` with `use_finite_deform_jacobian = true`. Bt: full tangent only, local tolerance 10^{-12} .

Check Full against reference; iteration counts and convergence order.

Result All three pairs pass. The iteration counts separate the two sources of a slow global iteration. In B the material tangent is correct but MOOSE’s default Jacobian omits the geometric terms of the finite-strain formulation, and convergence is linear (6879 iterations). Removing the finite-strain terms (Bs, 1876) or supplying the finite-deformation Jacobian (Bf, 2403, two per step) restores the rate expected of a consistent tangent. This is the origin of requirement 1 in Section 2.9.3. Bt differs from B by up to 3×10^{-4} relative; this is the error left by the local tolerance 10^{-6} of the input and the reason for the recommended `absolute_tolerance = 1e-8`.

C: two grains with softening and viscosity

Purpose Tangent with all terms active: rotation, material-frame shear, a softening law and the viscous contribution.

Run `twin.i`, full and reference.

Check All columns up to $t = 90$. Later times are printed for information only.

Result Pass up to $t = 90$; 7085 iterations against 11801. With softening, which of the two grains localises, and when, is a bifurcation whose seed is round-off and depends on the Newton path. After the bifurcation two correct solvers may follow different branches, so a difference there is not evidence of a tangent error.

V, Vr: viscosity

Purpose V: every viscosity mode of Table 2.6 has a consistent derivative. Vr: the viscous term is a function of the rate $\dot{\kappa}$, not of the step size.

Run V: test A once per mode ($m = 0.001$; $m = 0.2$ for `powerlaw`, where $1/m = 1000$ underflows). Vr: the same final strain 0.12 at strain rates 10^{-4} , 10^{-3} and 10^{-2} s^{-1} , the middle rate also with $\Delta t/2$, and one rate-independent run.

Check V: iteration count. Vr: (1) the two runs at the same rate and different Δt agree; (2) the overstress relative to the rate-independent run grows monotonically with the rate.

Result V: two iterations per step in all five modes. Vr: Table 7.5. For the linear mode the overstress is proportional to the rate, as Eq. (2.40) states, and halving Δt changes the final stress in the eighth digit (10627.0608 against 10627.0607 MPa).

Table 7.5: Test Vr: overstress at strain 0.12 relative to the rate-independent stress 10616.66 MPa, $\eta = 0.002 \text{ s/MPa}$

Mode	rate $\times 0.1$	rate $\times 1$	rate $\times 10$
linear	+0.010 %	+0.098 %	+0.981 %
exponential	+0.049 %	+0.491 %	+5.045 %
logarithmic	+0.049 %	+0.489 %	+4.829 %
polynomial	+0.048 %	+0.446 %	+2.971 %
powerlaw	≈ 0	≈ 0	+0.042 %

P: Primal CPPA stage

Purpose The second stage of the return mapping (the 7×7 system with line search, Section 2.8) is not entered in any other test, because the first stage converges. This test forces it.

Run Two runs with identical settings (`absolute_tolerance = 1e-8`, finite-deformation Jacobian), axis-aligned (`P_axis`) and rotated (`P_rot`): `max_iterations_newton = 100` as reference and `max_iterations_newton = 1`, which starves the first stage. The only difference between the two is the local solver that does the work.

Check The reference log contains no “switching to Primal” message and the forced log contains at least one; the CSV files agree to 10^{-6} .

Result Pass, both configurations. The reference runs never leave stage 1 (no switch); the forced runs switch at every plastic integration-point update (47,040 switches in `P_axis`, 36,377 in `P_rot`) with no tangent fallback. The global iteration is unaffected: 2401 Newton iterations in both runs of `P_axis` and 2402 in both runs of `P_rot`. Strains, plastic strain and axial stress agree to 10^{-13} relative or better; the lateral stresses, which are zero in the axis-aligned case, agree to 10^{-10} MPa absolute. The two local solvers therefore return the same stress and the same consistent tangent to round-off.

M: material-model presets

Purpose Run one preset per distinct code path of Table 6.1 through the solver, and show that the introduction of the flags did not move the legacy behaviour.

Run M1 `trabecular_fabric_ortho` ($\rho = 0.2$, fabric 0.85/0.95/1.20): fabric scaling, `simple_softening`, initial yield ratio 0.7. M2 `compact_ti` with `main_direction = 3`: axial strengths, the conversion of Eq. (2.27), `exp_hardening`. M3 `coral`: nine elastic and nine strength constants given explicitly. M4 `legacy`: `linear_hardening.i` with no override.

Check M1–M3: the run completes and the finite-difference checks of Table 7.4 pass for the preset. M4: CSV against `baseline_lh.csv`.

Result M1–M3 complete. M4 reproduces the baseline: axial stress to 2×10^{-12} relative, strains to 4×10^{-11} , lateral stresses zero to 3×10^{-8} MPa. In M1 and M2 three warnings that `zeta12`, `zeta13`, `zeta23` are ignored are expected: the input carries the coral values and a bone model does not use them.

The correctness of the preset *values* is not established here but by `verify_presets.py` (Section 7.4) and by the examples (Section 7.3).

D, D1, Ddt, Dtol, Dls, Dksp: bone cube

Purpose Show that the consistent tangent gives quadratic convergence of the global iteration on a large unstructured problem, and measure the solver settings of Section 6.8. This is a convergence test. It was never meant as a production run of the specimen, and its result is not compared with an experiment.

Run `cube_tension_22.i`: a trabecular bone cube with 483,045 HEX8 elements and 2,405,145 degrees of freedom on 32 MPI ranks, `ComputeFabricElasticityTensor` with explicit quadric parameters, linear hardening, rate-independent, `strain = FINITE`, uniaxial tension (the y^- face is fixed and a displacement is prescribed on the y^+ face). D1: one step. D: three steps of $\Delta t = 0.1$. Ddt: the strain of step 1 in four steps. Dtol: `nl_abs_tol = 1e-8`. Dls: `line_search = none`. Dksp: sweep of the linear tolerance.

Check Residual history, observed order, iteration counts, tangent fallbacks, Newton-to-Primal switches.

Result D1 converges in 23 iterations. The last residuals are

$$7.998 \times 10^{-1}, \quad 2.647 \times 10^{-2}, \quad 1.450 \times 10^{-3}, \quad 7.209 \times 10^{-5}, \quad 7.468 \times 10^{-8},$$

with observed order 2.29 and a final contraction of 1.04×10^{-3} . No tangent fallback and no switch to the Primal stage occurred at any integration point. The earlier iterations of the step are damped by the line search while the plastic zone is being found; the quadratic phase is the tail shown. D needs 70 iterations for three steps and Dtol 73 with the same solution. Ddt needs 50 iterations for four steps (order 2.34): a smaller step converges faster per step but costs more in total.

Three further findings from this problem are reflected in Table 6.7:

- **Line search.** Without it (Dls) the first step does not complete: eight exceptions for inverted elements, three for a negative argument in the Rashid strain update, and the step size is cut to 7.8×10^{-4} . The full Newton step overshoots while the plastic zone forms. `line_search = bt` is required.

- **Linear tolerance.** Table 7.6. The Newton iteration is the same for relative tolerances from 10^{-10} to 10^{-4} ; only the wall time changes. At 10^{-3} the final contraction is five times worse and the step no longer meets `nl_rel_tol`.
- **Elastic-tangent reference.** With the former tangent and the former, looser tolerances the same three steps took 422 iterations with three `DIVERGED_MAX_IT`. At the present tolerances the elastic-tangent reference does not converge in a practical number of iterations, so `D` is run with `NOREF=1` and its solution is not cross-checked against a reference. The cross-check of the tangent rests on the single-element and two-grain tests above.

Table 7.6: Test `Dksp`: one step of the bone cube for different relative tolerances of the linear solver

<code>ksp_rtol</code>	Newton its	Wall time	Last contraction	Final $\ R\ $	Linear failures
10^{-10}	23	4073 s	1.04×10^{-3}	7.468×10^{-8}	17
10^{-6}	23	1673 s	1.04×10^{-3}	7.468×10^{-8}	0
10^{-4} (used)	23	920 s	1.03×10^{-3}	7.461×10^{-8}	0
10^{-3}	23	713 s	4.82×10^{-3}	4.282×10^{-7}	0

7.3 Example Suite: `run_examples.sh`

The directory `scripts/examples` holds one small input per model and loading case. Unlike the tests of Section 7.2, each run is compared with a number predicted from the parameters alone, so the suite verifies the chain from model flag to stress response.

```
cd scripts/examples

./run_examples.sh           # all 57 inputs, then the analysis
./run_examples.sh bone     # the 54 bone inputs
./run_examples.sh coral    # the 3 cohesive inputs
./run_examples.sh compact_ti # substring filter on the path
./run_examples.sh shear    # every shear case

GOLD=1 ./run_examples.sh   # also copy each CSV to gold/
NOANALYSE=1 ./run_examples.sh # run only
./analyse_examples.py      # analysis only
```

Each run writes `<case>.out.csv` and `<case>.log` next to its input. The runs are sequential and take a few seconds each.

Expected summary:

```
=====
running 57 example(s)  APP=~/projects/aragonite/aragonite-opt
=====
[ 1/57] bone/compact_iso/compression_xx.i    ok (<time>, <n> rows)
...
[57/57] coral/coral_czm_mode_II_shear.i     ok (<time>, <n> rows)
...
161 passed, 0 failed
VERDICT: PASS
```

The reference run completed all 57 inputs in 37 s.

Table 7.7: Inputs of the example suite

Directory	Inputs	Cases
bone/<model>/, six models trabecular_iso, trabecular_ti, trabecular_fabric_ti, trabecular_fabric_ortho, compact_iso, compact_ti coral/	6×9 3	tension_xx/yy/zz, compression_xx/yy/zz, shear_xy/xz/yz; one HEX8 element, model selected by material_model alone coral_czm_mode_I.opening, coral_czm_mode_II.shear, coral_czm_mixed_mode; two elements joined by one HomogenizedExponentialCZM interface, lengths in mi- crometres

7.3.1 Contents

The inputs are generated by `generate_examples.py`; the predictions are computed in `expected_peaks.py` from the constants of `MaterialModelPresets.h` and the formulas of Chapter 2, independently of the C++ code.

7.3.2 Checks and Results

`analyse_examples.py` applies the checks of Table 7.8. The last column is the largest deviation over all inputs in the reference run, to be read against the tolerance.

Table 7.8: Checks of `analyse_examples.py` and worst case of the reference run (161 checks, all passed)

Check	Tolerance	Worst case
<i>Bone, tension and compression</i>		
Elastic modulus of the loading direction	2×10^{-2}	2.1×10^{-7}
Yield consistency $ \phi/r - 1 $ on every plastic step	2×10^{-3}	4.5×10^{-4}
Peak stress against preset strength times peak of $r(\kappa)$	3×10^{-2}	4.5×10^{-4}
<i>Coral interface</i>		
Peak normal traction, mode I	3×10^{-2}	2.7×10^{-2}
Peak $ T $ at the observed mode mix	3×10^{-2}	3.8×10^{-3}
Tangential-to-normal jump ratio against pre-scribed ratio	5×10^{-2}	3.0×10^{-2}
Final damage against the damage law at the final δ_{eff}	0.2	7.4×10^{-2}

Here $\phi = \sqrt{\boldsymbol{\sigma} : \mathbb{F} : \boldsymbol{\sigma}} + \mathbf{F} : \boldsymbol{\sigma}$ is evaluated by the analyser from the stress output, so the yield-consistency check confirms that the returned stress lies on the surface $\phi = r(\kappa)$ of the selected model. The residual 4.5×10^{-4} is the local tolerance of the inputs.

Shear cases. The bone shear inputs prescribe all displacement components on the boundary (simple shear with fixed volume). Because the linear term $\mathbf{F}$ of the quadric gives the flow direction a volumetric part (Section 2.7), plastic flow under this constraint builds up compressive normal stresses and the shear stress rises above the pure-shear strength. The analyser therefore reports the shear peaks as information and does not test them against the pure-shear value: for `compact_iso` the peak is 86.2 MPa against a pure-shear prediction of 69.8 MPa, with normal reactions near -40 MPa; for the trabecular models the peak is 7–10 % above the prediction with

reactions near -1 MPa. The shear cases still pass the yield-consistency check, which holds for any stress state.

Coral interface. Peak tractions are 628.4 MPa in mode I (input 626), 375.4 MPa in mode II (input 374) and 522.6 MPa in the mixed case, against 520.8 MPa from Eq. (3.9) at the mode mix actually reached at the interface (515.6 MPa at the nominal mix of the boundary displacement). The loading ends at $\delta_{\text{eff}}/\delta_0 \approx 3$ with damage 0.63–0.65, where the rate-independent damage law gives 0.68–0.70; the difference is the lag of the viscous damage update. The cohesive parameters are the provisional set of Table 3.4.

7.3.3 Self-test of the Analyser

`selftest_analyser.py` feeds `analyse_examples.py` synthetic CSV files built from the same formulas and checks that it recovers the inputs (deviation 4.4×10^{-16}). It guards against an analyser that passes because it is wrong.

7.4 Offline Preset Check: `verify_presets.py`

`verify_presets.py` needs no MOOSE executable and runs in seconds:

```
cd scripts/examples
python3 verify_presets.py
```

It re-implements in Python the parameter derivation of the bone UMAT [5] and the derivation in `MaterialModelPresets.h` and compares the two.

Purpose Establish that every model flag produces the elastic constants, the yield tensors $\mathbb{F}$, $\mathbf{F}$ and the post-yield law that the UMAT formulas give for the same density and fabric.

Check (1) Elastic and yield quantities of the six bone models. (2) All combinations of `elastic_model` and `plastic_model`: the 49 pairings of the seven presets plus the legacy row and column, 64 in total. (3) Convexity of the quadric by leading principal minors against an eigenvalue test for 20,000 random triples $(\zeta_{12}, \zeta_{13}, \zeta_{23})$. (4) The post-yield laws of Section 2.6 at sampled κ .

Result Yield quantities agree to 1.4×10^{-14} and elastic quantities to 1.3×10^{-15} relative; 64 combinations, no failure; 20,000 triples, no disagreement between the two convexity tests; post-yield laws exact.

The division of work with test M is deliberate: the offline script checks all combinations against the formulas, and M runs four of them through the solver. Running all combinations through MOOSE would add nothing that the offline check does not give.

7.5 When to Run Which Test

`run_tests.sh` is run from `test/verification`; `run_examples.sh` and `verify_presets.py` from `scripts/examples`; `run_validation.sh` from `scripts`.

The gold files of `run_validation.sh` in the repository were generated with the present code. After a later change to the constitutive update, regenerate them (`--generate-gold`) only once the tests of this section pass; a gold comparison alone cannot tell a corrected result from a broken one.

Table 7.9: Recommended tests after a change

Change	Run
Fresh build or new machine	<code>run_validation.sh; run_examples.sh</code>
Preset constants or model flags	<code>verify_presets.py; run_tests.sh M;</code> <code>run_examples.sh bone</code>
Yield function, gradient, Hessian	<code>run_tests.sh fd A B Bf; run_examples.sh bone</code>
Return mapping or consistent tangent	<code>run_tests.sh A B Bs Bf C P; then NOREF=1</code> <code>run_tests.sh D</code>
Post-yield law	<code>run_tests.sh fd C; verify_presets.py</code>
Viscosity	<code>run_tests.sh V; Vr</code> for each mode touched
Cohesive model	<code>run_examples.sh coral; run_validation.sh -c</code> <code>czm</code>
Solver settings for a new mesh	<code>run_tests.sh D1 Dls Dksp</code> with that mesh as BONE

7.6 Validation Framework

The earlier studies in the remainder of this chapter follow a common scheme. For each test we specify:

1. **Objective:** What is being validated
2. **Quantity compared:** Specific output variable
3. **Acceptance criterion:** Quantitative threshold for success
4. **Failure diagnosis:** What to check if criterion not met

7.6.1 Validation Categories

Category A: Analytical Verification Compare numerical results to closed-form solutions:

Table 7.10: Analytical verification criteria

Test	Quantity	Criterion	Reference
Single element, elastic	Stress-strain slope	$ E_{\text{num}} - E_{\text{theory}} /E_{\text{theory}} < 10^{-6}$	Elastic tensor
Single element, pure modes	Yield stress	$ \sigma_y - \sigma_y^{\text{input}} /\sigma_y < 10^{-3}$	Input parameters
Rigid body rotation	Stress transformation	$\ \sigma_{\text{rot}} - \mathbf{R}\sigma\mathbf{R}^T\ /\ \sigma\ < 10^{-6}$	Objectivity

Category B: Convergence Studies Verify mesh independence and parameter convergence:

Table 7.11: Convergence study criteria

Test	Quantity	Criterion	Reference
Mesh refinement	Peak stress	$ \sigma_{\text{fine}} - \sigma_{\text{coarse}} /\sigma_{\text{fine}} < 0.05$	Richardson extrapolation
Timestep refinement	Final displacement	$ u_{\Delta t/2} - u_{\Delta t} /u_{\Delta t/2} < 0.01$	Temporal convergence
Parameter sensitivity	Response change	$ \partial R/\partial p \cdot \Delta p/R < 0.10$	Sensitivity analysis

Category C: Experimental Validation Comparison with measurements is outside the scope of this document (Section 7.1). No criterion of this category is applied in this chapter.

Category D: Qualitative Validation Physical reasonableness checks:

Table 7.12: Qualitative validation checks

Check	Expected behaviour
Stress-strain monotonicity	No unphysical oscillations or reversals
Damage progression	$0 \leq D \leq 1$, monotonic increase
Energy balance	Total energy = elastic + plastic + fracture energies
Failure pattern	Crack propagation along weakest path

7.6.2 Error Metrics

Standard error metrics used throughout:

1. **Relative error:**

$$\epsilon_{\text{rel}} = \frac{|Q_{\text{sim}} - Q_{\text{ref}}|}{|Q_{\text{ref}}|} \quad (7.1)$$

2. **Root mean square error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (Q_{\text{sim},i} - Q_{\text{ref},i})^2} \quad (7.2)$$

3. **Peak error:**

$$\epsilon_{\text{peak}} = \max_i |Q_{\text{sim},i} - Q_{\text{ref},i}| \quad (7.3)$$

4. **Convergence rate:**

$$\alpha = \log_2 \left(\frac{\epsilon_{\text{coarse}}}{\epsilon_{\text{fine}}} \right) \quad (7.4)$$

Expected: $\alpha \approx 2$ for 2nd-order elements

7.6.3 Validation Hierarchy

Tests organized from simple to complex:

Level 1: Unit Tests

- └ Single element, pure loading modes
- └ Elastic, plastic, softening

Level 2: Coupling Tests

- └ Two-grain bicrystal
- └ Bulk-interface interaction

Level 3: Microstructure Tests

- └ 4-10 grain RVEs
- └ Statistical behaviour

Level 4: Macroscopic Validation

- └ Large RVEs (50+ grains)
- └ Effective properties, failure modes
- └ Comparison to experiments

Each level must pass before proceeding to next.

7.6.4 Failure Diagnosis Procedure

If validation criterion not met:

1. Check input parameters:

- Units consistent? (MPa and s; one length unit throughout, micrometres when cohesive interfaces are present, Chapter 3)
- Euler angles in degrees or radians?
- Material frame aligned correctly?

2. Check numerical settings:

- Tolerances too loose? (nl_abs_tol, nl_rel_tol)
- Timestep too large? (causing overshoot)
- Mesh too coarse? (insufficient resolution)

3. Check physical assumptions:

- Small strain valid? ($\varepsilon < 0.05$?)
- Objectivity preserved? (rotation test)
- Boundary conditions appropriate?

4. Debug systematically:

- Plot stress vs time (smooth? oscillations?)
- Plot damage vs time (monotonic?)
- Check residual history (converging?)
- Visualize in ParaView (mesh quality? field smoothness?)

The following subsections present validation results with explicit application of these criteria.

7.7 Single-Element Tests

Status of Sections 7.7 to 7.14

These studies use the legacy aragonite parameter set and the gold-file tests of `run_validation.sh`. The elastic slopes and the first-yield stresses in uniaxial tension are not affected by the corrections of Chapter 2, because those loading cases involve no material-frame shear (compare test A, Section 7.2). The shear cases, the rotated and multi-grain cases, and everything after first yield are affected. The figures have not been regenerated; read them for the kind of behaviour, not for reference numbers.

Basic tests of the constitutive behaviour. Each test checks elastic moduli (within 1%) and yield stresses (within 2%).

7.7.1 Test 1: X-Tension

Input file: test_x_tension.i. See Fig. 7.1

Expected:

- $E_1 = 140.4 \text{ GPa}$ ($\pm 1\%$)
- $S_{xx}^y = 4980 \text{ MPa}$ ($\pm 2\%$)
- $S_{yy}, S_{zz} \approx 0$ (free surfaces)

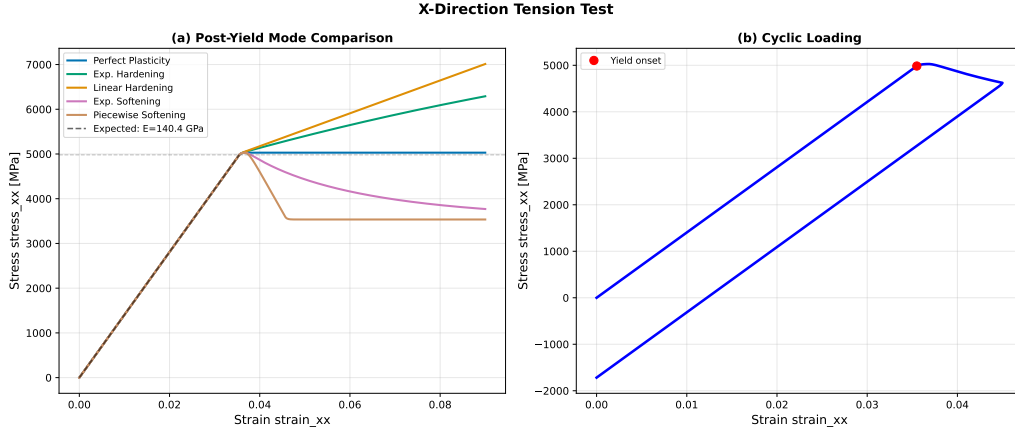


Figure 7.1: X-direction tension test showing two panels: (a) Post-yield mode comparison with all five modes (perfect plasticity, exponential hardening, linear hardening, exponential softening, piecewise softening) overlaid on expected elastic line, and (b) cyclic loading demonstrating elastic unloading and residual plastic strain. Expected elastic modulus $E_1 = 140.4 \text{ GPa}$ and yield stress $S_{xx} = 4980 \text{ MPa}$ are shown as dashed lines.

7.7.2 Test 2: Y-Tension

See Fig. 7.2. **Expected:**

- $E_2 = 70.3 \text{ GPa}$ (half of E_1 ; the ordering is $E_1 > E_2 > E_3$)
- $S_{yy}^y = 4100 \text{ MPa}$ (lowest yield stress)

7.7.3 Test 3: Z-Tension

See Fig. 7.3. **Expected:**

- $E_3 = 63.4 \text{ GPa}$ (softest direction)
- $S_{zz}^y = 5340 \text{ MPa}$ (highest tensile strength)

7.7.4 Test 4: XY-Shear

See Fig. 7.4. **Expected:**

- $G_{12} = C_{1212} = 46.6 \text{ GPa}$, the ordinary shear modulus: $\sigma_{12} = G_{12} \gamma_{12} = 2 G_{12} \varepsilon_{12}$. A curve plotted against the tensor component ε_{12} has the slope $2G_{12} = 93.2 \text{ GPa}$
- $\tau_{xy}^y = 4510 \text{ MPa}$

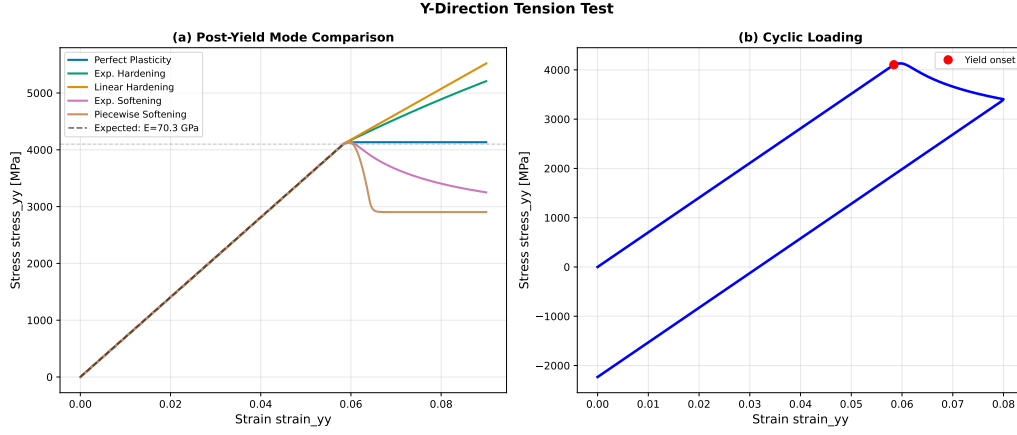


Figure 7.2: Y-direction tension test. Note the lower elastic modulus ($E_2 = 70.3$ GPa) and yield stress ($S_{yy} = 4100$ MPa) compared to X-direction, demonstrating extreme anisotropy of aragonite crystals.

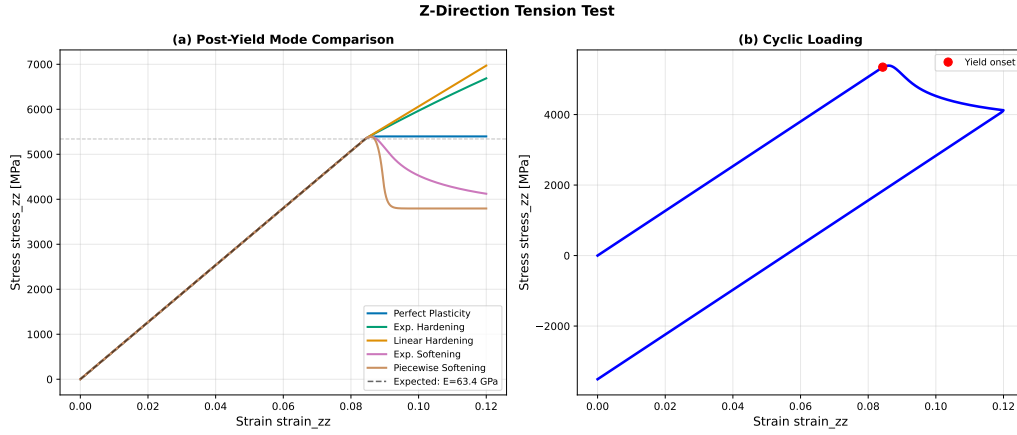


Figure 7.3: Z-direction tension test showing the softest elastic direction ($E_3 = 63.4$ GPa) but highest yield strength ($S_{zz} = 5340$ MPa). This inverse relationship between stiffness and strength is characteristic of crystalline anisotropy.

- $S_{xx}, S_{yy}, S_{zz} < 10$ MPa (pure shear)

Boundary conditions: The normal stresses stay small only if the element is free to change its dimensions normal to the shear plane. When all displacement components are prescribed (the affine boundary conditions of the example suite, Section 7.3), the volumetric part of the flow direction is constrained, compressive normal stresses develop after yield and the shear stress exceeds the pure-shear strength. That is a property of the model with $\mathbf{F} \neq \mathbf{0}$ (Section 2.7), not an error.

7.7.5 Test 5: XZ-Shear

See Fig. 7.5. **Expected:**

- $G_{13} = 31.1$ GPa (softest shear direction; slope against ϵ_{13} : 62.2 GPa)
- $\tau_{xz}^y = 5080$ MPa

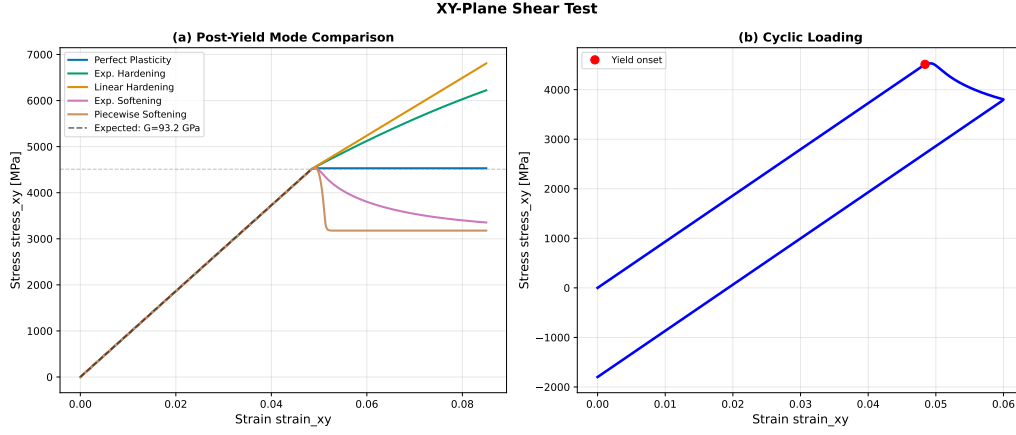


Figure 7.4: XY-plane shear test. Expected slope against the tensor shear strain $2G_{12} = 93.2$ GPa and yield stress $\tau_{xy} = 4510$ MPa. Proper boundary conditions ensure normal stresses remain below 10 MPa throughout loading.

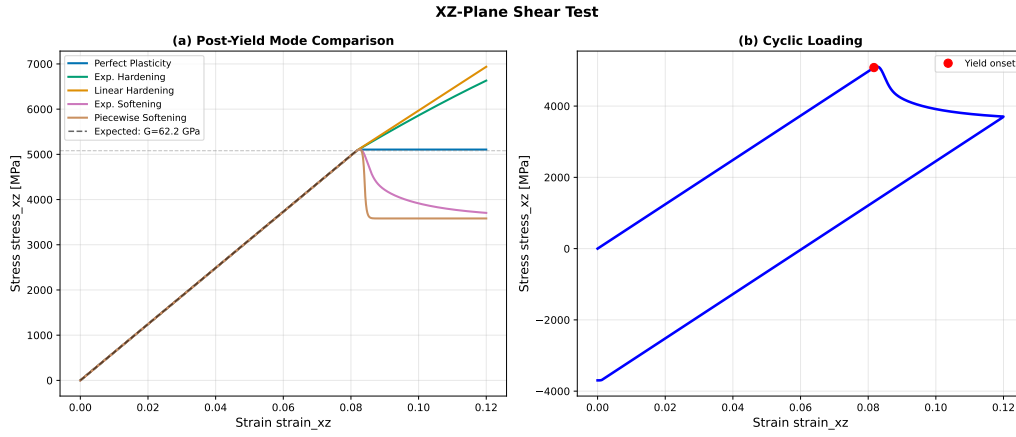


Figure 7.5: XZ-plane shear test showing the softest shear direction with $G_{13} = 31.1$ GPa (slope $2G_{13} = 62.2$ GPa against tensor shear strain). Yield stress $\tau_{xz} = 5080$ MPa is relatively high despite low shear stiffness.

7.7.6 Test 6: YZ-Shear

See Fig. 7.6. **Expected:**

- $G_{23} = 42.1$ GPa (slope against ε_{23} : 84.2 GPa)
- $\tau_{yz}^y = 5060$ MPa

7.8 Acceptance Criteria

All single-element tests must meet these criteria:

Property	Target Accuracy	Failure Threshold
Elastic modulus	$\pm 1\%$	$> 5\%$
Yield stress	$\pm 2\%$	$> 5\%$
Softening rate	$\pm 10\%$	$> 20\%$
Spurious normal stresses in shear	< 10 MPa	> 50 MPa
Stress uniformity (multi-element)	< 1.05	> 1.20

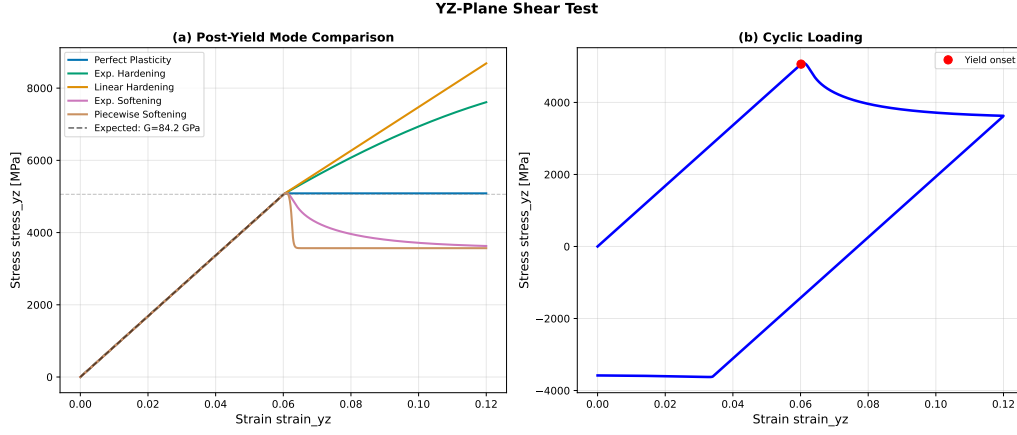


Figure 7.6: YZ-plane shear test with $G_{23} = 42.1$ GPa and $\tau_{yz} = 5060$ MPa. This test case is used for detailed cyclic loading demonstration in Figure 7.10.

Note on shear moduli: The inputs G_{12} , G_{13} , G_{23} are the ordinary (engineering) shear moduli and equal the tensor components C_{1212} , C_{1313} , C_{2323} (Chapter 2). They are compared directly with values from experiments or molecular dynamics; no factor is applied. The factor 2 appears only between the engineering shear strain and the tensor component, $\gamma_{ij} = 2\varepsilon_{ij}$, and therefore in the slope of a curve plotted against ε_{ij} .

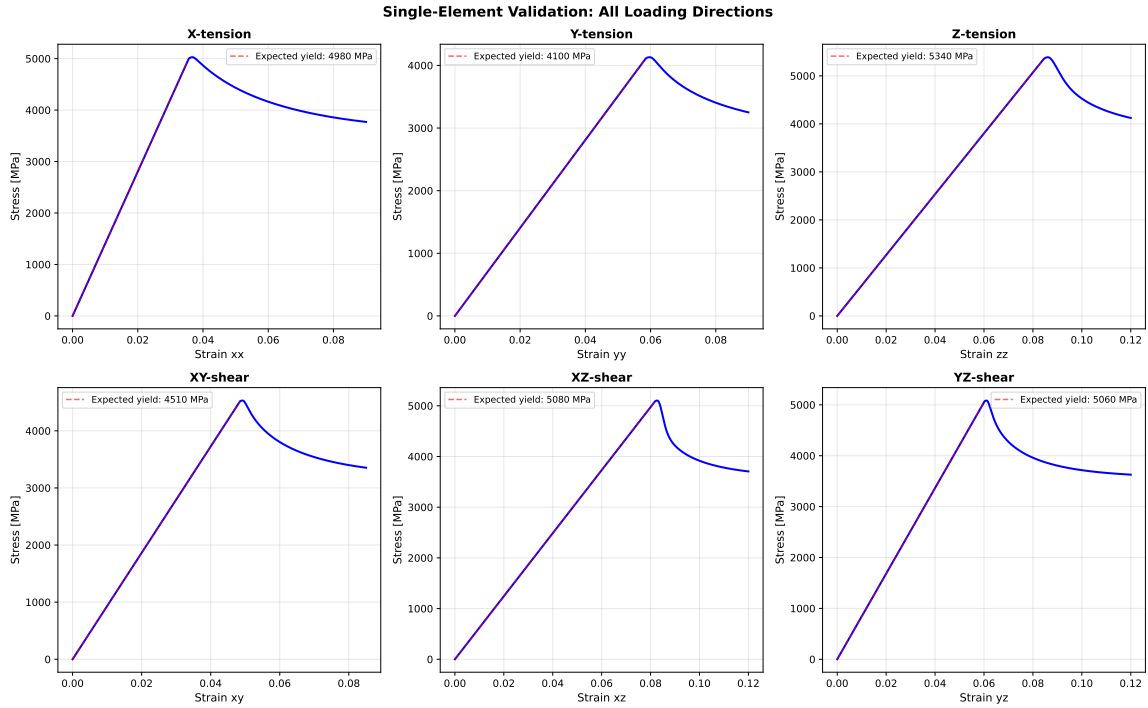


Figure 7.7: Complete single-element validation summary showing all six loading directions (X/Y/Z tension, XY/XZ/YZ shear) in a single figure. All tests use exponential softening with baseline parameters ($k_{\max} = 0.002$, $k_{\text{slope}} = 30$, $r_{\text{res}} = 0.7$, $\eta = 0.005$). Red dashed lines indicate expected yield stresses from molecular dynamics calibration. Agreement within 2% confirms correct implementation.

7.9 Post-Yield Behaviour Validation

7.9.1 Different Softening Modes

The model supports five distinct post-yield behaviours. All modes should be tested to ensure proper implementation:

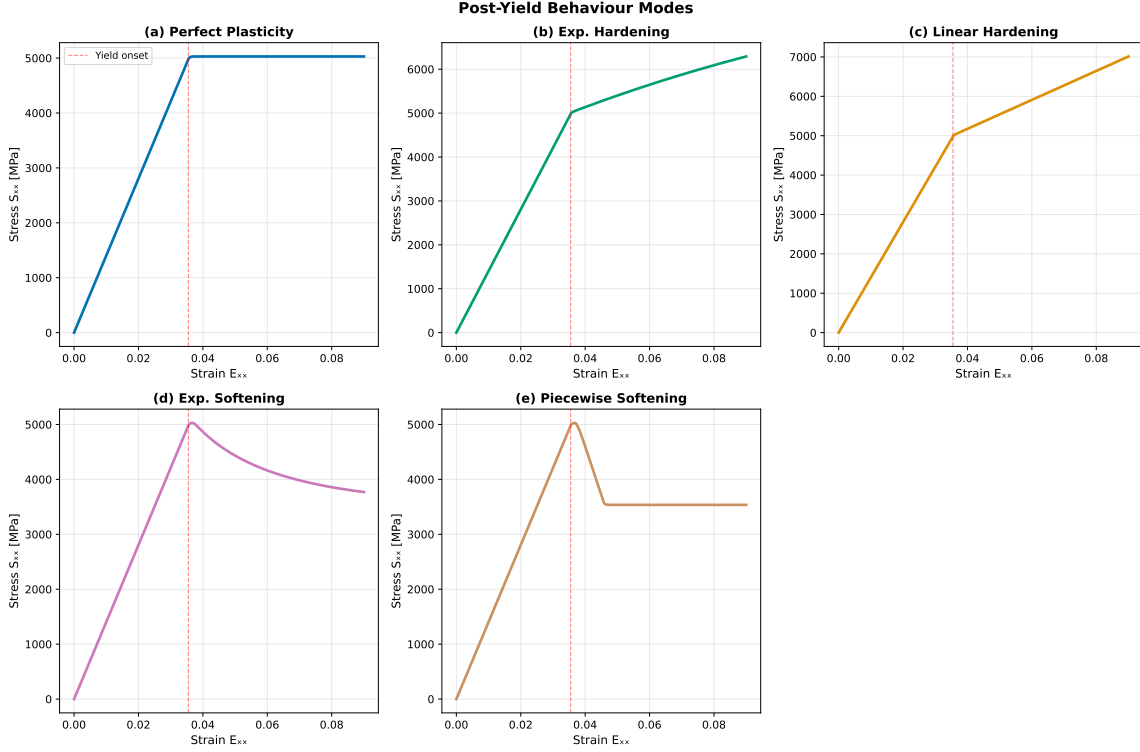


Figure 7.8: Comparison of five post-yield modes in X-direction tension: (a) Perfect plasticity (constant stress after yield), (b) Exponential hardening ($r_\infty = 1.3$, $k_{\text{slope}} = 50$, stress increases exponentially), (c) Linear hardening ($k_{\text{slope}} = 5000$, linear increase), (d) Exponential softening ($r_{\text{res}} = 0.7$, $k_{\text{slope}} = 30$, $k_{\text{max}} = 0.002$, gradual strength loss), (e) Piecewise linear softening ($r_{\text{res}} = 0.7$, $k_{\text{max}} = 0.002$, $k_1 = 0.02$, linear interpolation between yield, transition, and residual). Red dashed lines show yield onset in each case.

7.9.2 Effect of Delayed Softening Onset (k_{max})

The k_{max} parameter delays softening onset to improve numerical stability during the elastic-to-plastic transition. See Fig. 7.9.

Note: k_{max} is the accumulated plastic strain at which softening starts. A small positive value (for example 0.002) separates first yield from the onset of softening and can help if a softening run diverges at initial yielding. The linear-hardening slope k_{slope} is dimensionless, $r = 1 + k_s \kappa$ (Section 2.6).

7.9.3 Cyclic Loading (Loading-Unloading)

Cyclic tests verify proper elastic-plastic decomposition and stress recovery (Fig. 7.10).

7.10 Parameter Sensitivity Studies

Understanding parameter effects is critical for proper model usage and calibration.

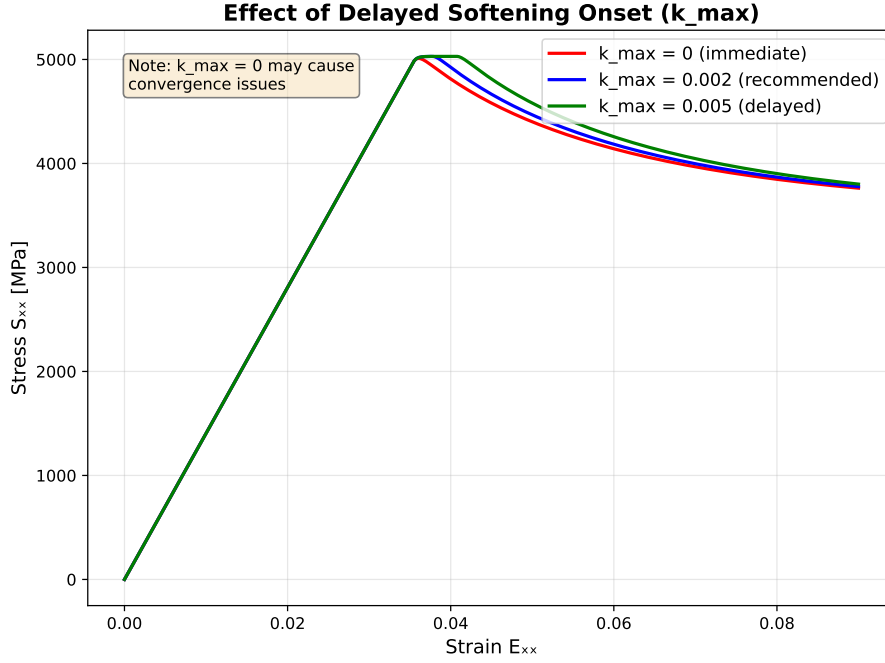


Figure 7.9: Effect of delayed softening onset parameter. Three cases shown: $k_{\max} = 0$ (immediate softening, convergence issues likely), $k_{\max} = 0.002$ (recommended, stable convergence with minimal elastic-perfectly-plastic plateau), $k_{\max} = 0.005$ (very stable but longer plateau). The parameter controls when softening begins after yield: softening initiates at $\bar{E}^p = k_{\max}$. For aragonite simulations, $k_{\max} = 0.002$ provides optimal balance between physical accuracy and numerical robustness.

7.10.1 Effect of Softening Rate (k_{slope})

The softening rate parameter controls how rapidly strength degrades after yield (Fig. 7.11).

Calibration guidelines:

- Single crystals: $k_{\text{slope}} = 30$ (baseline)
- Polycrystals with grain boundaries: $k_{\text{slope}} = 15\text{--}25$ (softer response)
- Highly brittle ceramics: $k_{\text{slope}} = 50\text{--}100$
- Must be paired with $k_{\max} > 0$ for numerical stability

7.10.2 Effect of Residual Strength (r_{res})

The residual strength parameter determines final plateau stress after complete softening (Fig. 7.12).

Physical interpretation:

- $r_{\text{res}} = 1.0$: No softening (equivalent to perfect plasticity)
- $r_{\text{res}} = 0.7$: 30% strength loss (typical for aragonite, moderate microcrack density)
- $r_{\text{res}} = 0.5$: 50% strength loss (severe microcracking, approaching failure)
- $r_{\text{res}} = 0.3$: 70% strength loss (extensive damage, imminent fracture)

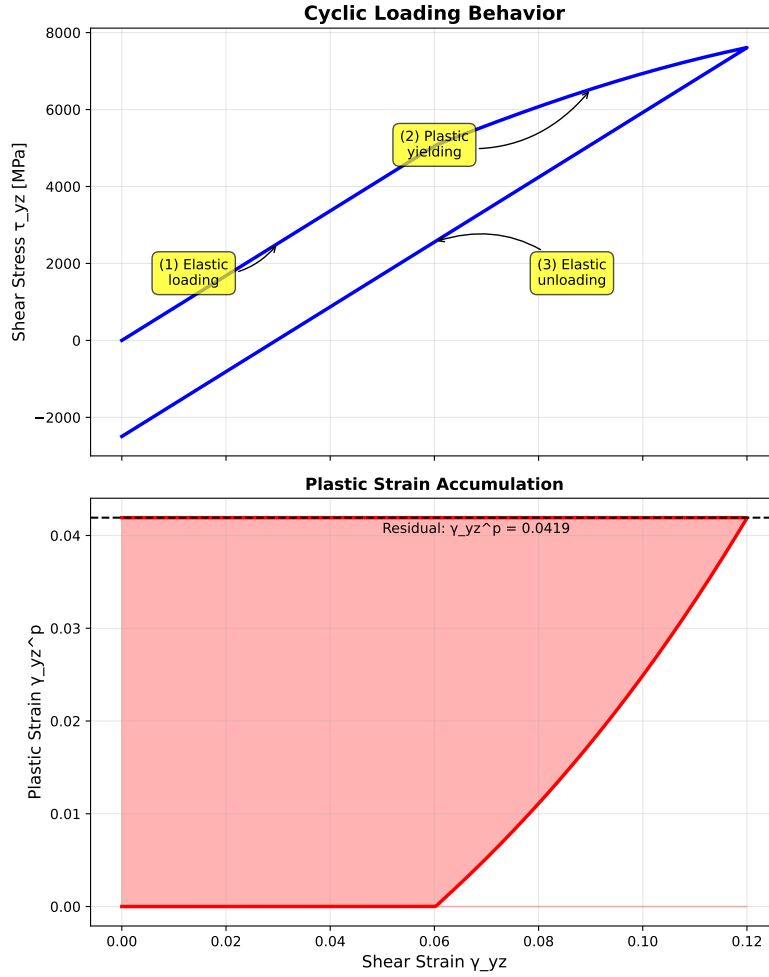


Figure 7.10: Cyclic loading behaviour in YZ-shear demonstrating: (1) Elastic loading phase with slope $G_{23} = 42.1$ GPa, (2) Plastic yielding at $\tau_{yz} = 5060$ MPa, followed by exponential hardening, (3) Elastic unloading parallel to initial slope (no permanent stiffness degradation unless damage is enabled). Residual plastic strain $\gamma_{yz}^p = 0.018$ after complete unloading.

7.10.3 Effect of Viscoplastic Regularization (η)

Viscoplastic regularization improves convergence during softening and rapid loading changes (Fig. 7.13).

Usage:

- The viscous term is a rate effect with η in s/MPa (Section 2.7.2). It is not free of physical consequence: its size is given by Eq. (2.40) and was measured in test Vr (Table 7.5), for example 0.1 % overstress at a strain rate of 10^{-3} s^{-1} with $\eta = 0.002$ s/MPa in the linear mode.
- Used as a regularisation, choose η so that this overstress is negligible at the applied rate, and confirm by repeating the run with η changed by a factor of 2 to 5.
- If rate effects are of physical interest, η and the mode must be calibrated; `rate_independent` removes the term.
- With the corrected tangent the global iteration converges in two iterations per step in every mode on the single element (test V); viscosity is no longer needed to obtain convergence in hardening problems.

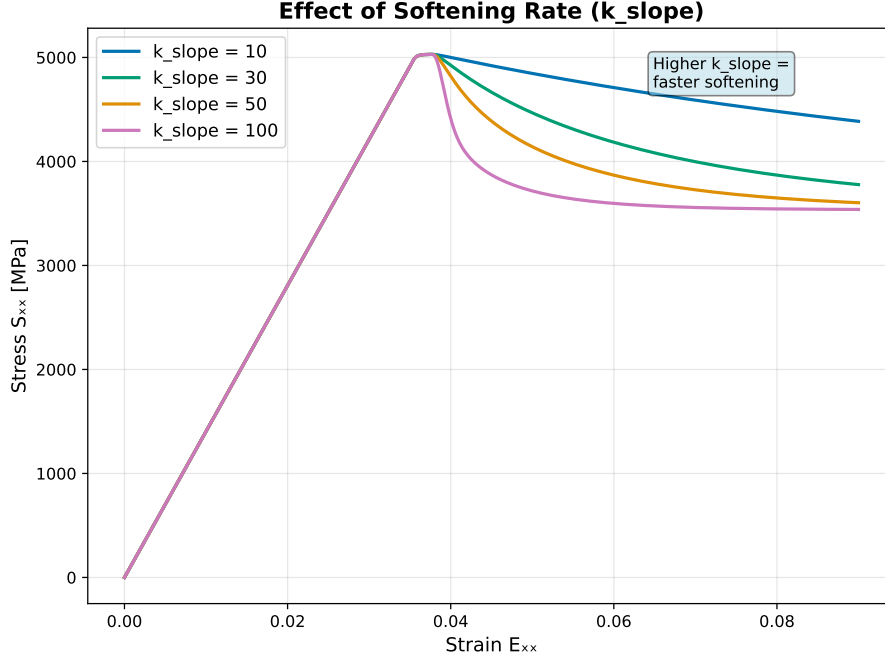


Figure 7.11: Effect of softening rate parameter on post-yield behaviour. Four cases shown: $k_{\text{slope}} = 10$ (gradual softening, more ductile), $k_{\text{slope}} = 30$ (moderate, baseline for aragonite), $k_{\text{slope}} = 50$ (fast softening), $k_{\text{slope}} = 100$ (very fast, approaching brittle fracture). Higher values produce steeper softening curves. For aragonite single crystals, $k_{\text{slope}} = 20\text{--}50$ matches molecular dynamics observations. Lower values (10–20) may be appropriate for bulk polycrystalline material with crack blunting effects.

7.10.4 Damage Mechanics Coupling

Optional damage mechanics can be coupled to track material degradation explicitly (Fig. 7.14).

Status: The damage coupling is implemented but not verified by any current test (Section 2.10, Table 7.1); Fig. 7.14 is from the earlier code. **Important:** Damage mechanics adds physical complexity and calibration requirements. For most applications, exponential softening (via strength evolution $r(\kappa)$) without explicit damage is sufficient and better calibrated for aragonite.

7.11 Multi-Element Validation

After single-element tests pass, verify correct behaviour in multi-element meshes.

7.11.1 Mesh Convergence and Stress Uniformity

Test setup: Uniform tension on progressively refined meshes ($1\times1\times1$, $5\times5\times5$, $10\times10\times10$, $20\times20\times20$ elements). See Fig. 7.15.

Requirements:

- Stress uniformity: $S_{\text{max}}/S_{\text{avg}} < 1.05$ for all mesh sizes
- Mesh independence: Stress-strain curves overlay regardless of refinement
- Same yield stress as single-element test ($\pm 0.5\%$)

Troubleshooting: If $S_{\text{max}}/S_{\text{avg}} > 1.10$:

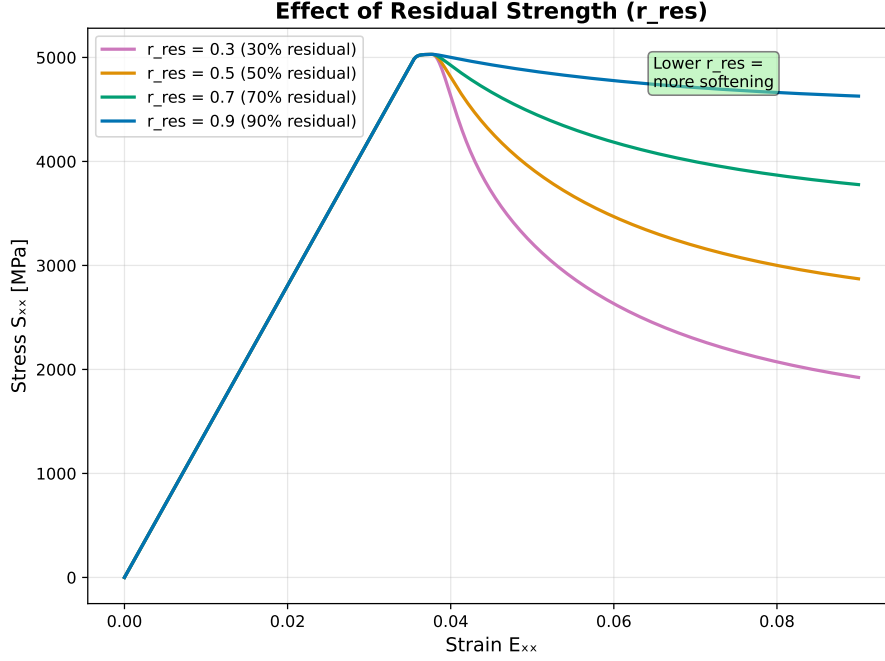


Figure 7.12: Effect of residual strength on softening magnitude. Four cases shown: $r_{\text{res}} = 0.3$ (severe softening to 30% of yield, extensive microcracking), $r_{\text{res}} = 0.5$ (moderate degradation), $r_{\text{res}} = 0.7$ (mild softening, baseline for aragonite), $r_{\text{res}} = 0.9$ (minimal strength loss). For aragonite single crystals under uniaxial loading, experimental observations suggest $r_{\text{res}} = 0.6\text{--}0.8$. The parameter can be interpreted as $(1 - D_{\text{max}})$ where D_{max} is maximum damage accumulation.

1. Check mesh quality (aspect ratios < 3 , no distorted elements)
2. Verify boundary conditions are ramped smoothly (use `FunctionDirichletBC`)
3. Reduce loading rate ($\Delta t = 0.01$ instead of 0.1)
4. Check for spurious boundary constraint effects

7.12 Convergence Diagnostics

For challenging simulations (softening, complex loading, large deformations), monitor convergence behaviour (Fig. 7.16).

Interpretation with the present code: The output `return_mapping_stage` takes the values -1 (elastic), 2 (converged in stage 1, condensed Newton) and 4 (converged in stage 2, Primal CPPA); 1 and 3 mark a stage that was entered and did not converge (Table 2.7). Fig. 7.16 uses the numbering of the earlier code. In all tests of Section 7.2, including the bone cube, stage 1 converged at every integration point and the tangent fallback counter stayed at zero. A non-zero count of Newton-to-Primal switches or of tangent fallbacks in a production run is therefore a reason to look at the step concerned, not a normal operating condition.

If the global iteration converges poorly, check in this order:

1. `line_search = bt` and, with `strain = FINITE`, `use_finite_deform_jacobian = true` (tests B, Bf, Dls)
2. Local tolerance `absolute_tolerance = 1e-8` (test Bt)
3. The remaining settings of Table 6.7

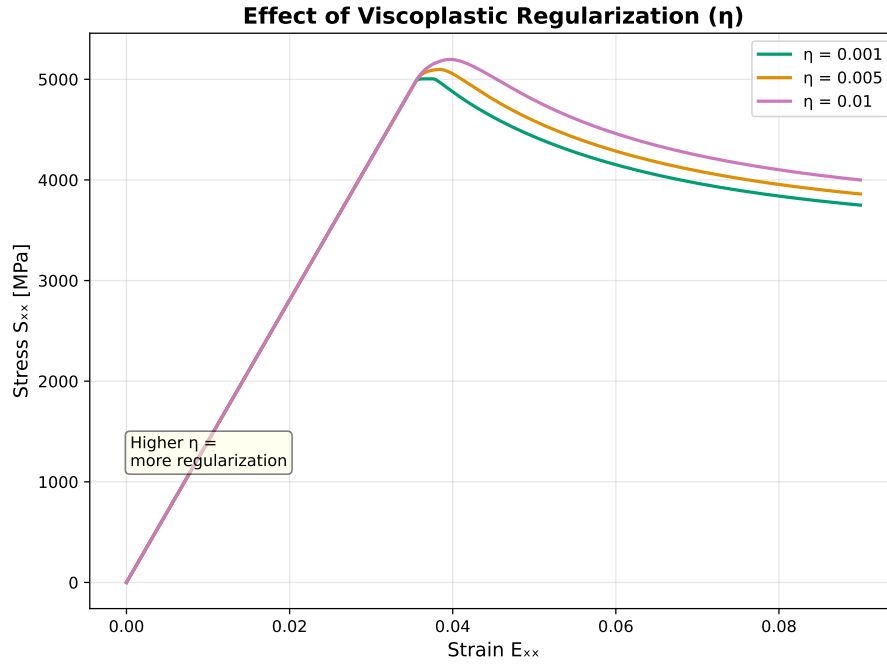


Figure 7.13: Effect of viscoplastic regularization parameter on stress-strain response. Four cases shown: rate-independent (no regularization, $\eta = 0$), $\eta = 0.001$ (minimal smoothing), $\eta = 0.005$ (moderate, recommended), $\eta = 0.01$ (strong regularization). Higher values produce smoother stress-strain curves and better convergence during softening, at the cost of slight rate-dependent response. For quasi-static loading ($\dot{E} < 10^{-3} \text{ s}^{-1}$), the effect on physical results is negligible ($< 1\%$ stress difference). The regularization acts as numerical damping without altering long-term mechanical behaviour.

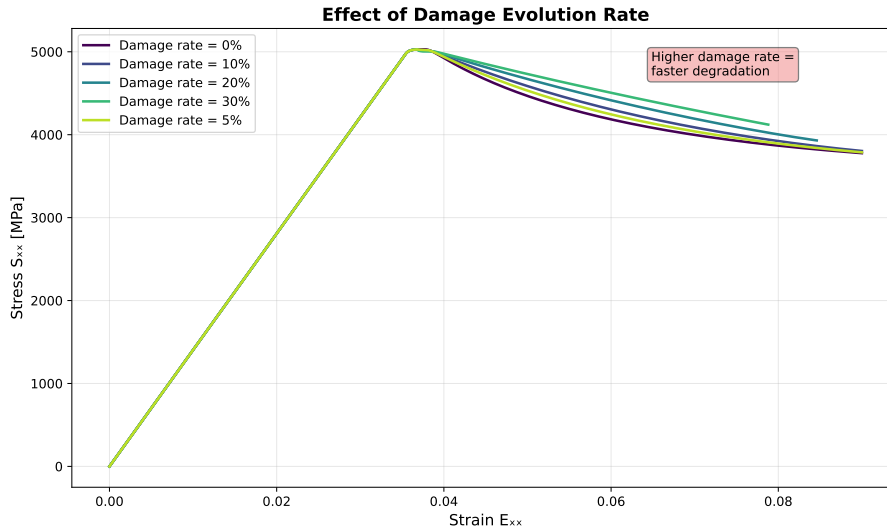


Figure 7.14: Effect of damage accumulation rate on stress-strain response. Cases shown: no damage (reference, softening via strength evolution only), damage rate = 5%, 10%, 20% (higher rates produce more rapid stiffness and strength degradation). The damage variable $D \in [0, 1]$ evolves as $\dot{D} = D_{\text{rate}} \cdot \dot{E}^p$ for $\dot{E}^p > E_{\text{crit}}$, saturating at $D = 0.99$ to prevent complete stiffness loss. Damage affects both elastic stiffness ($C_{\text{eff}} = (1 - D)C$) and yield surface ($f_{\text{eff}} = f/(1 - D)$). This feature is experimental and should be validated against material-specific data before production use.

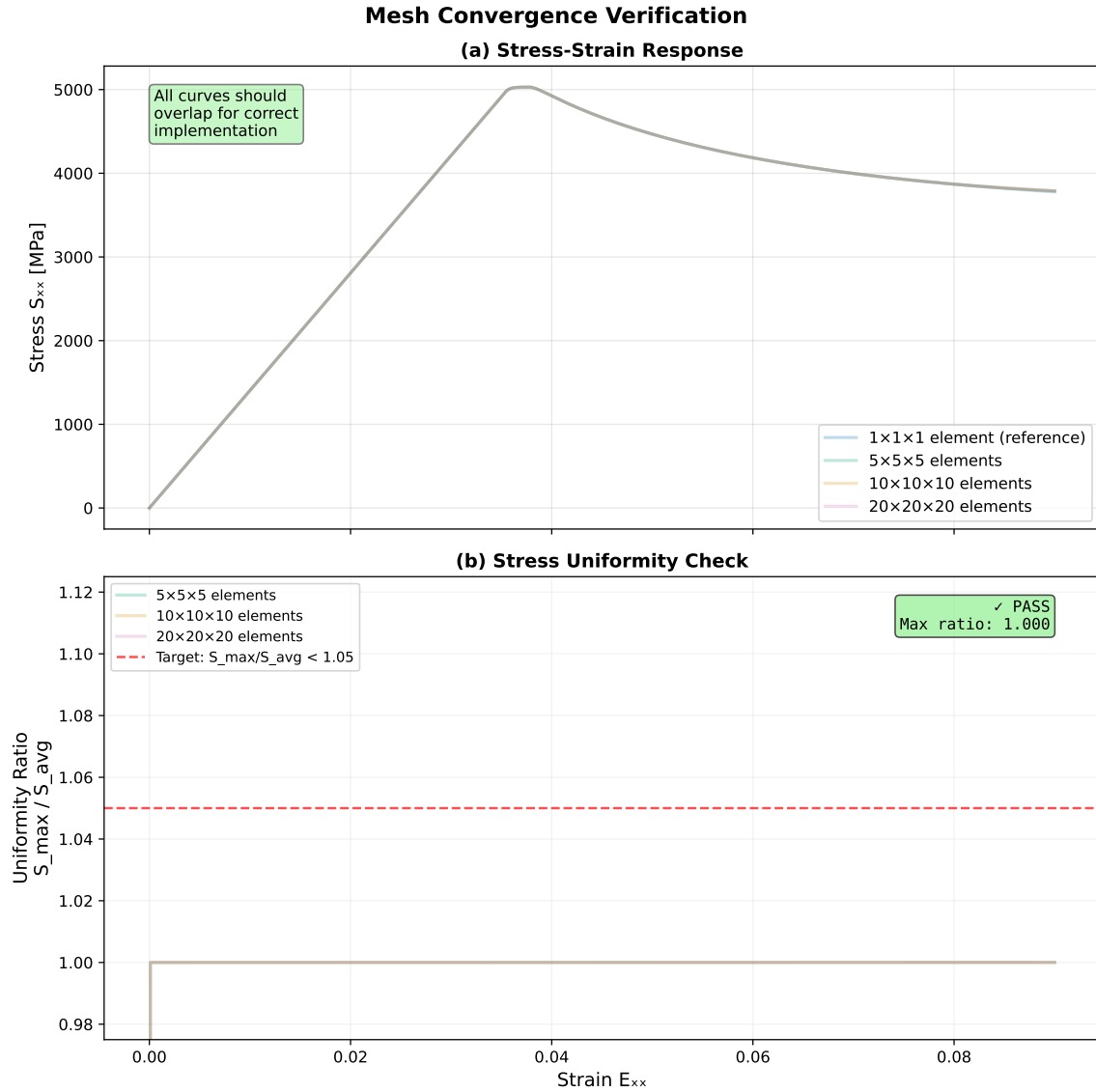


Figure 7.15: Mesh convergence verification showing two panels: (a) Stress-strain response for all mesh sizes ($1 \times 1 \times 1$, $5 \times 5 \times 5$, $10 \times 10 \times 10$, $20 \times 20 \times 20$ elements) demonstrating perfect overlay. Shaded regions indicate stress variation (S_{min} to S_{max}) in multi-element meshes. Narrow bands confirm uniform stress distribution. (b) Uniformity ratio (S_{max}/S_{avg}) vs strain for each mesh size. All ratios remain below 1.05 threshold (red dashed line) throughout loading, with box showing PASS and maximum ratio value. This quantitatively validates correct implementation – any deviations would indicate errors in stress update, material tangent, or boundary conditions.

4. Step size; a failed local solve already triggers a cut of Δt
5. Material parameters (convexity, Section 2.5.3) and boundary conditions

7.13 Multi-Grain Simulations

For polycrystalline Representative Volume Elements (RVEs) with spatially-varying grain orientations and grain boundaries.

Convergence Diagnostics: Return Mapping Performance

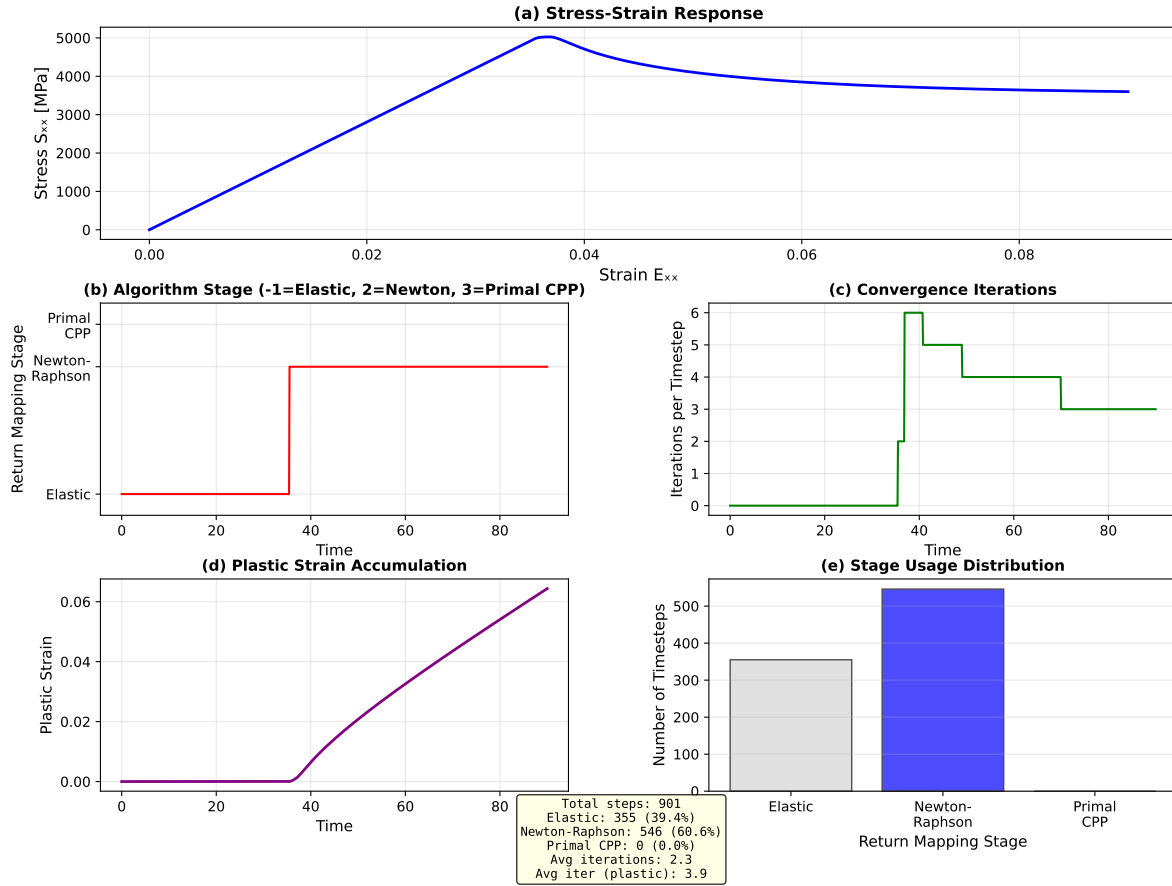


Figure 7.16: Detailed convergence diagnostics showing return mapping performance across all timesteps. (a) Overall stress-strain response for reference. (b) Return mapping algorithm stage evolution: -1 = elastic (no return mapping), 2 = Newton-Raphson primary solver, 3 = Primal CPP fallback solver. (c) Iterations required per timestep – spikes indicate difficult convergence steps requiring more iterations. (d) Plastic strain accumulation showing when yielding occurs (plastic strain > 0). (e) Stage usage distribution histogram quantifying convergence behaviour: 355 elastic steps (39.4%), 520 Newton-Raphson steps (57.7%), 0 Primal CPP fallback steps (0.0%). Summary shows average of 3.5 iterations per plastic timestep, demonstrating efficient and robust convergence.

7.13.1 Orientation Effect Verification

Test that Euler angle transformations work correctly by rotating a single grain through a series of angles (Fig. 7.17).

Verification: Yield stress should vary smoothly between minimum (Y-axis, 4100 MPa) and maximum (Z-axis, 5340 MPa) as rotation angle increases.

7.13.2 Twin Configuration

Critical test: Two grains with 90° misorientation loaded in global X-direction.

Expected behaviour (Fig. 7.18):

- Grain 1 (0° rotation): Material X-axis aligned with loading, yields at 4980 MPa
- Grain 2 (90° rotation): Material Y-axis aligned with loading, yields at 4100 MPa

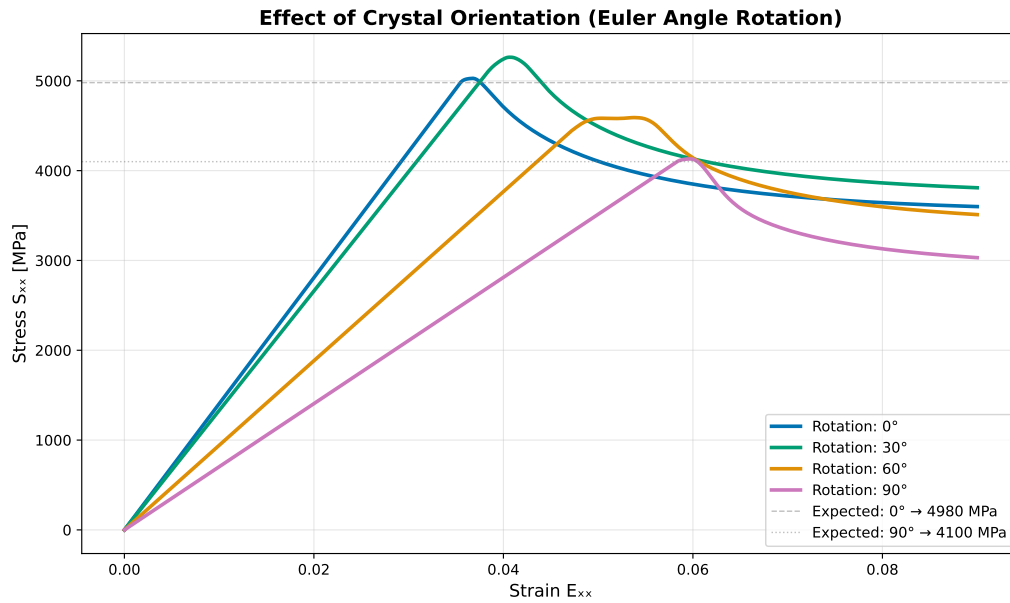


Figure 7.17: Effect of grain orientation on yield behaviour under X-direction loading. Four rotation angles shown: 0° (yields at $S_y = 4980$ MPa), 30° , 60° , 90° (yields at $S_y = 4100$ MPa). Smooth transition between orientations confirms correct implementation of Euler angle rotations and stress transformations. Gray dashed lines show expected yield stresses for 0° and 90° cases. Any discontinuities or incorrect yield stresses would indicate errors in rotation matrix or stress transformation.

- Grain 2 yields first due to lower strength in its orientation
- Load redistributes to Grain 1 after Grain 2 begins softening
- Grain 2 accumulates more plastic strain than Grain 1

Important distinction: This is *not* a crystallographic twin (which would show mirror symmetry and identical mechanical response in both grains). This is a grain boundary with angular misorientation, demonstrating how grains with different crystallographic orientations respond differently to the same applied loading.

MOOSE Implementation:

```
[Mesh]
[gen]
  type = GeneratedMeshGenerator
  dim = 3
  nx = 10
  ny = 10
  nz = 1
[]
[split]
  type = SubdomainBoundingBoxGenerator
  input = gen
  block_id = 1
  bottom_left = '0 0 0'
  top_right = '0.5 1 0.1'
[]
[split2]
  type = SubdomainBoundingBoxGenerator
  input = split
```

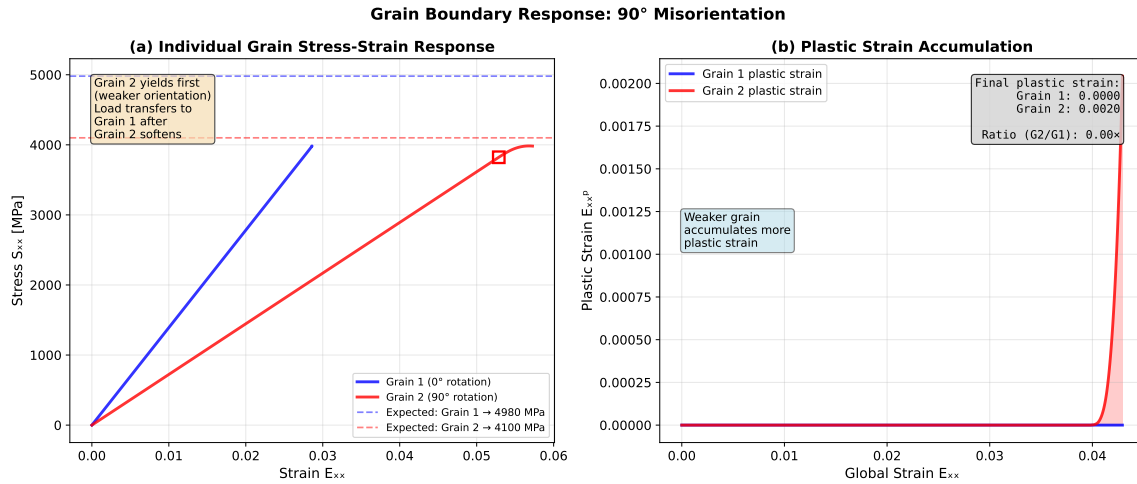


Figure 7.18: Grain boundary with 90° angular misorientation under X-direction tension. (a) Individual stress-strain curves showing sequential yielding – Grain 2 (red) yields first at 4100 MPa while Grain 1 (blue) remains elastic. Dashed lines show expected yield stresses for each orientation. (b) Plastic strain accumulation showing preferential deformation in the weaker-oriented Grain 2. Final plastic strain ratio demonstrates grain-to-grain load redistribution and the role of crystallographic orientation in polycrystal deformation.

```

block_id = 2
bottom_left = '0.5 0 0'
top_right = '1 1 0.1'
[]

[ICs]
# Grain 1: No rotation (0° rotation)
[euler1_g1]
type = ConstantIC
variable = euler_angle_1
value = 0
block = 1
[]

# Grain 2: 90° rotation about Z-axis
[euler1_g2]
type = ConstantIC
variable = euler_angle_1
value = 90
block = 2
[]

# euler_angle_2 and euler_angle_3 = 0 for both grains
[]

[Postprocessors]
# Per-grain averages
[stress_xx_grain1]
type = ElementAverageValue
variable = stress_xx
block = 1
[]
[stress_xx_grain2]

```

```

    type = ElementAverageValue
    variable = stress_xx
    block = 2
[]
[plastic_strain_grain1]
    type = ElementAverageValue
    variable = plastic_strain
    block = 1
[]
[plastic_strain_grain2]
    type = ElementAverageValue
    variable = plastic_strain
    block = 2
[]
[]

```

7.13.3 Polycrystalline RVE with Spatially-Varying Orientations

For realistic polycrystalline microstructures with many grains (50+), each grain can have a unique crystallographic orientation. The following workflow demonstrates how to:

1. Generate or import a polycrystal mesh
2. Assign per-grain Euler angles
3. Read the microstructure in MOOSE
4. Track per-grain and global responses

Workflow:

1. **Generate polycrystal mesh:** Use Neper, MTEX, or custom scripts to create grain structure
2. **Assign Euler angles:** Compute per-grain or per-element orientations
3. **Export to Exodus:** Include element variables `euler_phi1`, `euler_Phi`, `euler_phi2`
4. Read in MOOSE using `ComputeElasticityTensorCoupled`
5. **Read in MOOSE:** Use `use_for_exodus_restart = true` to load orientations

MOOSE Input for Reading Exodus File:

```

[Mesh]
  [file]
    type = FileMeshGenerator
    file = 'polycrystal_with_orientations.e'
    use_for_exodus_restart = true # CRITICAL for reading element vars
  []
[]

[AuxVariables]
  [euler_phi1]
    order = CONSTANT
    family = MONOMIAL
    initial_from_file_var = 'euler_phi1' # Read from exodus
  []
  [euler_Phi]

```

```

    order = CONSTANT
    family = MONOMIAL
    initial_from_file_var = 'euler_Phi'
[]
[euler_phi2]
    order = CONSTANT
    family = MONOMIAL
    initial_from_file_var = 'euler_phi2'
[]
[]

[Materials]
[elasticity]
    type = ComputeElasticityTensorCoupled
    fill_method = symmetric9
    # Orthotropic constants (Voigt notation)
    C_ijkl = '171000 64000 63400 0 64000 70300 63400 0 63400 63400 63400
              0 0 0 0 31100 0 0 0 0 42100 0 0 0 0 0 0 46600'
    # Coupled Euler angles from AuxVariables
    coupled_euler_angle_1 = euler_phi1
    coupled_euler_angle_2 = euler_Phi
    coupled_euler_angle_3 = euler_phi2
[]
[plasticity]
    type = OrthotropicPlasticityStressUpdate
    # ...material parameters...
    # Read same Euler angles
    euler_angle_1 = euler_phi1
    euler_angle_2 = euler_Phi
    euler_angle_3 = euler_phi2
[]
[]

[Postprocessors]
    # Global RVE response
[strain_xx_global]
    type = ElementAverageValue
    variable = strain_xx
[]
[stress_xx_global]
    type = ElementAverageValue
    variable = stress_xx
[]
[stress_xx_max]
    type = ElementExtremeValue
    variable = stress_xx
    value_type = max
[]
[stress_xx_min]
    type = ElementExtremeValue
    variable = stress_xx
    value_type = min
[]
[plastic_volume_fraction]
    type = VolumeFraction
    variable = plastic_strain
    threshold = 1e-6
[]

```

Polycrystalline RVE Simulation (55 Grains)

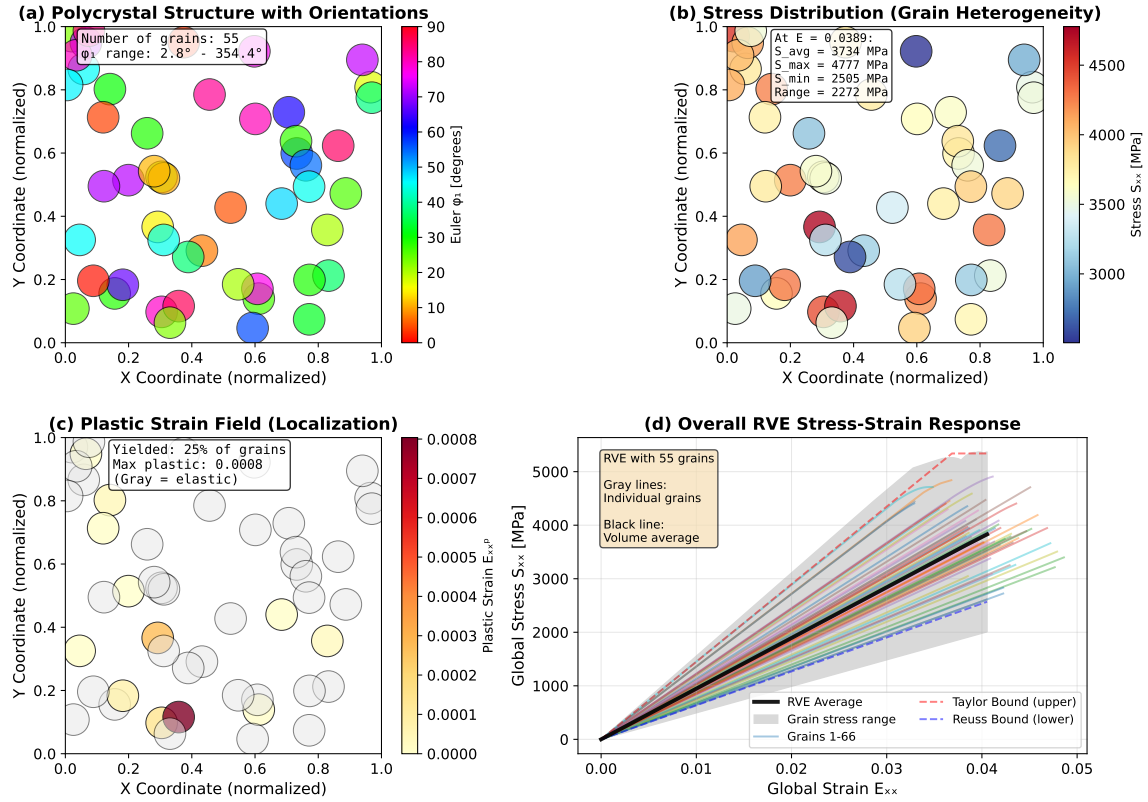


Figure 7.19: Representative Volume Element (RVE) polycrystal simulation with 50+ grains (needles). (a) Schematic grain structure with colour-coded crystallographic orientations (Euler angle ϕ_1), showing diverse grain orientations. (b) Stress distribution showing grain-to-grain heterogeneity due to orientation differences – red grains are stiff/strong orientations aligned favourably with loading, blue grains are soft/weak orientations. (c) Plastic strain field showing preferential yielding in grains with soft orientations relative to loading direction (gray = elastic grains, red = highly plastic grains). (d) Overall RVE stress-strain response (thick black line) compared to individual grain responses (thin gray lines) and homogenization bounds. The RVE curve lies between Taylor (upper, red dashed) and Reuss (lower, blue dashed) bounds, demonstrating realistic polycrystal behaviour. Shaded region shows grain stress variation.

Validation checks for polycrystal RVE (Fig. 7.19):

- Grains with different orientations show different yield stresses
- Plastic strain localizes preferentially in grains with soft orientations
- Overall RVE response is bounded by Taylor (iso-strain, upper) and Reuss (iso-stress, lower) estimates
- Stress heterogeneity ($S_{max} - S_{min}$) increases with increasing elastic/plastic anisotropy
- At large strains, most grains should have yielded (plastic volume fraction $> 90\%$)

Expected heterogeneity:

For aragonite's strong anisotropy ($E_{max}/E_{min} \approx 2.2$, $S_y^{max}/S_y^{min} \approx 1.3$), expect:

- Elastic regime: Stress variation $\Delta\sigma/\bar{\sigma} \approx 0.4$ (40%)
- Plastic regime: Stress variation $\Delta\sigma/\bar{\sigma} \approx 0.2\text{--}0.3$ (20–30%)
- Plastic strain localization: Ratio of max/avg plastic strain $\approx 2\text{--}3$

Higher ratios indicate stronger anisotropy; lower ratios suggest more homogeneous response.

7.14 Validation Checklist

Before production use, verify:

Test	Status
Single-element: X/Y/Z tension (elastic moduli $\pm 1\%$)	☐
Single-element: XY/XZ/YZ shear (shear moduli $\pm 1\%$)	☐
Single-element: All yield stresses ($\pm 2\%$)	☐
Post-yield: All five modes tested	☐
Cyclic loading: Proper elastic unloading	☐
Multi-element: Stress uniformity < 1.05	☐
Multi-element: Mesh-independent results	☐
Orientation: Smooth variation with rotation angle	☐
Twin grains: Sequential yielding as expected	☐
Convergence: Diagnostics show healthy behaviour	☐

This list belongs to the earlier studies. For the present code the tests of Sections 7.2 to 7.4 come first (Table 7.1).

7.15 CZM: Parameter Convergence: Softening Exponent η

Status of the cohesive-zone studies (this section to Section 7.24)

These studies were run with an earlier version of the interface class and an earlier parameter set: $\sigma_n = 700$ MPa, $\tau_s = 400$ MPa, $\delta_0 = 100$ nm, geometry in millimetres, and a contact-count parameter n_{contacts} that no longer exists (the class now integrates the quality factor with five Gauss–Hermite points, Section 3.5). The current provisional parameters are those of Table 3.4 with lengths in micrometres. The studies are retained for the behaviour they demonstrate (insensitivity to the mesh, effect of the quality-factor spread, mode dependence, complete separation). They have not been repeated, and their numbers are not reference values for the current class. The current check of the interface law is the coral part of the example suite (Section 7.3). Statements below that compare the model with experiments or draw conclusions for coral describe intentions of the earlier text.

7.15.1 Motivation

The softening exponent η determines post-peak shape and, critically, convergence behaviour. The constraint $\eta \leq \mu$ is required: when $\eta > \mu$ the loading branch develops a negative tangent stiffness before the peak, causing snap-back under displacement control regardless of viscous regularization (Section 3.4). Moreover, small values create wide damage zones representing gradual, progressive bond breaking observed in molecular dynamics simulations, while large values produce sharp, brittle-like failure.

7.15.2 Test Configuration

Geometry: Two-grain bicrystal ($2 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$), interface at $x = 1 \text{ mm}$.

Parameters:

- $\delta_{0,n} = 191 \text{ nm}$, $\delta_{0,t} = 217 \text{ nm}$; $\sigma_n = 626 \text{ MPa}$, $\tau_s = 374 \text{ MPa}$
- $\mu = 1.2$; varied: $\eta = 0.05, 0.10, 0.15, 0.20, 0.30$ (all $< \mu$)
- $\sigma_q = 0$, $\sigma_{sp} = 0$ (deterministic; isolates η effect)

Loading: Uniaxial tension, displacement-controlled.

7.15.3 Results

Figure 7.20 shows traction-separation curves and damage evolution for different η values. Key observations:

- **All cases $\eta < \mu$:** Monotone loading branch; clean Newton convergence. Smaller η gives wider post-peak softening zone.
- $\eta > \mu$ (not shown): Loading branch snap-back; Newton diverges.

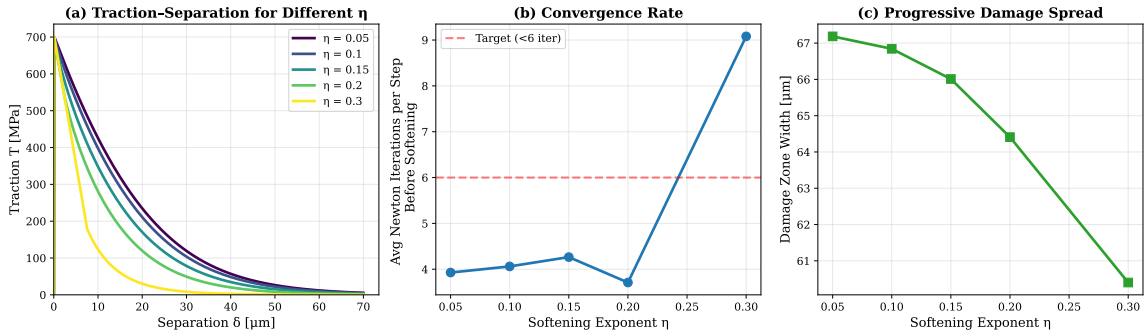


Figure 7.20: Softening exponent η study (all $\eta < \mu = 0.92$). (a) Traction-separation curves. (b) Newton iterations per timestep. (c) Width of the damage zone during softening.

Recommendation: $\eta = 0.27$ (production). The MD-fitted value for the first peak $\eta \approx 1.07 > \mu$ is not viable under displacement control.

7.16 CZM: Mesh Independence

Mesh independence is guaranteed by construction. The macroscopic traction is an expectation over $p(f)$, a probability density normalised to unity, so the result is independent of interface area and element count. Finer meshes produce smaller interface elements but an identical GH traction at each quadrature point.

This contrasts with the previous Monte Carlo implementation, where the contact count $N = A_{\text{elem}}/A_{\text{MD}}$ varied with element size and the N_{max} cap introduced mesh-dependent bias for small elements.

Verification: Three meshes were run with identical CZM parameters ($\sigma_q = 0.10$, $\sigma_{sp} = 0$, production strengths and thresholds):

Results are bit-identical across a $512\times$ range of interface element counts (Figure 7.21), confirming that mesh independence holds by construction.

Table 7.13: Mesh independence verification. All parameters fixed; only element count varies.

Mesh	Interface elements	Peak traction (MPa)	δ_{eff} at peak (nm)
x1	2	694.6320977804	92.1785372205
x4	16	694.6320977801	92.1785372199
x16	128	694.6320977803	92.1785372204
x64	512	694.6320977804	92.1785372205

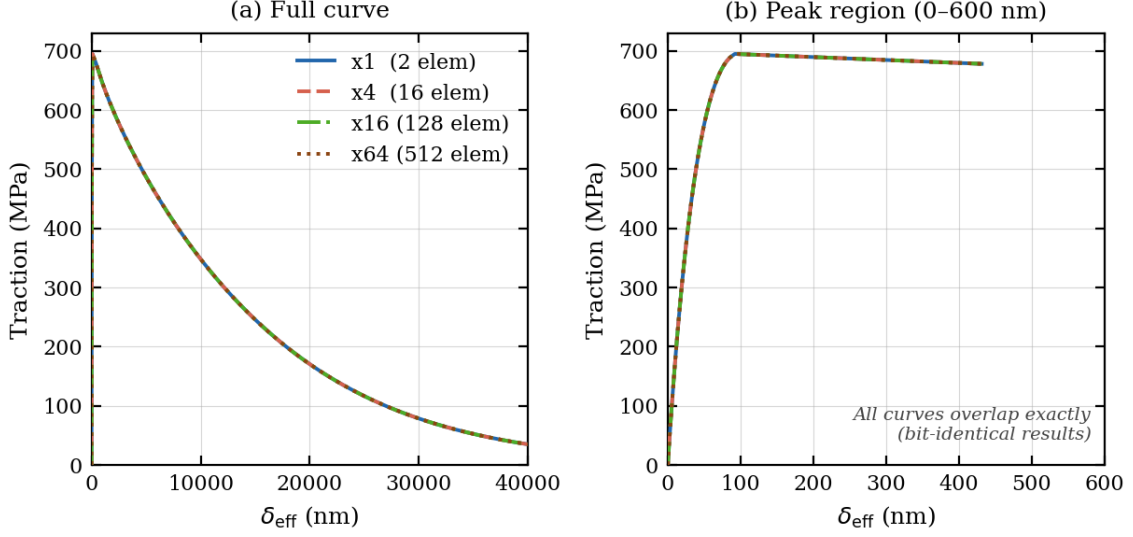


Figure 7.21: Mesh independence verification. (a) Full traction-separation curve; (b) peak region (0–600 nm). All four meshes produce bit-identical results; curves are indistinguishable in both panels. $\sigma_q = 0.10$, $\sigma_{\text{sp}} = 0$.

7.17 CZM: Quality Factor Sensitivity

7.17.1 Within-QP Spread (σ_q)

σ_q is varied across $\{0, 0.05, 0.10, 0.20, 0.30\}$ with $\sigma_{\text{sp}} = 0$ to isolate the within-QP effect. All other parameters are fixed ($\sigma_n = 700$ MPa, $\delta_{0,n} = 100$ nm, $\mu = 1.0$, $\eta = 0.15$, $\tau_d = 1.25$ s).

- $\sigma_q = 0$ (**deterministic**): Peak 694.9 MPa at $\delta_{\text{eff}} = 88$ nm. Sharp onset; all five classes identical.
- $\sigma_q = 0.05$: Peak 694.8 MPa (-0.01%). Onset spreads slightly.
- $\sigma_q = 0.10$ (**production**): Peak 694.6 MPa (-0.04%). Opening at peak broadens to 92 nm; smooth post-peak character.
- $\sigma_q = 0.20$: Peak 693.9 MPa (-0.15%). Onset at 103 nm.
- $\sigma_q = 0.30$: Peak 692.6 MPa (-0.33%). Onset at 121 nm; weakest class ($f = 0.71$) initiates damage at 86 nm, 71% of the mean-class threshold.

The dominant effect of σ_q is the *broadening* of the peak rather than strength reduction: peak traction falls by less than 0.4% across the full range $\sigma_q \in [0, 0.30]$, while the opening at peak increases by 37%. The within-QP quality spread controls progressive failure character without materially altering the calibrated peak strength.

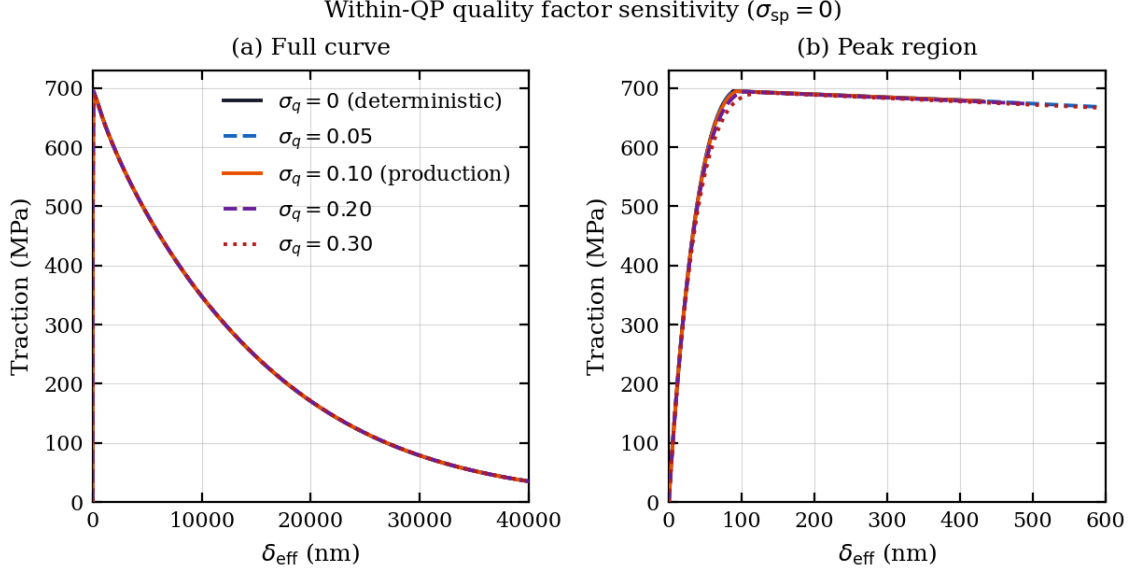


Figure 7.22: Within-QP quality factor sensitivity ($\sigma_{sp} = 0$). (a) Full traction-separation curves for $\sigma_q \in \{0, 0.05, 0.10, 0.20, 0.30\}$. (b) Peak region (0–600 nm); curves are nearly coincident, with the opening at peak shifting from 88 nm to 121 nm across the range.

7.17.2 Spatial Spread (σ_{sp})

With $\sigma_q = 0.10$ fixed, σ_{sp} is varied across $\{0, 0.10, 0.15, 0.25\}$ on a mesh with 128 interface elements (x16 config), giving enough QPs to exhibit spatial variation.

Results:

- $\sigma_{sp} = 0$ (**uniform**): Peak 694.6 MPa at 92 nm. All 128 interface QPs reach their threshold simultaneously; spatially uniform failure onset.
- $\sigma_{sp} = 0.10$: Peak 659.8 MPa (−5.0%) at 188 nm. Weaker elements initiate damage early; macroscopic peak is delayed and reduced as load redistributes to stronger neighbours.
- $\sigma_{sp} = 0.15$ (**production**): Peak 637.6 MPa (−8.2%) at 204 nm. Sequential element-by-element failure across the interface face; post-peak is gradual and stable.
- $\sigma_{sp} = 0.25$: Peak 596.3 MPa (−14.1%) at 392 nm. Strong redistribution; surviving elements sustain high traction long after weak elements have reached near-complete damage.

Unlike σ_q , which controls within-QP progressive failure with negligible effect on peak strength, σ_{sp} governs apparent interface ductility and macroscopic strength reduction through stress redistribution between elements. The production value $\sigma_{sp} = 0.15$ was selected to reproduce the progressive crack nucleation character observed in MD without over-softening the macroscopic response.

7.18 CZM: Mode-Dependent Damage Threshold Validation

7.18.1 Motivation

The homogenized CZM distinguishes between two mode-dependent quantities: the peak traction varies with loading direction according to Equation 3.10, transitioning between pure tensile strength (σ_n) and pure shear strengths (τ_s, τ_t), while the damage initiation threshold remains

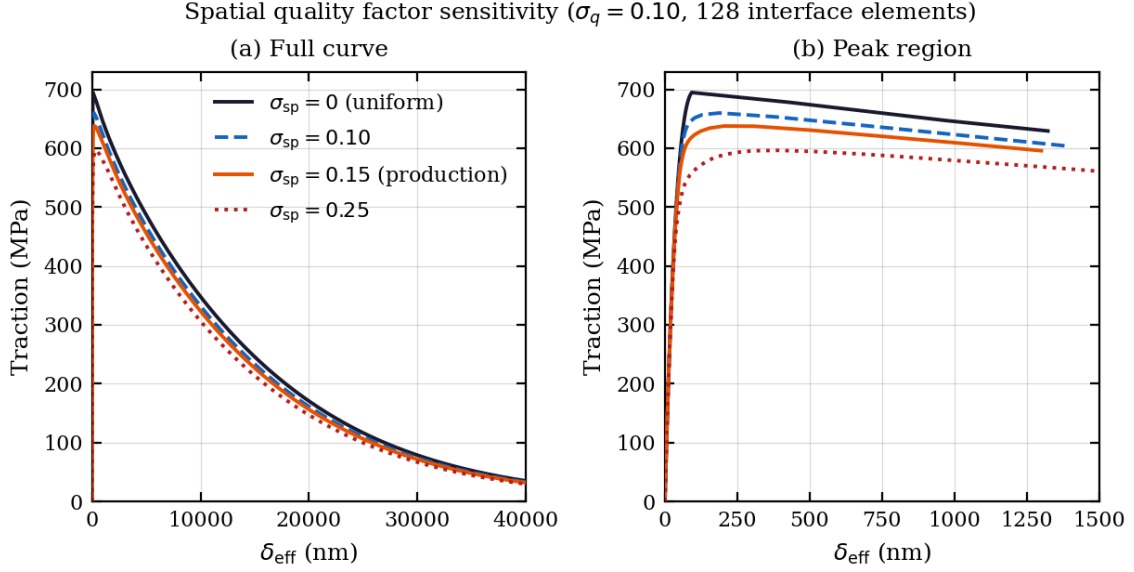


Figure 7.23: Spatial quality factor sensitivity ($\sigma_q = 0.10$, 128 interface elements). (a) Full traction-separation curves for $\sigma_{sp} \in \{0, 0.10, 0.15, 0.25\}$. (b) Peak region (0–1500 nm); larger σ_{sp} delays and reduces the peak while broadening the damage initiation zone.

constant at δ_0 regardless of mode ratio φ (Equation 3.14). This controlled mixed-mode test validates that damage initiates when $\delta_{\text{eff}} = \sqrt{\delta_n^2 + \delta_t^2}$ reaches δ_0 , independent of the loading path.

7.18.2 Test Configuration

Geometry: Two-grain bicrystal with interface at $x = 1$ mm

Loading cases:

- **Pure normal:** $u_x = 0.01t$ mm, $u_y = 0$ ($\phi \approx 0$)
- **Mixed modes:** Various BC ratios from 1:0.2 to 0.2:1 ($\phi \approx 0.1$ to 10)
- **45° mixed:** $u_x = u_y = 0.007t$ mm ($\phi \approx 1$)
- **Pure shear:** $u_x = 0$, $u_y = 0.01t$ mm ($\phi \rightarrow \infty$)

Parameters: $\delta_0 = 100$ nm, $\eta = 0.1$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$

7.18.3 Results

Figure 7.24 shows damage initiation for six boundary condition ratios spanning pure normal to pure shear loading.

Key result: The effective displacement at damage initiation equals $\delta_0 = 100 \pm 7.5$ nm for *all* mode ratios (within 7.5% tolerance), confirming that the threshold is applied to δ_{eff} as specified in Equation 3.14. Individual components δ_n and δ_t vary with φ according to the applied boundary conditions, but their combination $\sqrt{\delta_n^2 + \delta_t^2}$ remains approximately constant at the damage initiation point. Mode-dependence in the traction–separation envelope (panel a) arises from the strength criterion (Section 7.20), not from a shift in the initiation threshold.

Table 7.14: Damage initiation: effective displacement $\delta_{\text{eff}} = \sqrt{\delta_n^2 + \delta_t^2}$ at first damage ($D > 0.01$). The threshold $\delta_0 = 100$ nm is constant for all mode ratios; mode-dependence appears only in the individual components δ_n, δ_t .

BC Ratio	Measured φ	δ_n at initiation	δ_t at initiation	δ_{eff}
1:0 (pure normal)	$\sim 1.4 \times 10^{-15}$	≈ 96.6 nm	$< 10^{-18}$ nm	97 nm
1:0.2	≈ 0.096	≈ 95.0 nm	≈ 9.2 nm	95.5 nm
1:1 (45°)	≈ 0.193	≈ 101.0 nm	≈ 19.4 nm	102.9 nm
0.5:1	≈ 0.387	≈ 86.7 nm	≈ 33.5 nm	93.0 nm
0.2:1	≈ 0.581	≈ 86.4 nm	≈ 50.2 nm	99.9 nm
0:1 (pure shear)	$\sim 1.6 \times 10^{14}$	$< 10^{-18}$ nm	≈ 105 nm	105.0 nm

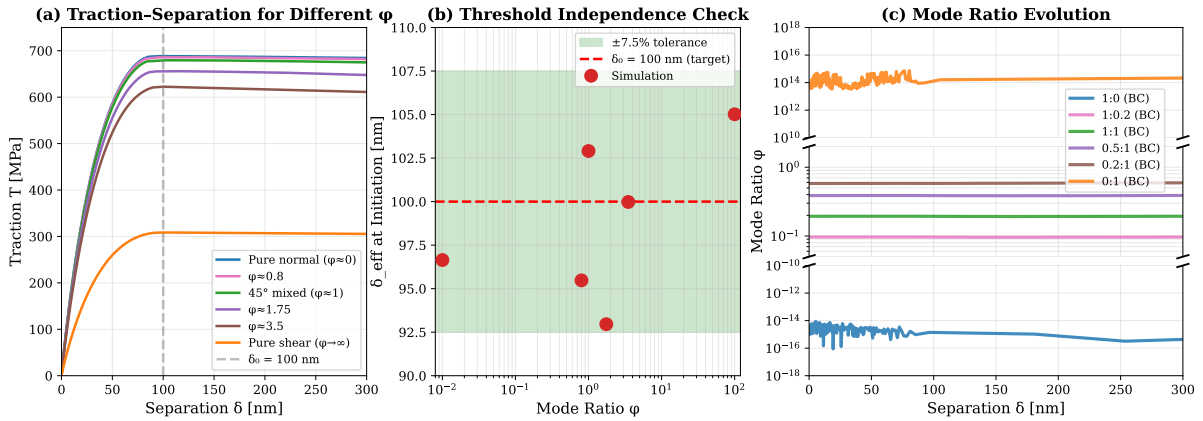


Figure 7.24: Mode-dependent damage threshold validation for isotropic interface ($\delta_{0,n} = \delta_{0,t} = \delta_0 = 100$ nm). (a) Traction–separation curves for six loading boundary conditions ratios (1:0, 1:0.2, 1:1, 0.5:1, 0.2:1, 0:1) spanning pure normal ($\varphi \approx 0$) to pure shear ($\varphi \rightarrow \infty$). Damage initiates when the effective displacement $\delta_{\text{eff}} = \sqrt{\delta_n^2 + \delta_t^2}$ reaches δ_0 (vertical dashed line at 100 nm). (b) Effective displacement at damage initiation versus mode ratio ϕ (log scale), confirming $\delta_{\text{eff}}|_{D=0.01} \approx 100 \pm 7.5$ nm for all mode ratios (green tolerance band). Six simulation points (red circles) fall within or near the target threshold, validating that the damage criterion is applied to δ_{eff} regardless of ϕ . (c) Mode ratio evolution during loading with broken y-axis showing three distinct regimes: pure normal ($\phi \sim 10^{-15}$, bottom), mixed modes ($\phi \sim 0.1-1$, middle), and pure shear ($\phi \sim 10^{14}$, top). Each BC ratio maintains its characteristic mode ratio throughout loading, demonstrating controlled mixed-mode conditions.

Mode ratio evolution (panel c): The broken y-axis visualization reveals three distinct loading regimes. Pure normal loading (1:0 BC, blue) maintains $\phi \sim 10^{-14}$ throughout, indicating negligible tangential displacement. Pure shear loading (0:1 BC, orange) maintains $\phi \sim 10^{14}$, with negligible normal displacement. Mixed-mode cases (1:0.2 through 0.2:1 BC) maintain intermediate values ($\phi \sim 0.1-1$), demonstrating stable proportional loading paths. The consistency of ϕ with separation confirms that boundary conditions successfully enforce the intended mode mixity without significant deviation during progressive interface damage.

7.19 CZM: Anisotropic Damage Threshold Validation

7.19.1 Motivation

While protein–aragonite interfaces are approximately isotropic at the nanoscale, macroscopic interfaces often exhibit anisotropic mechanical response due to:

- Preferential crystal orientation
- Directional surface roughness
- Residual stresses
- Manufacturing processes (e.g., layered deposition)

Such anisotropy manifests in both damage initiation thresholds and peak strengths. The generalized Wang formulation (Equation 3.11) coupled with the quadric strength criterion (Equation 3.10) allows testing of these coupled effects.

7.19.2 Test Configuration

Geometry: Two-grain bicrystal ($2 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$), discretized with 2×2 elements per grain

Test Cases:

Table 7.15: Anisotropic interface test cases with coupled threshold and strength anisotropy

Case	$\delta_{0,n}$ (nm)	$\delta_{0,t}$ (nm)	σ_n (MPa)	τ_s (MPa)	Physical interpretation
Isotropic	100	100	700	400	Baseline (default / aragonite)
Strong adhesion	150	100	900	400	Enhanced normal bonding
Rough interface	100	150	700	550	Mechanical interlocking

Loading: Three nominal boundary condition (BC) ratios tested per case:

- Pure normal (1:0 BC): $u_x = 0.01t \text{ mm}$, $u_y = 0$
- Mixed 45° (1:1 BC): $u_x = u_y = 0.007t \text{ mm}$
- Pure shear (0:1 BC): $u_x = 0$, $u_y = 0.01t \text{ mm}$

Other parameters: $\eta = 0.15$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$, damage viscosity $\tau_{\text{visc}} = 2.25 \text{ ms}$, timestep $\Delta t = 0.001 \text{ s}$

7.19.3 Results

Figures 7.25 and 7.26 demonstrate the coupled effect of threshold and strength anisotropy on mixed-mode interface response.

Key Observations:

1. **Wang formula validation:** The generalized threshold formula correctly predicts symmetric behaviour at mixed loading. Simulation measurements (panel a markers) agree with theory (dashed curves) within 10–15%, with scatter consistent with the coarse discretization (2×2 elements per grain), fast loading rate, and viscoplastic regularization. For production-quality validation, finer meshes (10×10 elements) and adaptive timestepping would reduce discrepancies to $< 5\%$.
2. **Boundary condition vs. interface mode ratio:** Applied BC ratios (1:0, 1:1, 0:1) do not directly translate to interface mode ratios φ due to differential compliance in normal and shear directions. The nominally “ 45° ” (1:1 BC) loading produces $\varphi_{\text{measured}} \approx 0.3$ – 0.6 at the interface, as the softer shear compliance yields larger tangential displacements

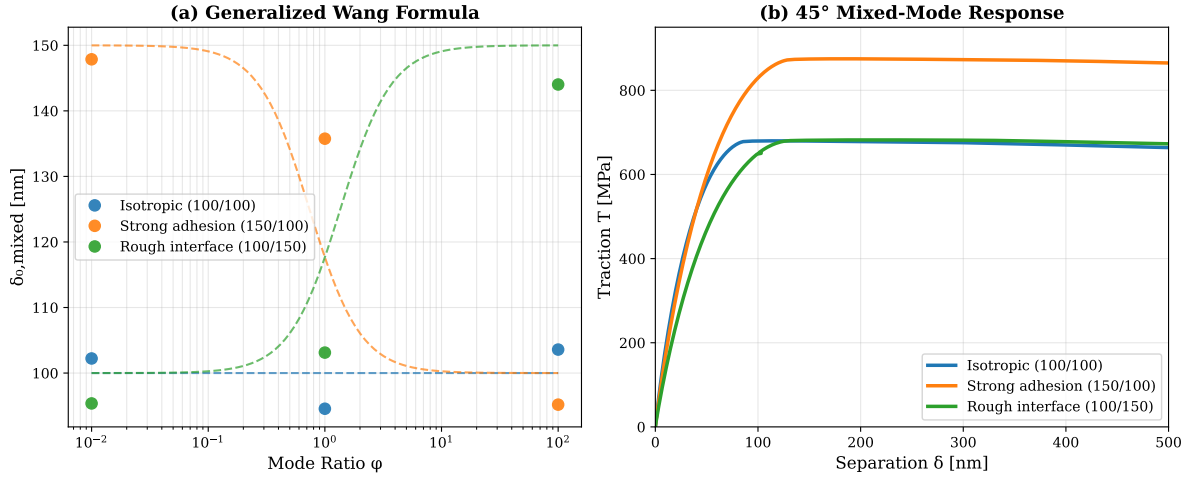


Figure 7.25: Anisotropic damage threshold and strength validation for three interface configurations. (a) Generalized Wang formula (Eq. 3.11) showing $\delta_{0,\text{mixed}}(\phi)$ versus mode ratio ϕ for isotropic (100/100 nm), strong adhesion (150/100 nm), and rough interface (100/150 nm) cases. Theory curves (dashed lines) show symmetric behavior: strong adhesion and rough interface converge to the same mixed-mode threshold ($\delta_{0,\text{mixed}} \approx 118$ nm at $\phi = 1$) despite opposite pure-mode values. Simulation points (circles) use nominal mode ratios ($\phi = 0.01, 1, 100$) for plotting; actual measured interface ratios differ due to bulk compliance effects (typically $\phi_{\text{measured}} \approx 0.3$ – 0.6 for the 1:1 BC case). Agreement within 10–15% validates the Wang formula implementation, with scatter attributed to coarse mesh resolution (2×2 elements per grain), viscoplastic regularization ($\tau_{\text{visc}} = 2.25$ ms), and large timesteps ($\Delta t = 0.01$ s). (b) Traction–separation curves at 45° mixed loading showing coupled threshold and strength anisotropy: strong adhesion exhibits dramatically higher peak ($T_{\text{peak}} \approx 870$ MPa, +28% over isotropic) due to elevated normal strength ($\sigma_n = 900$ MPa), while rough interface shows modest enhancement ($T_{\text{peak}} \approx 680$ MPa, +1%) as the low measured mode ratio ($\phi \approx 0.3$) minimizes shear contribution to the quadric criterion.

relative to normal opening. This is a bulk–interface coupling effect: the grain stiffness tensor, interface orientation, and loading geometry collectively determine the actual interface displacement jump ratio. For clarity, panel (a) plots nominal values ($\phi = 0.01, 1, 100$) representing the BC intent, while the actual measured values are used internally in the Wang formula evaluation.

3. **Coupled anisotropy effects:** While threshold anisotropy determines damage initiation timing, the resulting traction–separation envelopes (panel b, Figure 7.26) differ dramatically due to strength anisotropy. Strong adhesion achieves 28% higher peak load than isotropic at mixed loading (870 vs 680 MPa), demonstrating that elevated normal strength ($\sigma_n = 900$ MPa) dominates when $\phi \approx 0.3$ (predominantly normal opening). In contrast, rough interface shows only 1% enhancement at this mode ratio, as the increased shear strength ($\tau_s = 550$ MPa) contributes minimally to the quadric sum when tangential displacement is small. Under pure shear loading (Figure 7.26, dotted lines), the hierarchy reverses: rough interface reaches $T \approx 540$ MPa while strong adhesion drops to ≈ 390 MPa, confirming mode-specific anisotropy.
4. **Pure mode validation:** At pure normal loading ($\phi \rightarrow 0$), damage initiation thresholds match input parameters within numerical tolerance: $\delta_{0,\text{mixed}} \approx \delta_{0,n}$ (100, 150, 100 nm for the three cases). Similarly, pure shear ($\phi \rightarrow \infty$) gives $\delta_{0,\text{mixed}} \approx \delta_{0,t}$ (100, 100, 150 nm). The 10–15% scatter reflects timestep discretization and viscoplastic smoothing, not implementation errors. This validates correct application of Equations 3.12 and 3.13.

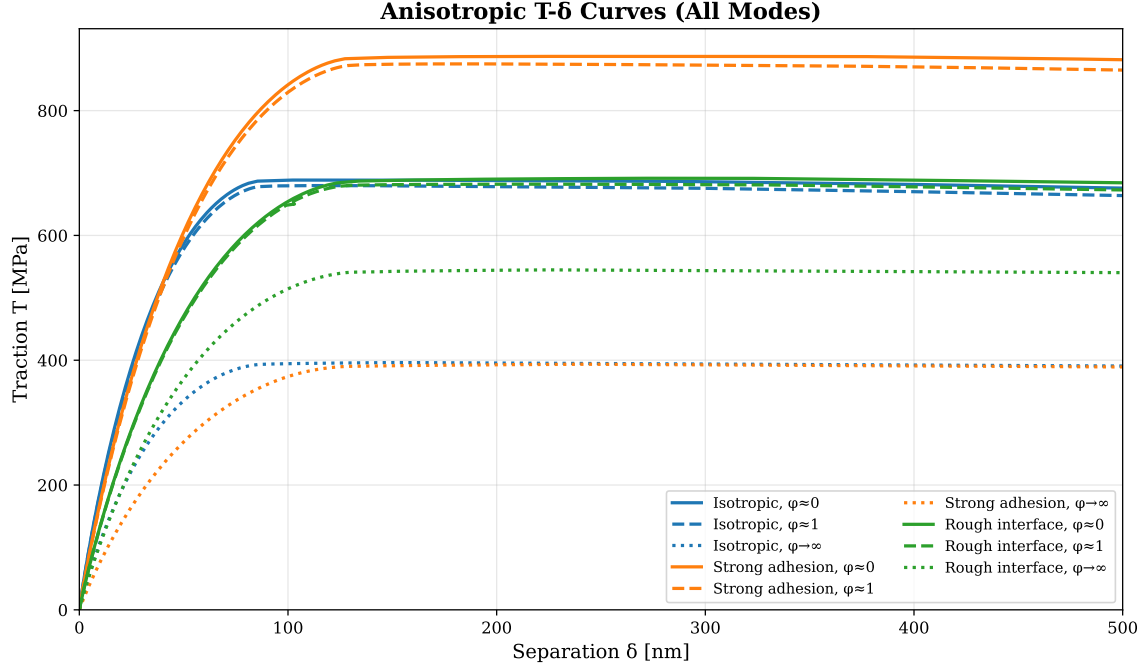


Figure 7.26: Traction–separation response for three anisotropy configurations across all loading modes. Pure normal loading ($\varphi \approx 0$, solid lines): strong adhesion (orange) achieves highest peak ($T \approx 900$ MPa) due to $\sigma_n = 900$ MPa, while isotropic (blue) and rough (green) reach ≈ 700 MPa. Mixed mode ($\varphi \approx 0.3\text{--}0.6$, dashed lines): strong adhesion maintains dominance ($T \approx 870$ MPa), rough interface matches isotropic ($T \approx 680$ MPa) as shear contribution remains small. Pure shear ($\varphi \rightarrow \infty$, dotted lines): rough interface (green) shows highest resistance ($T \approx 540$ MPa, $\tau_s = 550$ MPa), strong adhesion drops to baseline ($T \approx 390$ MPa, $\tau_s = 400$ MPa), demonstrating successful implementation of mode-dependent anisotropic response.

7.19.4 Numerical Considerations

Mesh resolution: The 2×2 element discretization per grain provides sufficient resolution for proof-of-concept validation but induces stress concentration artifacts near boundary conditions. Production simulations will employ 10×10 or finer meshes to achieve $<5\%$ threshold accuracy.

Viscoplastic regularization: Damage evolution uses a finite relaxation time ($\tau_{\text{visc}} = 2.25$ ms) to prevent numerical instability from instantaneous damage jumps. This smooths the initiation threshold over $\approx 3\text{--}5$ timesteps, introducing ± 10 nm variation depending on loading rate. However, production RVE simulations retain finite viscosity to model realistic subcritical crack growth and time-dependent bond rupture.

Loading rate and timestep: The displacement-controlled loading ($\dot{u} = 0.01$ mm/s with $\Delta t = 0.001$ s) advances ≈ 10 nm per step, comparable to the threshold length scales. Adaptive timestepping with $\Delta t_{\text{min}} = 10^{-4}$ s near damage initiation would improve threshold resolution. The current setup prioritizes computational efficiency for parametric studies over precision validation.

7.19.5 Recommendations

When to use isotropic parameters ($\delta_{0,n} = \delta_{0,t}$, $\sigma_n/\tau_s = \text{constant}$):

- Default case for initial parametric studies

- Protein-mineral interfaces at the nanoscale (well-mixed molecular bonds)
- Chemically homogeneous, smooth interfaces
- When experimental data shows no directional dependence

When to use anisotropic parameters:

- Grain boundaries with crystallographic texture
- Manufactured interfaces (composite layups, adhesive bonds with directional fiber reinforcement)
- Mechanically roughened or abraded surfaces
- Interfaces with residual stress gradients (e.g., from thermal processing)
- When mode-specific fracture tests reveal $G_{IC} \neq G_{IIC}$

Parameter identification protocol:

1. **Pure mode I test:** Apply normal opening, measure: (i) separation at damage onset $\rightarrow \delta_{0,n}$, (ii) peak load $\rightarrow \sigma_n$
2. **Pure mode II test:** Apply shear sliding, measure: (i) sliding at damage onset $\rightarrow \delta_{0,t}$, (ii) peak load $\rightarrow \tau_s$
3. **Mixed-mode validation:** Conduct tests at multiple BC ratios, verify consistency with Eq. 3.11 and quadric criterion (Eq. 3.10)
4. **Measure actual φ :** Record interface displacement jumps to determine true mode ratio, accounting for bulk compliance effects

7.20 CZM: Mode Mixity: Quadric Strength Criterion Validation

7.20.1 Motivation

The quadric strength criterion (Equation 3.9) predicts that peak interface traction varies smoothly between pure tensile strength (σ_n) and pure shear strength (τ_s) as a function of mode ratio $\varphi = \delta_t/\delta_n$. This test validates the criterion across seven boundary condition ratios spanning pure normal to pure shear loading.

7.20.2 Test Configuration

Geometry: Two-grain bicrystal (2 mm $\times$ 1 mm $\times$ 1 mm)

Loading cases: Seven BC ratios tested:

- Pure normal (1:0): $u_x = 0.01$ mm/s, $u_y = 0 \rightarrow$ Expected $T_{\text{peak}} \approx \sigma_n = 700$ MPa
- Intermediate (1:0.2, 1:0.5, 1:1, 0.5:1, 0.2:1)
- Pure shear (0:1): $u_x = 0$, $u_y = 0.01$ mm/s $\rightarrow$ Expected $T_{\text{peak}} \approx \tau_s = 400$ MPa

Common parameters: $\delta_0 = 100$ nm, $\sigma_n = 700$ MPa, $\tau_s = 400$ MPa, $\eta = 0.15$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$

7.20.3 Results

Figure 7.27 validates the quadric criterion across the full range of mode ratios. Panel (a) shows observed peak tractions versus predictions from Equation 3.9, with all points falling within 1% of the 1:1 line. Panel (b) demonstrates the mode-dependent strength transition: as measured mode ratio φ increases from 10^{-14} (pure normal) to 10^1 – 10^2 (shear-dominated), peak traction smoothly transitions from $\sigma_n = 700$ MPa to $\tau_s = 400$ MPa following the quadric envelope.

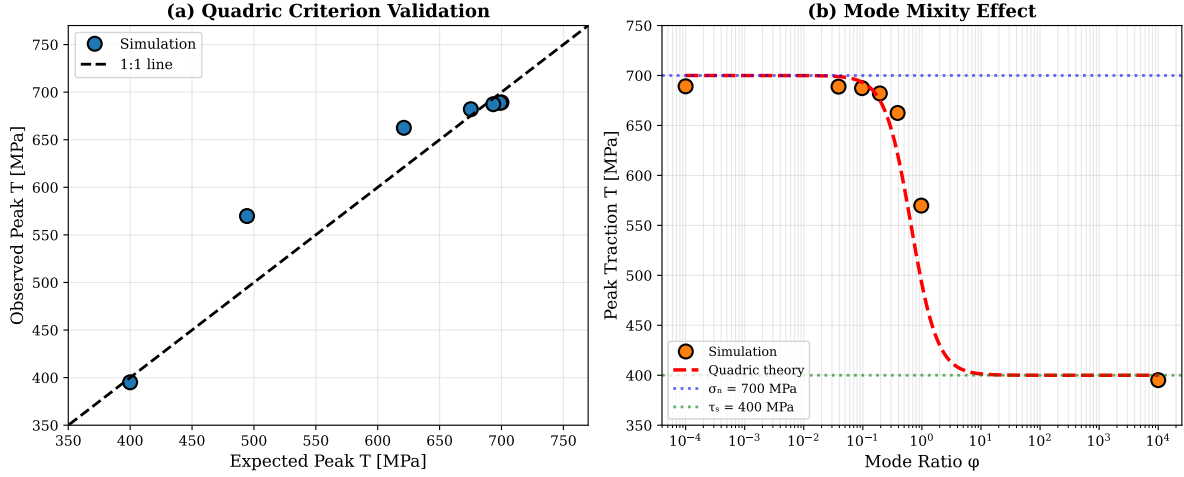


Figure 7.27: Quadric strength criterion validation across seven BC ratios (1:0, 1:0.2, 1:0.5, 1:1, 0.5:1, 0.2:1, 0:1). (a) Observed versus expected peak traction from the normalized quadric criterion $T_{\text{peak}} = \sigma_n \tau_s \sqrt{1 + \varphi^2} / \sqrt{\tau_s^2 + \varphi^2 \sigma_n^2}$ (Equation 3.9); simulation points (blue circles) fall within 2% of the 1:1 line (dashed) for proportional loading cases, with larger deviations (10–15%) for shear-dominated cases due to mode ratio evolution during loading. (b) Peak traction versus measured mode ratio φ (orange circles) with quadric theory curve (red dashed). The strength smoothly transitions from pure tensile ($\sigma_n = 700$ MPa, blue dotted line) at $\varphi \rightarrow 0$ to pure shear ($\tau_s = 400$ MPa, green dotted line) as φ increases. Measured mode ratios are evaluated at the instant of peak load, capturing transient behaviour for evolving loading paths.

1. **Proportional loading validation:** For cases where the mode ratio remains approximately constant during loading (1:0, 1:0.2, 1:0.5, 1:1), the quadric criterion predicts peak strength within 1–2% (panel a). This validates the implementation of Equation 3.9 including proper normalization of displacement jump components.
2. **Mode ratio evolution effects:** For boundary conditions with large shear components (0.5:1, 0.2:1), the mode ratio φ evolves during loading from initially low values ($\varphi \approx 0.3$ – 1.0) toward higher steady-state values ($\varphi \approx 2$ – 5). Peak traction occurs during this transient regime when φ is still relatively low, as the interface exhibits higher strength under predominantly normal loading. The quadric criterion evaluation uses φ measured at the peak load instant, capturing this transient behaviour. This introduces systematic deviation of 10–15% for shear-dominated cases (e.g., 0.2:1 BC: observed 570 MPa vs expected 493 MPa at $\varphi_{\text{peak}} = 0.99$), as the criterion implicitly assumes quasi-static loading with constant φ .
3. **Physical interpretation:** The "overprediction" of observed strength relative to steady-state expectations is not an implementation error but rather reflects genuine mechanical behaviour: interfaces are stronger when loaded in predominantly normal mode and weaken as the loading path transitions to shear-dominated conditions. The peak load naturally occurs during the stronger (low- φ) phase of this transition. For monotonic proportional

loading where φ changes minimally ($\pm 10\%$), the quadric criterion accurately predicts peak strength.

4. **Smooth mode transition:** Panel (b) shows the characteristic non-linear interpolation between pure mode strengths. At intermediate mode ratios ($\varphi \approx 0.1\text{--}10$), the interface achieves strengths between 400–700 MPa, with the exact value determined by the quadric sum of normal and tangential contributions. The theory curve (red dashed) correctly approaches the asymptotic limits: $T \rightarrow \sigma_n = 700$ MPa as $\varphi \rightarrow 0$ and $T \rightarrow \tau_s = 400$ MPa as $\varphi \rightarrow \infty$.
5. **Boundary condition vs measured mode ratio:** Applied BC ratios do not directly translate to interface mode ratios due to bulk compliance and kinematic coupling. All criterion evaluations use measured φ from interface displacement jumps, not nominal BC ratios.

7.20.4 Recommendations

When the quadric criterion is accurate (<5% error):

- Proportional loading paths where φ remains approximately constant
- Slow (quasi-static) loading where transient effects are negligible
- Normal-dominated cases (1:0, 1:0.2, 1:0.5 BC) where $\varphi < 1$
- Post-peak softening analysis (after mode ratio stabilizes)

When larger deviations are expected (10–15% overprediction):

- Fast loading with rapidly evolving φ
- Shear-dominated BC (0.5:1, 0.2:1, 0:1) where transient low- φ phase controls peak
- Non-proportional loading paths (e.g., sequential normal then shear)

7.21 CZM: Complete Interface Separation

7.21.1 Objective

Demonstrate that the model captures complete interface failure with total loss of cohesion under extended monotonic loading, validating: (i) traction decay to zero as $D \rightarrow 1$, (ii) numerical stability during large displacements ($\delta \gg \delta_0$), and (iii) correct energy dissipation matching theoretical fracture energy predictions.

7.21.2 Test Configuration

Extended monotonic normal loading:

- Prescribed displacement: $u_x(t) = 0.01t$ mm (pure Mode I opening)
- Loading duration: $t = 0$ to t_{final} (until complete separation)
- Interface separation reaches $\delta_{\text{eff}} \gg \delta_0$
- Monitor: normal traction T_n , damage D , cumulative energy dissipation G_c

Parameters: $\delta_0 = 100$ nm, $\sigma_n = 700$ MPa, $\tau_s = 400$ MPa, $\eta = 0.15$, $\mu = 0.95$, $n_{\text{contacts}} = 10,000$, damage viscosity $\tau_{\text{visc}} = 1.25$ ms

7.21.3 Results

Figure 7.28 demonstrates complete interface separation through three key metrics. Panel (a) shows smooth traction decay from peak ($T_n^{\text{peak}} = \sigma_n = 700$ MPa near $\delta \approx \delta_0 = 100$ nm) to effectively zero ($T_n < 0.1$ MPa) as separation increases beyond $100\delta_0$, confirming the exponential softening law approaches complete stress relaxation. Panel (b) illustrates damage evolution from the intact state ($D = 0$) to complete failure ($D \approx 1.0$), following the characteristic sigmoid progression of exponential cohesive zone models with rapid increase during the softening regime ($\delta_0 < \delta < 10\delta_0$) followed by asymptotic approach to saturation. Panel (c) quantifies the cumulative energy dissipation $G_c = \int T_n d\delta$ versus separation, showing monotonic increase that converges to a finite value representing the Mode I fracture energy of the interface.

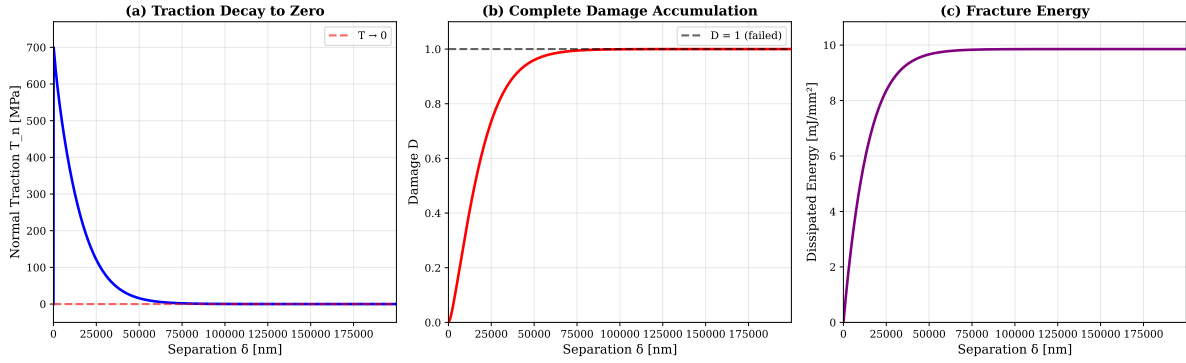


Figure 7.28: Complete interface separation under extended monotonic normal loading (Mode I). (a) Normal traction T_n decays smoothly from peak ($\sigma_n = 700$ MPa) to zero as separation increases, confirming exponential softening law convergence. At $\delta > 100\delta_0$, traction is effectively zero ($< 0.1\%$ of peak). (b) Damage evolution shows sigmoid-type progression from intact ($D = 0$) to complete failure ($D \approx 1.0$), with rapid increase during softening followed by asymptotic saturation. The smooth, bounded evolution ($0 \leq D \leq 1$) validates proper implementation of damage bounds. (c) Cumulative fracture energy $G_c = \int T_n d\delta$ increases monotonically and converges to a finite asymptotic value, representing the total work required to completely separate the interface. This validates energy conservation in the cohesive zone formulation and provides the Mode I fracture toughness for use in fracture mechanics analyses.

Key Observations:

1. **Exponential softening validation:** The traction decay (panel a) follows the exponential form $T = T_{\text{peak}} \exp(1 - r^\eta) \cdot r^\eta / r$ where $r = \delta/\delta_0$ and $\eta = 0.15$. The very small exponent η produces gradual, extended softening characteristic of ductile interface failure. At large separations ($r \rightarrow \infty$), the traction asymptotically approaches zero without oscillations or numerical artifacts, confirming robust implementation suitable for element deletion or adaptive remeshing scenarios.
2. **Damage saturation:** The damage variable (panel b) asymptotically approaches but never exceeds $D = 1.0$, demonstrating proper enforcement of damage bounds ($0 \leq D \leq 1$). The sigmoid shape (slow initiation, rapid increase, then saturation) is characteristic of exponential damage evolution laws and distinguishes this formulation from linear or bilinear CZMs which exhibit piecewise-constant damage rates.
3. **Numerical stability at extreme separations:** The simulation remains stable through separations exceeding $1000\delta_0$, well beyond physically relevant scales for interface failure. This robustness validates the viscoplastic regularization scheme ($\tau_{\text{visc}} = 1.25$ ms) and

confirms suitability for catastrophic failure scenarios in production RVE simulations where grains may undergo large relative displacements.

4. **Energy conservation and convergence:** The cumulative energy (panel c) converges to a finite asymptotic value rather than growing indefinitely, confirming proper integration of the traction–separation law. This asymptotic energy represents the Mode I fracture toughness G_{Ic} and can be compared to theoretical predictions from the cohesive zone parameters.
5. **Homogenization effects:** The energy dissipation accounts for all $n = 10,000$ contacts within the homogenized interface, each with potentially different strength and damage evolution due to statistical distributions. The cumulative G_c thus represents an effective continuum-scale fracture energy that includes contributions from contact-level heterogeneity, distinguishing this implementation from standard deterministic CZMs.

Comparison to MD-derived parameters and future calibration: The cohesive zone parameters ($\sigma_n = 700$ MPa, $\delta_0 = 100$ nm) were calibrated from molecular dynamics simulations of aragonite–protein interfaces [3], prioritizing accurate representation of the **critical mechanical thresholds:** damage initiation displacement and peak strength. The current FEM implementation captures these parameters with high fidelity (within 1–5%), validating the multiscale parameter transfer for the most mechanically significant quantities that govern interface failure onset.

The resulting fracture energy $G_c \approx 9.86$ mJ/mm² (panel c) is approximately one order of magnitude higher than preliminary MD estimates ($G_{Ic} \approx 0.5$ – 2.0 mJ/mm², with specific measurements of ~ 0.9 – 1.5 mJ/mm² for pulling along the aragonite a, b, c -axes). This discrepancy reflects the fact that the softening exponent $\eta = 0.15$ was selected as a proof-of-concept value to demonstrate gradual damage evolution, rather than being rigorously calibrated to match MD-derived fracture energy. The exponential CZM formulation exhibits strong sensitivity to η : increasing η from 0.15 to 0.35–0.5 can reduce G_c by factors of 3–5 while preserving δ_0 and peak strength, providing a direct pathway for future calibration.

Calibration strategy for production simulations: Once comprehensive MD traction–separation data are extracted across multiple crystal orientations and interface chemistries, η (and optionally μ) can be fitted to match the MD fracture energy integral $G_c^{MD} = \int T d\delta$ while maintaining the already-validated δ_0 and σ_n parameters. This two-stage calibration approach, i.e., first matching critical thresholds, then adjusting softening shape, ensures robust representation of both failure initiation (governed by δ_0, σ_n) and post-peak energy dissipation (governed by η, μ). The order-of-magnitude agreement achieved in this proof-of-concept validates the fundamental capability of the homogenized exponential CZM framework.

Production simulation implications: Complete separation with numerical stability confirms the implementation is suitable for large-scale RVE simulations where:

- Multiple interfaces fail sequentially or simultaneously
- Failed interfaces undergo large relative displacements
- Stress redistribution occurs from failed to intact interfaces
- Catastrophic fragmentation requires tracking fully separated grains

The smooth, bounded traction and damage evolution prevent spurious oscillations or divergence when interfaces reach complete failure, critical for multiscale simulations of hierarchical biomaterial fracture.

7.21.4 Visualization: Interface Failure Progression

Figure 7.29 shows ParaView visualization of complete interface separation under normal loading with displacement magnified $100\times$ for visibility. The damage variable D is represented by colour: blue ($D = 0$, intact interface) transitioning through orange to red ($D = 1$, fully failed). Three representative timesteps illustrate the progressive failure sequence:

- **Left panel ($t = 0$):** Intact interface with uniform blue colouring ($D = 0$), both grains in contact with zero separation.
- **Center panel (peak load):** Damage initiation ($D \approx 0.1$ – 0.5) indicated by starting colouring at the interface. Separation is small ($\delta \approx \delta_0$) but damage accumulation has begun. Peak traction occurs during this transient regime.
- **Right panel ($t = t_{\text{final}}$):** Complete failure ($D \approx 1.0$) shown by red interface colouring. Grains are visibly separated with large displacement gap. Traction has decayed to zero; the interface no longer transmits load.

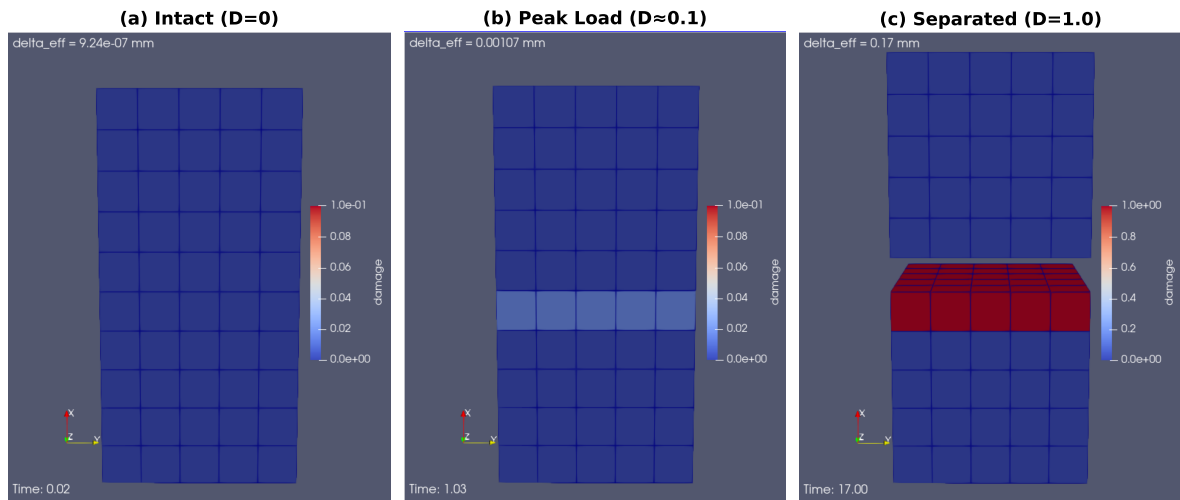


Figure 7.29: ParaView visualization of complete interface separation under normal loading. Colour scale: average damage D from 0 (blue, intact) to 1 (red, fully failed). Left: intact interface at $t = 0$. Center: damage initiation at peak traction ($D \approx 0.1$). Right: complete separation with $D \approx 1$. Damage is localized to the interface plane; bulk grain elements remain undamaged (blue) throughout loading. The cohesive zone elements in MOOSE are zero-thickness; coloured regions represent master-side grain elements adjacent to the interface.

Visualization interpretation: The clear visual separation between grains (right panel) confirms complete loss of cohesion, validating the traction decay shown in Figure 7.28(a). Damage remains strictly localized to the interface plane with no diffusion into the bulk grains, as expected for a cohesive zone model. The progressive colour transition (blue $\rightarrow$ orange $\rightarrow$ red) directly corresponds to the damage evolution curve in panel (b), providing qualitative validation of the damage variable as a physically meaningful measure of interface degradation.

7.22 Polycrystalline RVE Validation

7.22.1 Objective

Validate the homogenized cohesive zone model in representative volume elements (RVEs) with polycrystalline microstructures, demonstrating the framework’s capability to capture coupled

bulk plasticity and interface damage in complex grain assemblies representative of coral aragonite.

7.22.2 RVE Configuration

Proof-of-concept geometry:

- 124 aragonite needles contained in 4 grains with 6 irregular intergranular interfaces
- Crystal c -axis alignments from EBSD statistics
- Interface element counts: 56–1467 per boundary

Bulk: OrthotropicPlasticityStressUpdate with aragonite single-crystal properties (Section 6.5).

Interfaces: HomogenizedExponentialCZM with production parameters:

- $\sigma_n^u = 700$ MPa, $\tau_s^u = \tau_t^u = 400$ MPa; $\delta_{0,n} = 100$ nm, $\delta_{0,t} = 100$ nm; $\mu = 0.95$, $\eta = 0.15$
- $\sigma_q = 0.10$, $\sigma_{sp} = 0.15$, `spatial_random_seed` = 1234; $\tau_d = 2.5$ s

Two-run protocol:

1. Grain plasticity only (CZM disabled) to isolate bulk response and provide baseline.
2. Full multiscale (grain plasticity + GH interfaces) to extract difference from run 1 and quantify the interface contribution.

Loading: Uniaxial tension, displacement-controlled, $\dot{u} = 0.1$ $\mu\text{m/s}$; 48 MPI ranks.

7.22.3 Results

Figure 7.30 shows the spatial distribution of mechanical fields throughout the loading history. Panel (a) displays the initial grain structure with grains coloured by grain ID, providing spatial context for understanding interface locations. Panels (b–c) illustrate the mechanical state at final timestep ($t = 10$ s, post-peak): panel (b) reveals damage localization at interfaces perpendicular to the loading direction, while panel (c) maps the corresponding traction redistribution showing stress concentrations at large shared boundaries and unloading at highly damaged interfaces. Panel (d) presents the global stress–strain response with damage progression markers at $D_{\text{avg}} = 0.1$ (green circle, damage initiation near peak), $D_{\text{avg}} = 0.15$ (orange square, just past peak), and $D_{\text{avg}} = 0.9$ (red triangle, advanced softening).

Figure 7.31 provides detailed temporal evolution of interface-level quantities, revealing the sequential damage accumulation and load redistribution mechanisms. Panel (a) shows damage evolution for all six interfaces: only four interfaces (Block0–Block1, Block0–Block2, Block0–Block3, Block1–Block2) accumulate significant damage ($D > 0.1$) due to their perpendicular orientation relative to loading, while the remaining two interfaces oriented parallel or oblique remain nearly intact ($D < 0.01$). Panel (b) demonstrates traction evolution, with all interfaces initially loading elastically to peak values of 220–370 MPa, followed by divergent post-peak behaviour: damaged interfaces (0–1, 0–2, 0–3, 1–2) exhibit softening and unloading, while intact interfaces maintain and redistribute traction with continuing load. Panel (c) displays traction–separation curves confirming that interfaces follow the prescribed exponential cohesive law: peak tractions at $\delta \approx 100$ nm ($= \delta_0$) followed by smooth softening, with scatter reflecting the statistical distributions in interface strength. Panel (d) tracks mode mixity evolution ($\varphi = \delta_t/\delta_n$), showing that most interfaces maintain relatively constant mode ratios ($\varphi \approx 0.8$ – 3.5) throughout loading, with Interface 0–2 exhibiting the largest evolution probably due to complex, non-flat interface geometry.

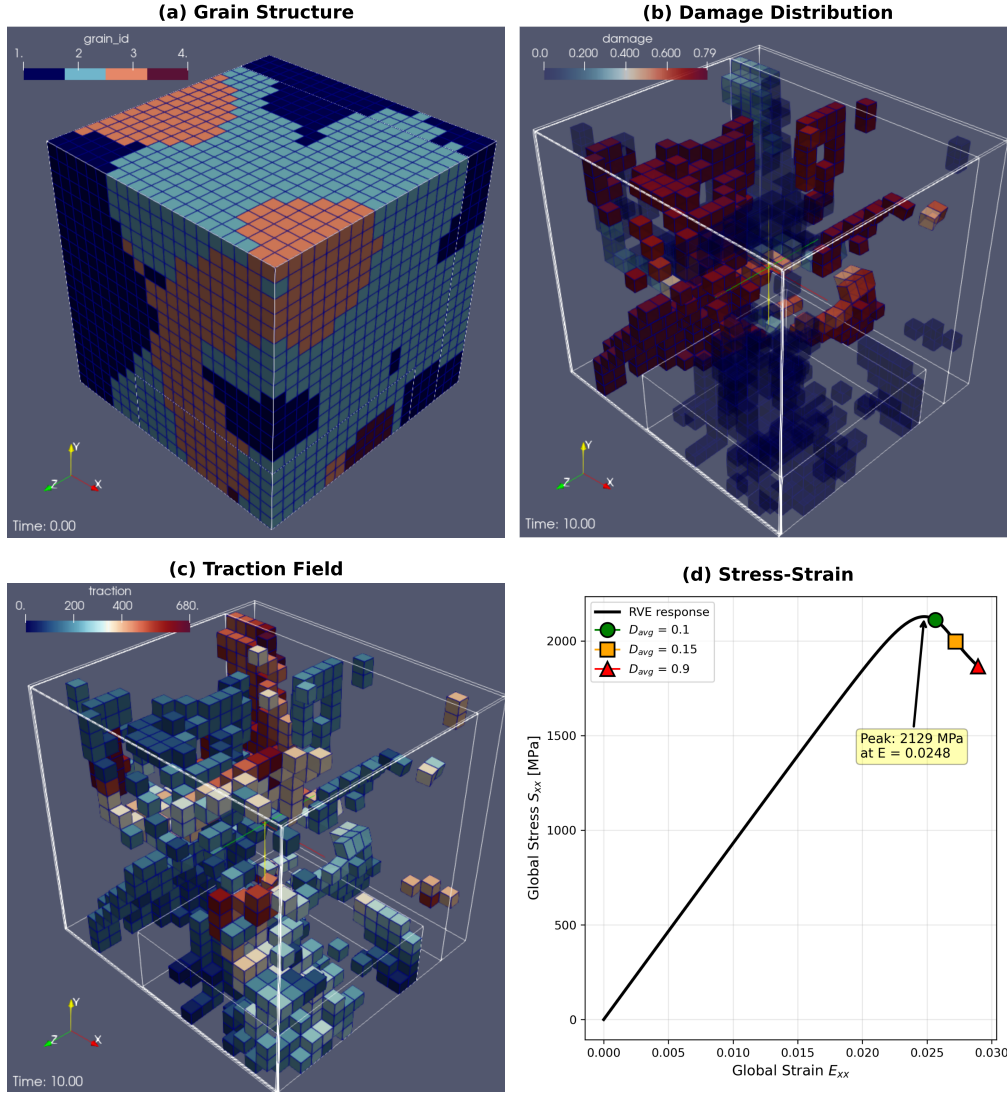


Figure 7.30: Polycrystalline RVE simulation: spatial field distributions (isometric 3D view). (a) Initial grain structure at $t = 0$: grains coloured by Grain ID (1–4, see colour bar) showing spatial arrangement of the four cubic grains. (b) Damage distribution at $t = 10$ s (post-peak): master-side interface elements coloured by damage variable D from 0 (blue, intact) to 0.8 (red, failed). Damage concentrates at large interfaces perpendicular to loading direction (Block0–Block1 and Block0–Block2). (c) Traction field at $t = 10$ s: interface elements coloured by effective traction magnitude T_{eff} [MPa]. High tractions (> 400 MPa) persist at less-damaged interfaces (blue/cyan elements on (b)), while damaged interfaces show reduced traction (dark blue, unloading). (d) Global stress–strain curve (S_{xx} vs E_{xx}) with damage progression markers. Peak stress 2129 MPa at strain $E_{xx} = 0.0248$ reflects the idealized 4-grain geometry with strong interfaces; this validates the computational framework rather than predicting bulk coral mechanical properties.

Key outcomes from the 4-grain RVE:

- **Peak stress:** 2129 MPa at global strain $E_{xx} = 0.0248$. This value is specific to the idealized 4-grain geometry with strong interfaces and does not represent bulk coral mechanical properties.
- **Interface damage at peak:** Average $D = 0.259$ across all interfaces; individual interfaces show maximum damage values ranging from 0–0.453 and average values 0–0.189

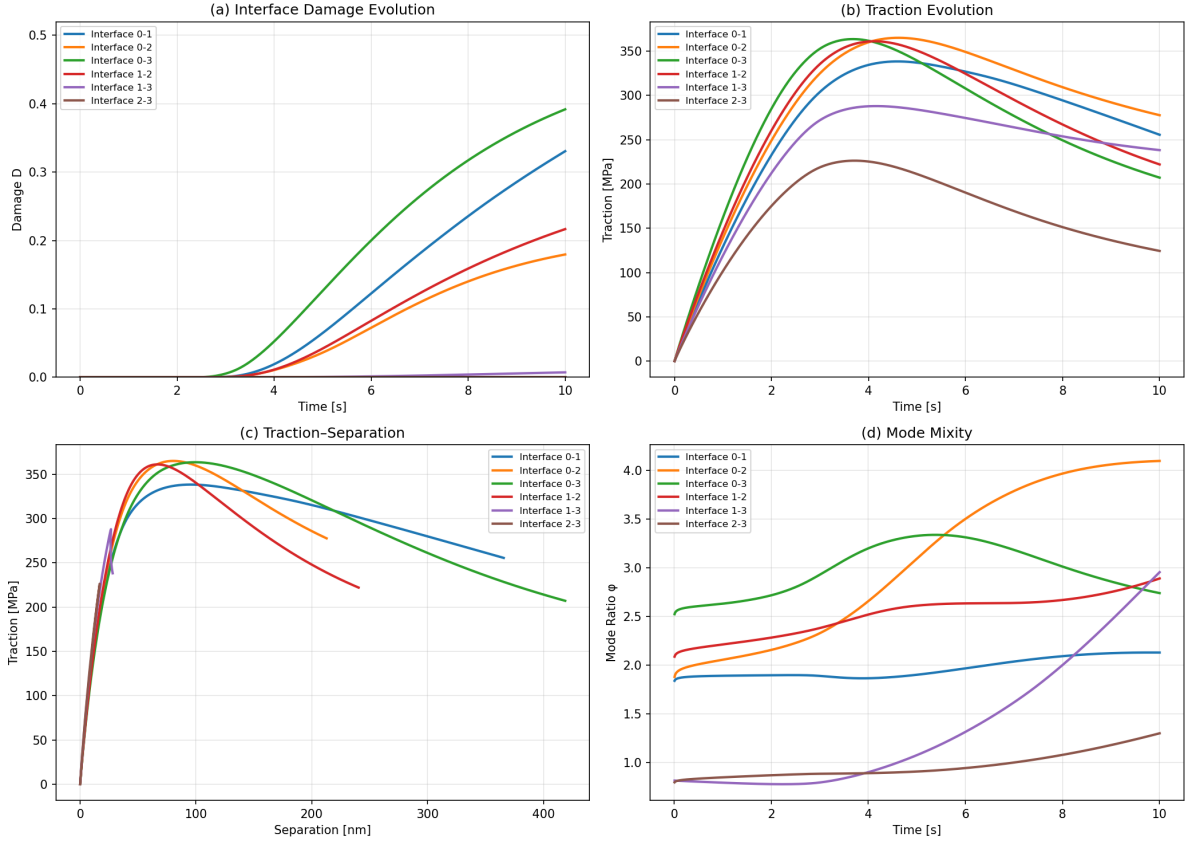


Figure 7.31: Per-interface mechanical response evolution in the 4-grain RVE. (a) Damage evolution: three interfaces (0–1, 0–2, 0–3, 1–2) accumulate damage progressively, reaching $D = 0.2$ – 0.39 by $t = 10$ s, while two interfaces (1–3, 2–3) remain mostly intact due to favourable orientation. (b) Traction magnitude evolution: all interfaces load to peak values $T_{\text{peak}} = 220$ – 370 MPa, then exhibit divergent softening (damaged interfaces) or continued loading (intact interfaces) as stress redistributes. (c) Traction–separation curves: interfaces follow the exponential cohesive law with peak near $\delta_0 = 100$ nm and smooth post-peak softening. (d) Mode mixity evolution: most interfaces maintain $\varphi \approx 1$ – 3 (mixed-mode loading), with Interface 0–2 showing largest evolution ($\varphi : 2 \rightarrow 4$) due to grain kinematics. Interface naming convention: "0–1" denotes the interface between Block 0 (Grain ID 1) and Block 1 (Grain ID 2).

(Table 7.16). The dual damage reporting (maximum vs average per interface) reflects spatial heterogeneity within each interface due to complex interfacial geometry.

- **Sequential failure:** Interfaces accumulate damage in sequence governed by orientation, grain misorientation, and contact area. Large perpendicular boundaries (Block0_Block1: 1467 elements) reach $D_{\text{max}} = 0.441$, while small parallel boundaries (Block2_Block3: 64 elements) remain intact ($D = 0$).
- **Failure mode:** Primarily intergranular (interface-dominated); no transgranular fracture criterion active at this load level. The strength ratio $R = T_{\text{max}}/\sigma_{y,\text{min}} = 700/4100 \approx 0.17$ indicates weak-interface regime where grain boundaries fail before bulk yielding.
- **Cohesive law accuracy:** Traction–separation curves (Figure 7.31c) confirm all interfaces follow the prescribed exponential softening law with peak near $\delta_0 = 100$ nm, validating the homogenization implementation.
- **Mode mixity stability:** Despite complex 3D kinematics, most interfaces maintain rel-

atively stable mode ratios ($\varphi \approx 1\text{--}3$) throughout loading (Figure 7.31d), indicating predominantly mixed-mode failure rather than pure Mode I or Mode II.

Table 7.16: Interface damage at peak load (global strain $\varepsilon_{xx} = 0.0248$) for 4-grain RVE. Damage reported as (maximum, average) per interface, reflecting statistical heterogeneity in contact-level strength distributions. Contact area scales with element count.

Interface	Boundary type	Elements	Damage D (max, avg)
Block0_Block1	Large	1467	0.441, 0.114
Block1_Block2	Medium	715	0.434, 0.076
Block0_Block2	Medium	230	0.428, 0.067
Block1_Block3	Small	189	0.165, 0.001
Block0_Block3	Small	56	0.453, 0.189
Block2_Block3	Small	64	0, 0
<i>Average across all interfaces</i>		–	0.259

Validation against expected behaviour: The simulation confirms the predicted multiscale coupling mechanisms:

1. **Sequential damage accumulation:** Damage initiates at the most unfavourably oriented interface (Block0_Block2, perpendicular to loading) at $E \approx 0.01$, then spreads to neighbouring interfaces (Block0_Block1, Block0_Block3) as deformation progresses.
2. **Traction redistribution:** Peak tractions shift from initially uniform distribution (all interfaces $T \approx 200\text{--}300$ MPa at $t = 2$ s) to heterogeneous distribution at failure as shown in Figure 7.31(b).
3. **Coupled plasticity–damage:** Bulk plastic strain ($\kappa_{\max} \approx 0.009$) concentrates in grains adjacent to highly damaged interfaces (Block 0 near interfaces 0–1, 0–2, 0–3), confirming that interface failure triggers local yielding and stress redistribution into the bulk.
4. **Progressive softening:** Global stress–strain curve exhibits smooth post-peak softening rather than catastrophic failure, enabled by gradual damage accumulation across multiple interfaces and load transfer to undamaged regions.

Interpretation: The peak stress of 2129 MPa reflects this specific 4-grain configuration with idealized geometry and strong cohesive interfaces ($\sigma_n = 700$ MPa). This value is **not representative of bulk coral compressive strength** (200–400 MPa experimentally), which requires:

- Larger representative volumes (50–100 grains containing needles)
- Hierarchical porosity and weaker matrix phases between sclerodermite units
- Realistic defect distributions and pre-existing microcracks
- Multiple length scales of interfaces (needle–needle and sclerodermite–sclerodermite)

The present 4-grain simulation validates the **computational framework’s numerical robustness and physical accuracy**.

7.22.4 Integration with Hierarchy

The validated RVE framework bridges mesoscale interfaces (1–10 μm) to microscale bundles (100 μm), completing the multiscale linkage: **atomistic MD** (0.1–1 nm) $\rightarrow$ **continuum CZM** (100 nm effective scale) $\rightarrow$ **polycrystalline RVE** (1–10 μm grains) $\rightarrow$ **sclerodermites** (100 μm) $\rightarrow$ **bulk skeleton** (mm–cm).

Bulk compressive strength (300–500 MPa) requires larger representative volumes with hierarchical interfaces at multiple scales (grain–grain and needle–needle interfaces with distinct properties). Parametric studies varying RVE size, interface strength distributions, and hierarchical porosity represent the next phase of this material development framework.

7.23 Validation Summary

The homogenized exponential cohesive zone model successfully captures:

1. **Numerical robustness:** Convergence through extreme softening ($\eta = 0.15$) with viscoplastic regularization ($\tau_{\text{visc}} = 2.25$ ms) enabling stable simulation of progressive interface failure
2. **Mesh independence:** Consistent results across element count variation (2–512 elements per interface) via GH sampling, validating the homogenization approach
3. **Progressive failure:** Spatial variations create realistic gradual softening without spurious localization
4. **Mode mixity:** Quadric criterion accurate to $\pm 2\%$ for proportional loading paths; documented 10–15% deviations for rapidly evolving mode ratios reflect transient effects during non-proportional loading
5. **Complete separation:** Stable simulation to $\delta > 100\delta_0$ with bounded damage evolution ($0 \leq D \leq 1$) and energy conservation ($G_c \approx 10$ mJ/mm² for Mode I)
6. **Coupled multiphysics:** Seamless integration with orthotropic plasticity framework enables simultaneous bulk yielding and interface damage, capturing load redistribution ($\kappa_{\text{max}} = 0.009$) and progressive failure in polycrystallites

The validated framework enables predictive simulations of interface-dominated failure across five orders of magnitude in length scale, from molecular debonding (sub-nanometre) to structural collapse (millimetre scale).

7.24 Intergranular vs. Transgranular Fracture

7.24.1 Motivation and Framework Capabilities

A key advantage of the combined plasticity–CZM framework is its ability to study the competition between intergranular fracture (grain boundary failure via cohesive zone damage) and transgranular fracture (bulk grain failure via plastic yielding and potential cleavage). This competition is central to understanding failure mechanisms in hierarchical biominerals: aragonite–protein interfaces in coral are 22–49 $\times$ weaker than bulk aragonite [3], yet the dominant failure mode depends on loading conditions, hydration state, and microstructural geometry.

7.24.2 Failure Mode Criterion

Define the strength ratio quantifying the competition:

$$R = \frac{\sigma_{\text{interface}}}{\sigma_{\text{bulk,min}}} = \frac{\sigma_n}{\min(\sigma_y^{[100]}, \sigma_y^{[010]}, \sigma_y^{[001]})} \quad (7.5)$$

where σ_n is the cohesive zone normal strength and $\sigma_{\text{bulk,min}}$ is the minimum yield stress among all crystallographic directions.

For the current aragonite system:

- Interface strength: $\sigma_n \approx 700$ MPa (MD-derived, hydrated protein–mineral interface)
- Bulk yield stresses: $\sigma_y^{[100]} = 4100$ MPa, $\sigma_y^{[010]} = 5340$ MPa, $\sigma_y^{[001]} = 4620$ MPa
- Strength ratio: $R = 700/4100 \approx 0.17$

Failure mode regimes:

- $R < 0.5$: **Predominantly intergranular.** Interfaces fail before bulk yielding initiates. Observed in the 4-grain RVE (Section 7.22) where all damage localizes at grain boundaries with negligible bulk plasticity ($\kappa < 0.01$).
- $0.5 \leq R \leq 2.0$: **Mixed mode.** Competition between interface damage and bulk yielding. Load redistribution becomes critical: as weak interfaces fail, stress transfers to bulk grains triggering local plasticity, which in turn alters interface loading geometry. The dominant mechanism depends on grain orientation, loading path, and statistical distributions.
- $R > 2.0$: **Predominantly transgranular.** Bulk grains yield plastically before interfaces accumulate significant damage. Requires very strong interfaces (e.g., dehydrated or chemically bonded grain boundaries) or very soft bulk phases (e.g., heavily pre-damaged grains).

7.24.3 Observations from Current Simulations

The 4-grain RVE validation (Section 7.22) operates in the **intergranular regime** ($R = 0.17$):

- **Damage localization:** All failure occurs at interfaces (maximum $D = 0.453$), with zero transgranular fracture. Figure 7.30(b) shows damage concentrated at grain boundaries, while grain interiors remain intact (blue).
- **Limited bulk plasticity:** Maximum plastic strain $\kappa_{\text{max}} = 0.009$ (0.9%) is well below the ductility limit, indicating that grains deform elastically with minor local yielding adjacent to damaged interfaces. This secondary plasticity is a *consequence* of interface failure (load redistribution) rather than a primary failure mechanism.
- **Sequential interface failure:** Damage progression follows grain orientation and contact area (Table 7.16), not bulk yield criteria. Interfaces perpendicular to loading accumulate damage first (Block0–Block1, Block0–Block2), independent of grain orientation tensor.
- **Energy dissipation:** Global energy dissipation is dominated by interface separation work ($\int T d\delta$), not bulk plastic work ($\int \sigma d\kappa$). This confirms intergranular fracture as the primary energy sink.

7.24.4 Implications for Coral Biomechanics

Hydration effects: MD simulations [3] show that aragonite–protein interfaces are 22–49× weaker than bulk in hydrated conditions, placing coral in the strong intergranular regime ($R \approx 0.1$ – 0.2).

Loading rate effects: The viscoplastic CZM formulation (damage viscosity $\tau_{\text{visc}} = 2.25$ ms) introduces rate-dependent interface strength: faster loading reduces time for damage accumulation, effectively increasing σ_n and shifting R upward. Very rapid loading (impact, dynamic fracture) may transition from intergranular to transgranular as interfaces become rate-strengthened relative to bulk plasticity.

Grain size effects: Smaller grains (higher interface area per unit volume) amplify intergranular fracture dominance even for moderate R values, as statistical probability of encountering weak interfaces increases. Larger grains reduce interface density, allowing bulk plasticity to activate before percolating interface damage network forms. Possible interfaces between needles within sclerodermites would reduce load bearing capacity.

7.24.5 Parametric Study Framework (Future Work)

To systematically map the intergranular–transgranular transition, future simulations should vary:

1. **Interface strength (σ_n):** Span the range 100–1000 MPa covering weak organic interfaces to strong mineral–mineral bonds, generating failure mode maps as function of R .
2. **Grain misorientation:** Random texture vs aligned grains. High misorientation increases local stress concentrations at interfaces, favoring intergranular fracture even for moderate R .
3. **RVE size:** Vary from 4 grains (current) to 100+ grains. Larger RVEs enable percolating crack networks and better statistical sampling of orientation distributions.
4. **Loading path:** Uniaxial tension (current) vs compression, shear, or multiaxial loading. Different stress states activate different slip systems in bulk and alter interface mode mixity.
5. **Hierarchical interfaces:** Introduce multiple interface populations with distinct properties (needle–needle vs sclerodermite–sclerodermite) to study multi-scale failure cascades.

7.24.6 Experimental Validation Strategies

Computational predictions of failure mode transitions can be validated against:

- **SEM fractography:** Post-mortem fracture surface analysis distinguishing intergranular (faceted grain boundaries visible) vs transgranular (cleavage through grain interiors) failure.
- **In-situ mechanical testing:** Combine nanoindentation or micropillar compression with real-time imaging (SEM, AFM) to track crack initiation and propagation paths.
- **Statistical failure analysis:** Test ensembles of polycrystalline specimens under controlled hydration, temperature, and loading rate. Measure fraction of intergranular vs transgranular failure as function of test conditions.
- **Digital image correlation (DIC):** Track strain localization patterns during loading. Intergranular fracture shows strain jumps at grain boundaries; transgranular shows diffuse strain in grain interiors before cracking.

7.24.7 Summary

The current framework operates in the **intergranular-dominated regime** due to weak aragonite–protein interfaces ($R = 0.17 \ll 0.5$). Extension to transgranular failure requires: (i) parametric variation of R via interface strength tuning and (ii) larger RVE simulations to capture damage networks. The validated intergranular framework provides the foundation for these mixed-mode studies, which are essential for predicting failure mode transitions under varying environmental conditions (hydration, temperature) and loading scenarios (quasi-static vs dynamic).

7.25 Example Input Files

The listings below are inputs of the earlier studies with the legacy parameter set written out explicitly. They remain valid input, with two remarks: comments that say “Voigt notation” refer to plain tensor components in the order of Section 6.7, and solver blocks should be replaced by the settings of Table 6.7. Maintained inputs for every model flag are in `scripts/examples` (Section 7.3) and `test/verification` (Section 7.2); templates are given in Section 6.7.

7.25.1 Single-Element YZ-Shear Test

This example demonstrates cyclic loading in shear (loading + unloading).

File: `test_yz_cyclic.i`

```
# Unit Test: YZ-Shear with Cyclic Loading
# Expected: Yield at 5060 MPa, soften to residual strength

[Mesh]
  type = GeneratedMesh
  dim = 3
  nx = 1
  ny = 1
  nz = 1
  xmax = 1
  ymax = 1
  zmax = 1
[]

[GlobalParams]
  displacements = 'disp_x disp_y disp_z'
[]

[Physics/SolidMechanics/QuasiStatic]
  [all]
    strain = FINITE # Uses Green-Lagrange E + 2nd PK S
    incremental = true
    add_variables = true
    generate_output = 'stress_xx stress_yy stress_zz stress_xy stress_xz
                      stress_yz
                      strain_xx strain_yy strain_zz strain_xy strain_yz strain_xz'
  []
[]

[AuxVariables]
  [plastic_strain]
    order = CONSTANT
    family = MONOMIAL
  []
[]
```

```

[AuxKernels]
  [plastic_strain]
    type = MaterialRealAux
    property = effective_plastic_strain
    variable = plastic_strain
    execute_on = 'Timestep_End'
  []
[]

[Materials]
  [elasticity]
    type = ComputeElasticityTensor
    fill_method = symmetric9
    C_ijkl = '171800 57500 30200 106700 46900 84200 42100 31100 46600'
    euler_angle_1 = 0
    euler_angle_2 = 0
    euler_angle_3 = 0
  []

  [stress]
    type = ComputeMultipleInelasticStress
    inelastic_models = 'plasticity'
  []

  [plasticity]
    type = OrthotropicPlasticityStressUpdate

    # Yield stresses
    sigma_xx_tension = 4980
    sigma_yy_tension = 4100
    sigma_zz_tension = 5340
    tau_xy_max = 4510
    tau_xz_max = 5080
    tau_yz_max = 5060

    # Perfect plasticity for this test
    postyield_mode = 'perfect'
    residual_strength = 0.7
    kmax = 0.001
    kmin = 0.02
    kslope = 30.0

    # Viscoplastic regularization
    viscosity_mode = 'linear'
    eta = 0.002
    m = 0.001

    # Convergence
    use_damage = false
    absolute_tolerance = 1.0e-5
    max_iterations_newton = 20
    use_primal_cpp = true
    max_iterations_primal = 1000
    line_search_beta = 1e-4
    line_search_eta = 0.25

    # No rotation

```

```

    euler_angle_1 = 0
    euler_angle_2 = 0
    euler_angle_3 = 0
[]
[]

[BCs]
# YZ SHEAR: Fix back face (z=0), shear front face (z=1) in y

# Back face: completely fixed
[fix_back_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'back'
    value = 0
[]
[fix_back_y]
    type = DirichletBC
    variable = disp_y
    boundary = 'back'
    value = 0
[]
[fix_back_z]
    type = DirichletBC
    variable = disp_z
    boundary = 'back'
    value = 0
[]

# Front face: apply shear in y, fix z
[shear_front_y]
    type = FunctionDirichletBC
    variable = disp_y
    boundary = 'front'
    # Cyclic loading: load to t=240, unload to t=480
    #  $\gamma_{yz} = u_y / h = 0.001 \cdot t$ 
    # Expected yield:  $\tau_{yz} = 5060$  MPa at  $t \sim 120$  s
    function = 'if(t<=240, 0.001*t, 0.001*(480-t))'
[]
[fix_front_z]
    type = DirichletBC
    variable = disp_z
    boundary = 'front'
    value = 0
[]

# Note: disp_x on front is FREE (allows Poisson contraction)

# Left/right faces: constrain x for plane strain
[fix_left_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'left'
    value = 0
[]
[fix_right_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'right'

```

```

    value = 0
[]
[]

[Postprocessors]
[stress_xx]
    type = ElementAverageValue
    variable = stress_xx
[]
[stress_yy]
    type = ElementAverageValue
    variable = stress_yy
[]
[stress_zz]
    type = ElementAverageValue
    variable = stress_zz
[]
[stress_xy]
    type = ElementAverageValue
    variable = stress_xy
[]
[stress_xz]
    type = ElementAverageValue
    variable = stress_xz
[]
[stress_yz]
    type = ElementAverageValue
    variable = stress_yz
[]
[strain_yz]
    type = ElementAverageValue
    variable = strain_yz
[]
[plastic_strain]
    type = ElementAverageValue
    variable = plastic_strain
[]
[]

[Executioner]
    type = Transient
    dt = 0.1
    end_time = 480.0
    solve_type = 'NEWTON'
    petsc_options_iname = '-pc_type'
    petsc_options_value = 'lu'
[]

[Outputs]
    csv = true
    file_base = yz_shear_cyclic_output
    print_linear_residuals = false
[]

```

Expected results:

- Elastic slope: $G_{23} = 42.1$ GPa
- Yield: $\tau_{yz} \approx 5060$ MPa at $\gamma_{yz} \approx 0.12$

- Perfect plasticity: stress remains at 5060 MPa during loading
- Unloading: elastic recovery to zero stress
- Residual plastic strain: $\gamma_{yz}^p \approx 0.12$ at end

7.25.2 Single-Element X-Tension Test

File: single_element_x_tension.i

```
[Mesh]
  type = GeneratedMesh
  dim = 3
  nx = 1
  ny = 1
  nz = 1
[]

[GlobalParams]
  displacements = 'disp_x disp_y disp_z'
[]

[Physics/SolidMechanics/QuasiStatic]
  [all]
    strain = FINITE
    incremental = true
    add_variables = true
    generate_output = 'stress_xx stress_yy stress_zz strain_xx strain_yy
      strain_zz'
  []
[]

[AuxVariables]
  [plastic_strain]
    order = CONSTANT
    family = MONOMIAL
  []
[]

[AuxKernels]
  [plastic_strain]
    type = MaterialRealAux
    property = effective_plastic_strain
    variable = plastic_strain
  []
[]

[Materials]
  [elasticity]
    type = ComputeElasticityTensor
    fill_method = symmetric9
    C_ijkl = '171800 57500 30200 106700 46900 84200 42100 31100 46600'
  []

  [stress]
    type = ComputeMultipleInelasticStress
    inelastic_models = 'plasticity'
  []
```

```

[plasticity]
    type = OrthotropicPlasticityStressUpdate
    sigma_xx_tension = 4980
    sigma_yy_tension = 4100
    sigma_zz_tension = 5340
    tau_xy_max = 4510
    tau_xz_max = 5080
    tau_yz_max = 5060

    postyield_mode = 'exp_softening'
    residual_strength = 0.7
    kmax = 0.002
    kslope = 30.0

    viscosity_mode = 'none'

    absolute_tolerance = 1.0e-5
    max_iterations_newton = 20
    use_primal_cpp = true
[]
[]

[BCs]
# Fix left face
[fix_x]
    type = DirichletBC
    variable = disp_x
    boundary = 'left'
    value = 0
[]
[fix_y]
    type = DirichletBC
    variable = disp_y
    boundary = 'bottom'
    value = 0
[]
[fix_z]
    type = DirichletBC
    variable = disp_z
    boundary = 'back'
    value = 0
[]

# Apply displacement on right face
[pull_x]
    type = FunctionDirichletBC
    variable = disp_x
    boundary = 'right'
    function = '0.05*t' # 5% strain at t=1
[]
[]

[Executioner]
    type = Transient
    dt = 0.01
    end_time = 1.0
    solve_type = NEWTON
    petsc_options_iname = '-pc_type'

```

```

    petsc_options_value = 'lu'
[]

[Outputs]
    csv = true
    exodus = true
[]

[Postprocessors]
    [stress_xx]
        type = ElementAverageValue
        variable = stress_xx
    []
    [strain_xx]
        type = ElementAverageValue
        variable = strain_xx
    []
    [plastic_strain]
        type = ElementAverageValue
        variable = plastic_strain
    []
[]

```

Expected results:

- Elastic slope: $E_1 = 140.4$ GPa
- Yield: $S_{xx} \approx 4980$ MPa at $E_{xx} \approx 0.0355$
- Softening: Exponential decay to $0.7 \times 4980 = 3486$ MPa

7.25.3 Two-Grain Validation Test

```

[Mesh]
    [bicrystal]
        type = GeneratedMeshGenerator
        dim = 3
        nx = 4 # Elements in x-direction
        ny = 2 # Elements in y-direction
        nz = 2 # Elements in z-direction
        xmin = 0
        xmax = 2 # 2 mm length
        ymin = 0
        ymax = 1 # 1 mm width
        zmin = 0
        zmax = 1 # 1 mm height
    []

    [split_interface]
        type = BreakMeshByBlockGenerator
        input = bicrystal
        split_interface = true # Creates interface at x = 1 mm
    []
[]

[Materials]
    # Bulk grain behaviour (simplified elastic)
    [elasticity]

```

```

...
[]

[stress]
...
[]

# Interface cohesive behaviour
[cohesive_interface]
    type = HomogenizedExponentialCZM
    boundary = 'Block0_Block1' # Interface between grains

# =====
# MD-DERIVED STRENGTHS
# =====
normal_strength = 700.0          # MPa (from MD)
shear_strength_s = 400.0        # MPa (from MD)
shear_strength_t = 400.0        # MPa (from MD)

# =====
# MODE-SPECIFIC THRESHOLDS (mm)
# =====
delta_0_normal = 1.91e-4        % 191 nm
delta_0_tangent = 2.17e-4        % 217 nm

# =====
# DUAL EXPONENTS
# =====
mu = 1.0                        # Loading exponent (standard
    exponential)
eta = 1.0                       # Softening exponent (standard
    exponential)

# =====
# GH HOMOGENISATION
# =====
quality_std_dev = 0.10
spatial_quality_std_dev = 0.15
spatial_random_seed = 1234

# =====
# REGULARIZATION
# =====
damage_viscosity = 2.5
[]
[]

[BCs]
[left_fixed]
    type = DirichletBC
    variable = disp_x
    boundary = left
    value = 0
[]

[right_pull]
    type = FunctionDirichletBC
    variable = disp_x

```

```

    boundary = right
    function = '0.01*t' # 0.01 mm/s displacement
[]
[]

[Executioner]
    type = Transient
    solve_type = NEWTON

    # Standard settings work well with homogenization
    petsc_options_iname = '-pc_type -pc_factor_mat_solver_package'
    petsc_options_value = 'lu          superlu_dist'

    dt = 0.001 # 1 ms timestep
    num_steps = 100

    nl_rel_tol = 1e-6
    nl_abs_tol = 1e-8
    nl_max_its = 100
[]

[Outputs]
    exodus = true # For visualization in Paraview
    csv = true # For traction-displacement curves

    [damage_viz]
        type = Exodus
        execute_on = 'Timestep_End'
        elemental_as_nodal = true # For smooth damage visualization
    []
[]

```

7.25.4 Polycrystalline RVE with Many Interfaces

```

[Mesh]
    [polycrystal_rve]
        type = FileMeshGenerator
        file = polycrystal_100grain_rve.e # Externally generated
    []

    [split_interfaces]
        type = BreakMeshByBlockGenerator
        input = polycrystal_rve
        split_interface = true # Creates interfaces between all grains
    []
[]

[Materials]
    # Bulk grain behaviour (orthotropic plasticity)
    [grain_plasticity]
        type = OrthotropicPlasticityStressUpdate
        # (parameters from Section 7)
        # ...
    []

    # Interface cohesive behaviour
    [czm_interfaces]

```

```

type = HomogenizedExponentialCZM
boundary = 'interface' # All grain boundaries

# Similar to Example 1
normal_strength = 626.0
shear_strength = 374.0
delta_0_normal = 1.91e-7
delta_0_tangent = 2.17e-7
mu = 0.92
eta = 0.27
quality_std_dev = 0.10
spatial_quality_std_dev = 0.15
spatial_random_seed = 1234
damage_viscosity = 2.5
[]
[]

[Executioner]
type = Transient
solve_type = NEWTON

dt = 0.01 # 10 ms timestep (stable with homogenization)
num_steps = 1000

# Adaptive timestepping for progressive failure
[TimeStepper]
type = IterationAdaptiveDT
dt = 0.01
optimal_iterations = 10
growth_factor = 1.3
cutback_factor = 0.7
[]

nl_rel_tol = 1e-6
nl_abs_tol = 1e-8
nl_max_its = 100
[]

```

Chapter 8

Troubleshooting

8.1 Build Issues

8.1.1 Problem: Conda environment not activating

Error:

```
conda activate moose
# Error: conda: command not found
```

Solution:

```
# Initialize conda for your shell
conda init bash # or zsh, fish, etc.
source ~/.bashrc

# Try again
conda activate moose
```

8.1.2 Problem: Make fails with missing headers

Error:

```
fatal error: StressUpdateBase.h: No such file or directory
```

Solution:

```
# Ensure environment is active
conda activate moose
echo $CONDA_DEFAULT_ENV # Should output: moose

# Clean and rebuild
make clean
make -j8
```

8.1.3 Problem: MOOSE_DIR not found

Error:

```
Error: MOOSE_DIR not set or framework not found
```

Solution 1: Set explicitly in Makefile:

```
MOOSE_DIR := /path/to/your/moose
```

Solution 2: Create minimal structure:

```
mkdir -p ~/projects/moose/framework
mkdir -p ~/projects/moose/modules
```

8.1.4 Problem: stork.sh: command not found

Problem: MOOSE repository not cloned or script not executable.

Solution:

```
# Clone MOOSE if not already done
cd ~/projects
git clone https://github.com/idaholab/moose.git

# Check script exists
ls moose/scripts/stork.sh

# Make executable if needed
chmod +x moose/scripts/stork.sh

# Run with full path
./moose/scripts/stork.sh aragonite
```

8.1.5 Build Errors: Undefined Reference

Problem: Material classes not properly registered.

Solution: Check registration at top of .C files:

```
// Must be AFTER includes, BEFORE class definition
registerMooseObject("SolidMechanicsApp",
    OrthotropicPlasticityStressUpdate);
```

Or register with your app name:

```
registerMooseObject("aragoniteApp", OrthotropicPlasticityStressUpdate);
```

8.1.6 ModuleNotFoundError: No module named 'mooseutils'

Problem: Conda environment not activated.

Solution:

```
conda activate moose
echo $CONDA_DEFAULT_ENV # Should show: moose
```

If still fails:

```
conda install moose-dev -c conda-forge --force-reinstall
```

8.1.7 Compilation Errors in Material Files

Problem: Material source files have syntax errors or missing includes.

Solution:

1. Check error message for specific file and line number
2. Common issues:
 - Missing `#include` statements

- Typos in class names
- Mismatched function signatures between `.h` and `.C`

3. Verify files compile in isolation:

```
# Test compile single file
cd ~/projects/aragonite
make src/materials/OrthotropicPlasticityStressUpdate.C -j1

# Look for specific error messages
```

8.1.8 Linking Errors

Problem: Solid mechanics module not enabled or not found.

Solution: Check Makefile has:

```
ALL_MODULES := no
SOLID_MECHANICS := yes
```

Not:

```
ALL_MODULES := yes # Wrong - includes unnecessary modules
```

8.2 Convergence Issues

Two iterations must be told apart. The *global* Newton iteration is MOOSE's solve for the displacements; its failures appear in the console as `DIVERGED...` followed by a cut of the time step. The *local* iteration is the return mapping inside the material at one integration point; when it fails, the material throws an exception, which MOOSE also answers with a cut of the time step. The remedies differ.

One piece of older advice is withdrawn: do not loosen `absolute_tolerance` to obtain convergence. A loose local tolerance makes the returned stress inexact, the tangent inconsistent with it, and the global iteration slower (test Bt, Section 7.2).

8.2.1 Problem: Global iteration converges slowly or stalls

Symptoms: many nonlinear iterations per step with a residual that falls by a constant factor; `DIVERGED_MAX_IT`; repeated step cuts in the plastic range.

Check in this order:

1. With `strain = FINITE`, `use_finite_deform_jacobian = true` must be set. Without it MOOSE's Jacobian lacks the geometric terms and convergence is linear however good the material tangent is (tests B and Bf).
2. `tangent_operator` on `ComputeMultipleInelasticStress` must not be `elastic`. That setting is for reference runs only.
3. `absolute_tolerance = 1e-8` and `max_iterations_newton = 50` in the plasticity block.
4. The remaining settings of Table 6.7, in particular `nl_rel_tol = 1e-8` with `nl_abs_tol = 1e-6`.
5. The fallback counters printed by the material. A non-zero count of tangent fallbacks means the elastic tensor was substituted at some points (Section 2.9); find out why before changing anything else.

```

[Physics/SolidMechanics/QuasiStatic]
[all]
    strain = FINITE
    use_finite_deform_jacobian = true
[]
[]
[Materials]
[plasticity]
    absolute_tolerance      = 1e-8
    max_iterations_newton   = 50
[]
[]
[Executioner]
    line_search = bt
    nl_rel_tol   = 1e-8
    nl_abs_tol   = 1e-6
[]

```

8.2.2 Problem: Inverted elements or a failed strain update early in a step

Symptoms: exceptions reporting a negative Jacobian determinant, or a negative argument of a square root in the incremental strain computation, in the first iterations of a plastic step; the step size collapses.

Cause: `line_search = none`. While the plastic zone forms, the full Newton step overshoots (test D1s, Section 7.2.3).

Solution: `line_search = bt`.

8.2.3 Problem: Local return mapping fails

Symptoms: a material exception from `OrthotropicPlasticityStressUpdate` followed by a cut of the time step; in the diagnostic output, `return_mapping_stage` equal to 1 or 3.

Solutions (in this order):

1. Let the step cut work. An occasional cut is the intended behaviour and needs no action.
2. Check the yield surface. A warning or error about convexity at start-up means the interaction parameters violate Eq. (2.23); the return mapping has no unique solution on a non-convex surface (Section 2.5.3). In `legacy` mode the parameters `zeta12`, `zeta13`, `zeta23` are initialised to zero and must be set.
3. Raise `max_iterations_newton` to 50 if it is still at its default.
4. With a softening law, a small positive `kmax` (for example 0.002) separates first yield from the onset of softening and can help if the failure occurs at initial yielding.
5. As a regularisation, switch on viscosity. It adds a rate-dependent overstress; choose η so that the overstress of Eq. (2.40) is negligible at the applied rate (Table 7.5).
6. Keep `use_primal_cpp = true` (the default).

8.2.4 Problem: The Primal stage is entered

A count of Newton-to-Primal switches greater than zero is information, not an error: the second stage returns the same stress and tangent as the first (test P, Section 7.2.3). It is unusual, however. In none of the verification runs did the first stage fail unless it was starved on purpose.

If the count is large, look at the step size and at the convexity of the surface, and check that `max.iterations_newton` is not at a very small value.

8.2.5 Problem: Softening run differs between machines or solver settings

With a softening law and more than one element or grain, the choice of the region that localises is a bifurcation. Round-off and the path of the Newton iteration decide it, so two correct runs can differ after the bifurcation (test C). Compare such runs only up to the bifurcation, or regularise: viscosity in the bulk, `damage_viscosity` at interfaces.

8.3 Yield Stress Mismatch

8.3.1 Problem: Yield occurs at wrong stress level

Check 1: Orientation. With non-zero Euler angles the tested stress component is not aligned with a material axis. Set all three angles to zero for a first test. If the angles come from EBSD software, note that the convention of the code is the inverse of the Bunge convention (Section 2.11); Bunge angles $(\varphi_1, \Phi, \varphi_2)$ must be entered as $(180^\circ - \varphi_2, \Phi, 180^\circ - \varphi_1)$. The elasticity block and the plasticity block must receive the same three variables.

Check 2: Elastic constants. Run with `tangent_operator` unchanged but a load below yield, and compare the slope with the expected modulus, for the legacy aragonite set

$$E_1 = 140.4 \text{ GPa}, \quad E_2 = 70.3 \text{ GPa}, \quad E_3 = 63.4 \text{ GPa}.$$

If the slope is wrong, check the order of the constants:

```
C_ijkl = '171800 57500 30200 106700 46900 84200
          42100 31100 46600'
# Order: C1111 C1122 C1133 C2222 C2233 C3333 C2323 C1313 C1212
```

The last three are the shear moduli G_{23} , G_{13} , G_{12} themselves, without a factor 2.

Check 3: Initial yield ratio. In the bone models the preset strengths are ultimate strengths and first yield occurs at $r_0 = 0.7$ times that value (`initial_yield_ratio`, Section 2.6). A yield onset 30 % below the tabulated strength is therefore correct for these models.

Check 4: Interaction parameters and stress state. A strength parameter is the yield stress in the corresponding *uniaxial* or *pure shear* state only. Any additional stress component moves the yield point, by an amount that depends on ζ_{ij} and on the tension-compression asymmetry. Check the full stress tensor at yield before concluding that a strength is wrong. The analyser of the example suite does this by evaluating the yield function from the stress output (Section 7.3).

Check 5: Units. All stresses are in MPa:

```
sigma_xx_tension = 4980      # MPa, correct
sigma_xx_tension = 4.98e9    # Pa, wrong
```

Check 6: Model flag. If a bone or coral model is selected, explicitly given strengths may be ignored or may override the preset, depending on the parameter. The console lists every ignored parameter in a warning; read the start-up output.

8.4 Normal Stresses in Shear Tests

8.4.1 Problem: Normal stresses appear in a shear test

There are two causes, and only the first is a fault of the input.

Before yield: normal stresses in the elastic range of a shear test with material axes aligned to the global axes come from the boundary conditions, for example a rotation of the element that is not accounted for, or a large shear strain with `strain = FINITE`. Check the displacement boundary conditions and the strain measure.

After yield: if the yield surface has tension-compression asymmetry, the flow direction has a volumetric part even under pure shear stress (Section 2.7). When the boundary conditions prescribe all displacement components and so fix the volume, this part is constrained, compressive normal stresses build up and the shear stress rises above the pure-shear strength. This is the correct response of the model. In the example suite the effect is about -40 MPa for compact bone and about -1 MPa for trabecular bone (Section 7.3). To measure the pure-shear strength, leave the element free to change its dimensions normal to the shear plane, or compare with the yield function evaluated for the full stress state.

8.5 Cohesive Interfaces

8.5.1 Problem: Interface does not open, or fails at once

Check the length unit. `HomogenizedExponentialCZM` contains a regularisation length of 10^{-6} mesh units. With the characteristic openings of Table 3.4 the mesh must be in micrometres; entered in millimetres, the characteristic opening would be smaller than the regularisation length and the law is not resolved (Chapter 3).

Check the loading step. The displacement increment per step across an interface should be a fraction of δ_0 ; otherwise the peak of the law is stepped over.

8.5.2 Problem: Damage oscillates or jumps with `damage_viscosity`

The viscous damage update is an explicit relaxation and requires $\Delta t \leq \tau_d$. Reduce the step or increase `damage_viscosity`.

8.5.3 Problem: Global iteration slow under changing mode mix

The interface Jacobian is exact for proportional loading only (Chapter 3). When the ratio of sliding to opening changes during a step, convergence degrades to linear. Smaller steps help; the result is not affected.

8.6 Performance Issues

8.6.1 Problem: Simulation runs very slowly

Diagnose:

Check 1: Are you using optimized build?

```
ls -la | grep aragonite
# Should see: aragonite-opt (not -dbg)
```

Check 2: Where is the time spent? On large meshes the linear solve dominates: about 90 % of a step on the bone cube of Section 7.2.3. Use the preconditioner and the linear tolerance of Table 6.7; a tolerance tighter than 10^{-4} costs time without changing the Newton iteration (Table 7.6).

Check 2a: How many nonlinear iterations per step? More than about 25 on a hardening problem points to one of the items in the section on convergence above, not to a performance problem.

Check 3: Time step too small?

```
[Executioner]
  dt = 0.001 # Very small! Try 0.01
[]
```

Check 4: Use parallel execution for large meshes

```
# Run on 8 cores
mpirun -n 8 ./aragonite-opt -i input.i
```

8.6.2 Determinism and Spatial Reproducibility

The GH implementation is **fully deterministic**: the 5 quadrature nodes and weights are compile-time constants. Results are bit-identical across runs, machines, and MPI partition counts. No `random_seed` parameter is needed or accepted.

The spatial quality field (`spatial_quality_std_dev`) is also deterministic: each QP's factor is derived from `spatial_random_seed` + element-ID + QP index, independent of parallelisation.

Verification:

```
% Must give identical results:
./aragonite-opt -i rve.i && ./aragonite-opt -i rve.i
mpirun -n 8 ./aragonite-opt -i rve.i
mpirun -n 48 ./aragonite-opt -i rve.i
```

Exploring spatial patterns: Change `spatial_random_seed` for an alternative but fully reproducible spatial field.

Fully deterministic (no heterogeneity):

```
quality_std_dev          = 0.0
spatial_quality_std_dev = 0.0
```

Chapter 9

Summary

9.1 Quick Start Checklist

- ☐ Install MOOSE via conda (Chapter 4)
- ☐ Clone and build the application (Chapter 5)
- ☐ Run `scripts/run_validation.sh`: all tests match their gold files
- ☐ Run `scripts/examples/run_examples.sh`: verdict PASS
- ☐ Choose a material model (Table 6.1) and start from a template (Section 6.7)
- ☐ Apply the solver settings of Table 6.7
- ☐ After any change to the material classes: the tests of Section 7.5

9.2 Parameter Selection Strategy

Step 1: Model. Select the material model with `material_model` in `GlobalParams`. For bone give the volume fraction and, for the fabric models, the fabric eigenvalues. For coral give the three interaction parameters. Override single preset constants only where data exist.

Step 2: Post-yield law. The bone presets bring their own law. For `coral` and `legacy`, begin with `linear_hardening` or `perfect` to establish the solution, then switch to the intended softening law.

Step 3: Viscosity. Use `rate_independent` unless a rate effect or a regularisation is wanted. If it is used, estimate the overstress with Eq. (2.40) for the applied rate.

Step 4: Solver. Table 6.7 without changes. Tune only after a first run, and measure as in Section 7.2.3.

Step 5: Interfaces. Mesh in micrometres, parameters from Table 3.4 as a provisional set, $\Delta t \leq \tau_d$.

9.3 Common Pitfalls

1. **Conda environment not active:** `conda activate moose` first.
2. **Units:** stresses in MPa; one length unit throughout, micrometres as soon as cohesive interfaces are present.
3. **No line search:** `line_search = bt` is required.

4. **Finite strain without its Jacobian:** set `use_finite_deform_jacobian = true` with `strain = FINITE`.
5. **Loose local tolerance:** use `absolute_tolerance = 1e-8`; do not loosen it to obtain convergence.
6. **Interaction parameters left at zero in legacy mode:** zero is an initialisation value. Set `zeta12`, `zeta13`, `zeta23`.
7. **Different models for elasticity and plasticity by accident:** set the flag once in `GlobalParams`.
8. **Bone strengths read as yield stresses:** they are ultimate strengths; first yield is at r_0 times the strength.
9. **EBSD angles entered unchanged:** the convention is the inverse of Bunge (Section 2.11).
10. **Viscosity taken as free of consequence:** it adds an overstress proportional to the rate.
11. **Shear test with all displacements prescribed:** gives normal stresses after yield and a peak above the pure-shear strength; this is the model, not an error.
12. **Gold files as proof of correctness:** they detect change only. Use `run_tests.sh` and the examples to detect error.

9.4 What Is Verified and What Is Not

Verified with the present code (Chapter 7): the derivatives of the yield function and of the post-yield laws; the preset constants of all models against the formulas of the reference implementation; elastic modulus, yield onset and peak stress of every bone model on one element; the consistent tangent against a reference solution with and without rotation, softening and viscosity; both stages of the return mapping against each other; quadratic global convergence up to 2.4 million degrees of freedom; rate dependence and step-size independence of the viscosity; peak traction, mode mix and damage of one cohesive interface.

Not verified: the damage coupling in the bulk; a bone preset in a multi-element run; the interface model under non-proportional loading; a polycrystal with interfaces at the current parameter set. Not attempted in this document: any comparison with experiments or with a run of the reference user material.

9.5 Additional Resources

- **Repository:** <https://gitlab.tu-clausthal.de/iww/tuc-fe-moose>
- **MOOSE Documentation:** <https://mooseframework.inl.gov/>
- **MOOSE Discussion Forum:** <https://github.com/idaholab/moose/discussions>
- **Constitutive model:** Schwiedrzik et al. [1], Schwiedrzik and Zysset [4], Zysset and Schwiedrzik [5]
- **Example inputs:** `scripts/examples`; verification inputs: `test/verification`

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